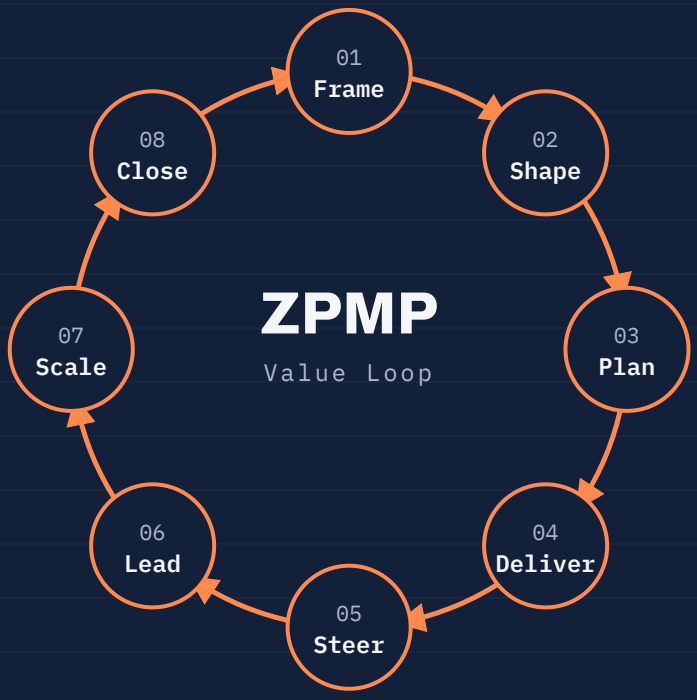
ZPMP

ZyeemZia™ Project Management Protocol

Deliver value, not just plans.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 39 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A modern, practical certification track for people who run projects: predictive, agile and hybrid delivery, risk and earned value, leadership, and AI-powered project management, organised around one memorable framework.

01
Learn
Work through 39 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 40-question online exam at zpmp.zyeemzia.com. Pass mark 80%.

04
Get certified
Download your certificate online. 90%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Week 1	m1 Frame	Read lessons, build your project charter	☐
Week 2	m2 Shape	Stakeholder map and backlog for your capstone	☐
Week 3	m3 Plan	Estimates, critical path and budget	☐
Week 4	m4 Deliver	Run a two-week sprint or Kanban board	☐
Week 5	m5 Steer	Risk register and EVM practice	☐
Week 6	m6 Lead	Team charter, RACI, communication plan	☐
Week 7	m7 Scale	AI-use plan and change-management plan	☐
Week 8	m8 Close	Practice exam, capstone pack, certification exam	☐

Learner name .....

Start date ..... Target exam date .....

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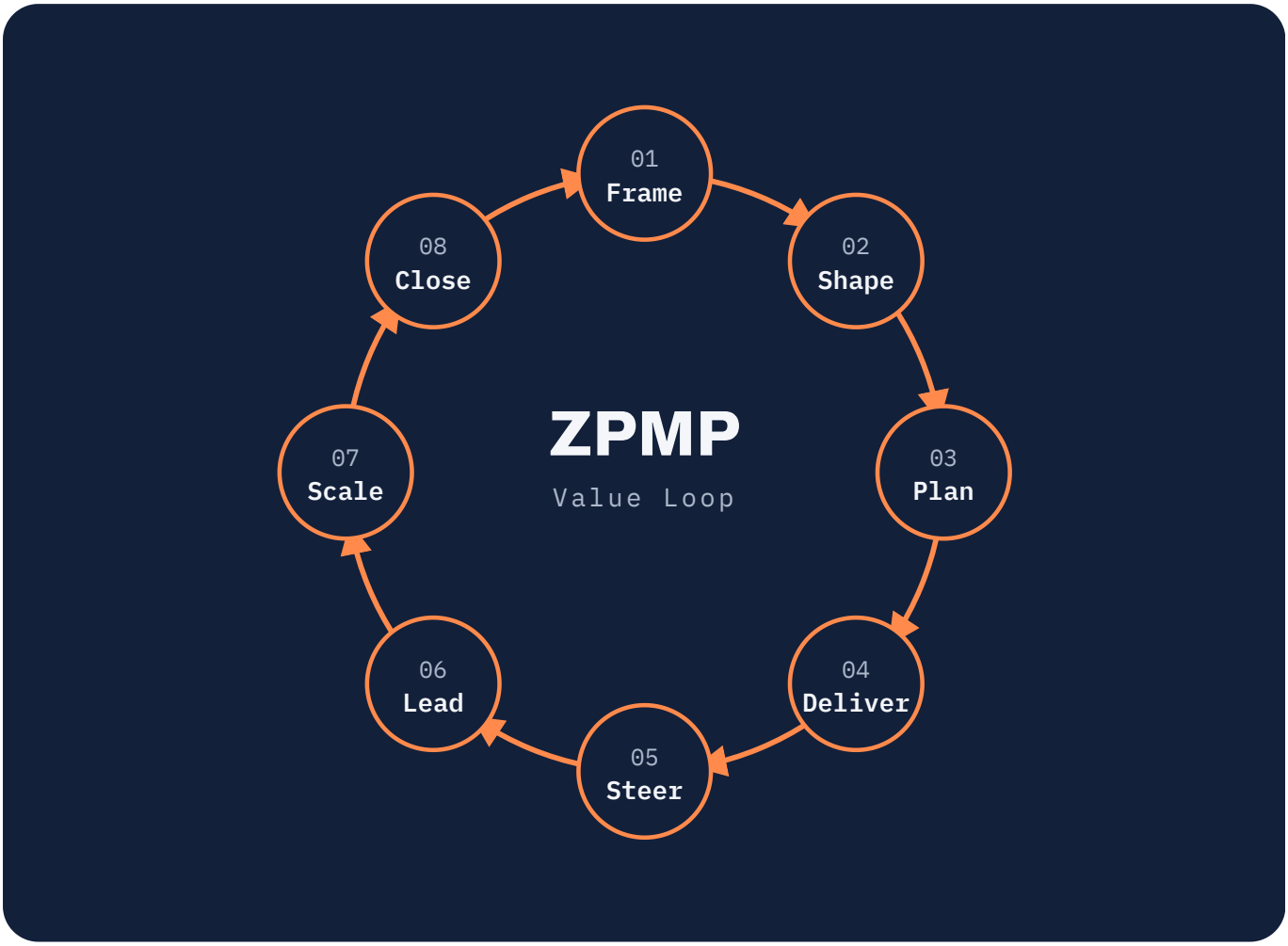
+ **Glossary · Capstone guide · Practice exam & answers · Certificate · Notes**

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THE FRAMEWORK

The ZPMP Value Loop

Eight stages every successful project moves through, again and again: frame the value, shape the work, plan, deliver, steer, lead, scale and close.



01 Frame
Frame the work: what projects are, how value is created and how to choose the right approach

02 Shape
Shape the work: understand the people, discover the real needs and keep scope under control

03 Plan
Plan the work: honest estimates, realistic schedules and budgets that hold up under uncertainty

04 Deliver
Deliver the work: run Scrum and Kanban well, design hybrids on purpose and build quality in

05 Steer
Spot uncertainty early, measure progress truthfully and forecast the finish with confidence.

06 Lead
Grow teams that trust each other, resolve conflict well and communicate with purpose.

07 Scale
Multiply your impact with AI and automation, lead adoption and connect projects to strategy responsibly.

08 Close
Finish well, secure the value, act with integrity and prepare for the ZPMP exam and capstone.

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01

MODULE 1 · FRAME

Foundations of Modern Project Management & Value

Frame the work: what projects are, how value is created and how to choose the right approach

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Distinguish projects, operations, products, programs and portfolios and explain how they connect to strategy
- Apply the principles and performance domains of modern project management to real decisions
- Select and tailor a predictive, agile or hybrid life cycle using an uncertainty matrix
- Build a lean business case, benefits register and OKR link for a project
- Draft a one-page project charter and describe the modern project manager's role

LESSONS

1. What a Project Is Today · 22 min
2. The Value Mindset, Principles & Performance Domains · 25 min
3. Life Cycles & Tailoring the Approach · 28 min
4. Business Case, Benefits & OKRs · 26 min
5. The Project Charter & the Modern Project Manager · 25 min

MODULE 1 · FRAME · LESSON 1 · 22 MIN

What a Project Is Today

Projects, operations, products, programs and portfolios — and where the ZPMP Value Loop fits

- Define a project in modern, value-focused terms
- Distinguish projects from operations and long-lived products
- Explain how projects, programs and portfolios connect to strategy
- Describe the eight stages of the ZPMP Value Loop

Every organisation runs on two engines. One keeps today's business working: payroll, customer service, production lines. The other changes the business: a new app, a new hospital wing, a new way of working. That second engine is **project work**, and in 2026 it is where most competitive advantage is won or lost.

DEFINITION · PROJECT

A **project** is a temporary, purposeful effort that creates a unique result — a product, service, capability or change — so that an organisation can move from where it is now to a better future state. The effort ends; the result is meant to keep creating value.

Four quick tests: is it a project?

- **Temporary** — it has a planned start and a planned finish (the result may last for decades).
- **Unique** — the result, context or team has not been combined in exactly this way before.
- **Purposeful** — it exists to deliver a change that someone values, not just to keep busy.
- **Progressively elaborated** — details become clearer as you learn, so plans are refined over time.

Projects vs operations vs products

Project

- Temporary effort with an end date
- Creates something new or changes something
- Team often assembled for the effort
- Success = outcomes and benefits enabled
- Example: building a mobile wallet's savings feature

Operations

- Ongoing and repetitive
- Sustains the business as it is
- Stable, permanent teams and processes
- Success = efficiency, reliability, service levels
- Example: processing 50,000 wallet transactions a day

Product

- Long-lived offering with its own life cycle
- Evolves through many projects or continuous releases
- Often owned by a stable, cross-functional team
- Success = customer and business value over time
- Example: the wallet app itself, improved every sprint

INSIGHT · PROJECT THINKING MEETS PRODUCT THINKING

Many digital teams no longer 'finish' a project and walk away. They fund a **stable product team** and deliver change continuously. Project management skills still matter here — scope, risk, stakeholders, money and time do not disappear — they are simply applied in shorter, repeating loops.

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Where projects live: from strategy to daily work

Strategy	`Vision and mission` `Strategic objectives and OKRs` `Risk appetite`
Portfolio	`Chooses and balances investments` `Funds what best serves strategy` `Stops what no longer fits`
Program	`Coordinates related projects` `Manages shared dependencies` `Delivers combined benefits`
Projects and products	`Create the deliverables` `Enable outcomes` `Hand over to operations`
Operations	`Run the new capability` `Realise most of the benefits` `Feed back improvement ideas`

Project, program and portfolio at a glance

ASPECT	PROJECT	PROGRAM	PORTFOLIO
Main question	Are we building this right?	Are these related efforts adding up to the intended benefits?	Are we investing in the right things?
Scope	A defined result	A set of related projects and activities	All investments in a business area or company
Time frame	Defined start and finish	Usually longer, ends when benefits are secured	Ongoing
Measured by	Outcomes enabled, delivered within agreed constraints	Benefits realised across components	Strategic return and balance of the whole mix
Typical leader	Project manager or delivery lead	Program manager	Portfolio manager or investment board

EXAMPLE · ONE COMPANY, EVERY LEVEL

A Dhaka-based mobile financial services company runs **customer support** (operations) for its **wallet app** (product). This quarter it is building a **micro-savings feature** (project) as part of a **financial inclusion program** that also includes agent training and a rural marketing campaign. The executive team decides how much money goes to inclusion versus a new merchant-payments line — that is **portfolio** management.

The ZPMP Value Loop — eight stages

1

Frame

Clarify the problem, the value sought, the approach and the authority to proceed.

2

Shape

Understand stakeholders, define scope and requirements, decide what matters most.

3

Plan

Estimate, sequence, schedule and budget at a level of detail that fits the uncertainty.

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4
Deliver
Build and release value through agile, predictive or hybrid execution with built-in quality.

5
Steer
Measure performance, manage risk and correct course using evidence.

6
Lead
Build the team, communicate, resolve conflict and keep people aligned.

7
Scale
Use digital and AI tools, manage organisational change and connect to the portfolio.

8
Close
Hand over, confirm benefits, capture lessons and feed them into the next loop.

PRO TIP · WHY A LOOP, NOT A LINE

Stages overlap and repeat. A two-week sprint runs through Plan, Deliver and Steer many times; Lead never stops; what you learn in Close becomes input to the next Frame. Use the loop as a **thinking checklist**, not as a rigid sequence of phases.

SCENARIO · DECIDE · PROJECT OR OPERATIONS?

You work at a Chattogram logistics company. Your manager says: "Please project-manage our monthly payroll run from now on — make a Gantt chart and run it as a project every month." Payroll has run the same way for six years without major problems. What is the best response?

(A) Agree and create a new project plan, charter and Gantt chart every month.

(B) Explain that payroll is operations best managed with a stable process, checklist and service levels — and offer to run a separate project if the goal is to improve or automate payroll.

(C) Refuse, because project managers should not touch operational work.

(D) Turn payroll into a Kanban board and call it an agile project.

Circle your choice. Answer and feedback: module answer key.

What makes project work different in 2026

- Value and outcomes matter more than finishing a list of tasks.
- Teams are often distributed across cities and time zones, working through digital tools.
- AI-assisted tools draft plans, summarise meetings and flag risks — people still own the judgement (covered in Scale).
- Regulation, data privacy and sustainability expectations change mid-project.
- Most organisations blend predictive and agile methods rather than picking one.

Key terms

Project
A temporary, purposeful effort that creates a unique product, service, capability or change to move an organisation toward a better future state.

Operations
Ongoing, repetitive work that sustains the organisation's existing products and services.

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Program
A group of related projects and activities managed together so their combined benefits are greater than if they were managed separately.

Portfolio
The full set of projects, programs and operational investments an organisation manages to achieve its strategy, selected and balanced as a whole.

KNOWLEDGE CHECK

1. Which of these is the clearest example of a project?

   (A) Answering customer calls at a bank's contact centre every day
   (B) Producing 20,000 shirts per week on an existing production line
   (C) Migrating a bank's core system to a cloud platform by next March
   (D) Running the monthly payroll for all staff across the company's branches

2. A company groups an app launch, agent training and a marketing campaign so that together they increase rural financial inclusion. This group is best described as a:

   (A) Portfolio
   (B) Program
   (C) Product
   (D) Operation

3. What does 'progressively elaborated' mean?

   (A) The project grows larger every month
   (B) Details are refined as more becomes known
   (C) The plan is fixed on day one and never changed
   (D) Stakeholders are added gradually

4. In the ZPMP Value Loop, which stage focuses on measuring performance, managing risk and correcting course?

   (A) Steer
   (B) Shape
   (C) Deliver
   (D) Close

5. Why do many digital organisations fund stable product teams rather than one-off projects?

   (A) Because project management is no longer needed in digital firms
   (B) Because product work does not need budgets or funding decisions
   (C) To deliver change continuously and keep product knowledge in one team
   (D) To reduce how often teams need to talk directly to customers

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 2 · 25 MIN

The Value Mindset, Principles & Performance Domains

Why modern project management starts with value — and the principles and domains that guide judgement

- Trace the chain from outputs to outcomes, benefits and value
- Explain the twelve principles of modern project management in your own words
- Describe the eight performance domains and what good looks like in each

A project can finish on time, on budget and to specification — and still be a failure. If nobody uses the new system, if the clinic is built where patients cannot reach it, if the feature solves a problem customers do not have, the organisation has spent money and gained nothing. Modern project management therefore measures success by **value delivered**, not tasks completed.

From work to value

01	→	02	→	03	→	04	→	05
Activities The work the team does		**Outputs** Deliverables: the app, the building, the report		**Outcomes** What changes because people use the outputs		**Benefits** Measurable gains for the organisation or society		**Value** Worth to stakeholders after cost, risk and time

EXAMPLE · A HOSPITAL APPOINTMENT APP

Output: a booking app is live. **Outcome:** 60% of patients book online instead of queuing. **Benefit:** phone calls fall by a third and average waiting time drops by 25 minutes. **Value:** a calmer front desk, staff time redeployed to patient care and better patient experience — worth far more than the build cost.

Organisations create value through a connected **value delivery system**: portfolios choose investments, programs and projects create capabilities, products evolve them and operations run them. Information must flow both ways — strategy guides projects, and evidence from projects corrects strategy.

Twelve principles of modern project management (the PMBOK 7th edition set, in ZPMP words)

P1
Act as a responsible steward
Look after money, people, trust and the environment as if they were your own, and act with integrity.

P2
Build a team that works together
Create a safe, collaborative environment where people share accountability for results.

P3
Engage stakeholders genuinely
Involve the people affected early and often, and understand their interests and influence.

P4
Keep value at the centre
Constantly check that the work still leads to the intended outcomes and benefits.

P5
Think in systems
See the project as a set of interacting parts within a wider organisation; changes ripple.

P6
Lead, whatever your title
Motivate, influence and set direction — leadership is a behaviour, not a job title.

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P7
Tailor the approach
Fit methods, processes and documents to the context instead of applying one recipe everywhere.

P8
Build quality in
Prevent defects through good design and clear acceptance criteria rather than inspecting them out later.

P9
Navigate complexity
Expect surprises from human behaviour, ambiguity and technology; use feedback to find a way through.

P10
Respond to risk wisely
Manage threats and opportunities so the overall risk exposure supports the goals.

P11
Stay adaptable and resilient
Absorb setbacks, recover quickly and adjust plans as conditions change.

P12
Enable change
Help people adopt the new way of working so the future state is actually reached.

INSIGHT · PRINCIPLES GUIDE JUDGEMENT; PROCESSES PRESCRIBE STEPS
A process says *what to do*. A principle says *what matters* when the process does not fit. When you meet a situation no template covers — and you will — fall back on the principles.

Eight performance domains — where effort must go well

D1
Stakeholders
Productive relationships and agreement with the people who influence or are affected by the project.

D2
Team
Shared ownership, high performance and healthy team culture.

D3
Development approach and life cycle
A delivery approach and phase structure that suit the product and context.

D4
Planning
Enough organised, coordinated planning to deliver — no more, no less.

D5
Project work
Efficient processes, resource and supplier management, and continuous learning.

D6
Delivery
Scope and quality that achieve the intended outcomes.

D7
Measurement
Reliable data to understand performance and act early (developed in Steer).

D8
Uncertainty
Awareness of risk, ambiguity, complexity and volatility, with responses that fit (developed in Steer).

WATCH OUT · EDITIONS CHANGE — IDEAS LAST
Professional standards are revised from time to time and may regroup principles and domains under new names. Learn the **ideas** here, then check the exact edition and outline used by any external exam you plan to take.

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Output mindset vs value mindset

Output mindset

- "Did we deliver every feature on the list?"
- Success declared at go-live
- Change requests seen as threats
- Reports show % of tasks complete
- PM acts as task tracker

Value mindset

- "Did the change achieve the outcome we needed?"
- Success confirmed when benefits appear
- Change welcomed when it increases value
- Reports show outcomes, risks and decisions needed
- PM acts as value navigator and integrator

SCENARIO · DECIDE · DELIVERED — BUT WILL IT BE USED?

You are managing a field-sales app for a Dhaka consumer-goods company. The sponsor's main concern is go-live on 1 March. In user tests, sales officers say the order screen is too slow on low-end phones in poor-signal areas. Fixing it would take two extra weeks. What should you do FIRST?

(A) Go live on 1 March as promised; performance can be fixed later.

(B) Quietly add two weeks to the schedule and inform the sponsor afterwards.

(C) Bring the evidence to the sponsor with options — e.g. a smaller release for good-signal regions on 1 March plus the performance fix two weeks later — and let the sponsor decide based on value.

(D) Ask the developers to work weekends until it is fixed by 1 March.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a project you know that finished on time and on budget but disappointed its users or sponsor. Which link in the chain — output, outcome, benefit or value — broke, and which principle would have helped most?

KNOWLEDGE CHECK

6. Patients now book appointments online instead of queuing. In the value chain this is best described as:
   (A) An output (B) An activity
   (C) A constraint (D) An outcome
7. Which principle is most directly applied when a PM adapts documents and ceremonies to a small, low-risk project?
   (A) Build quality in (B) Stay adaptable and resilient
   (C) Tailor the approach (D) Think in systems
8. A change in one team's API causes delays in two other teams. Which principle best helps the PM anticipate this?
   (A) Act as a responsible steward (B) Lead, whatever your title
   (C) Stay adaptable (D) Think in systems

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9. Which performance domain focuses on choosing between predictive, adaptive and hybrid delivery?

   (A) Development approach and life cycle
   (B) Planning
   (C) Delivery
   (D) Project work

10. Why are principles especially useful to a project manager?

   (A) They replace the need for any defined process or methodology
   (B) They are only needed on agile projects, not on predictive ones
   (C) They guarantee success when every principle is applied in full
   (D) They guide judgement when no process or template fits

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 3 · 28 MIN

Life Cycles & Tailoring the Approach

Predictive, iterative, incremental, agile and hybrid — and how to choose with confidence

- Compare predictive, iterative, incremental, agile and hybrid life cycles
- Use an uncertainty matrix to select a starting approach
- Tailor practices to the project, team and organisation

A **project life cycle** is the series of phases a project passes through from start to finish. The **development approach** is how the product is created within those phases. Choosing the wrong approach is one of the most expensive early mistakes a project can make — and one of the easiest to avoid.

Five life cycles compared

APPROACH	HOW IT WORKS	REQUIREMENTS	WHEN VALUE ARRIVES	BEST FIT
Predictive	Plan the whole effort up front, then execute in sequence	Fixed early	Mostly at the end	Stable scope, high cost of change, e.g. building construction
Iterative	Repeat cycles to refine the solution through feedback	Clarified through cycles	At the end, but solution improves each cycle	Unclear solution, e.g. new service design
Incremental	Deliver usable pieces one after another	Defined per increment	Each increment adds value	Value can be split, e.g. district-by-district rollout
Agile	Short iterative and incremental cycles with constant feedback	Evolve continuously	Frequently, often every 1–4 weeks	High uncertainty, fast learning needed, e.g. digital products
Hybrid	Combine elements deliberately	Mixed: some fixed, some evolving	Depends on the design	Mixed work types, regulation plus innovation

DEFINITION · ITERATIVE VS INCREMENTAL

Iterative = making the *same thing better* through repeated cycles (sketch → prototype → refined prototype). **Incremental** = adding *more pieces* of finished value over time (module 1, then module 2). Agile combines both.

A typical hybrid life cycle

01 **Initiate** — Charter and business case approved →

02 **Design and approvals** — Architecture, compliance and vendor contracts fixed →

03 **Build in sprints** — Features delivered and demoed every two weeks →

04 **Integration and sign-off** — Regulatory and security acceptance gate →

05 **Rollout in waves** — Region by region, learning between waves →

06 **Close and transition** — Handover to operations, lessons captured

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To choose a starting approach, ask two questions. **How clear and agreed are the requirements?** And **how certain is the way we will build it** — the technology or method? Plotting your project on these two axes is a practical version of the well-known Stacey-style complexity matrix.

The ZPMP uncertainty matrix

ZONE	REQUIREMENTS	TECHNOLOGY OR METHOD	SUGGESTED STARTING APPROACH
Clear	Known and agreed	Proven	Predictive; use standard plans and checklists
Complicated	Mostly known	Needs expert analysis	Predictive or hybrid with expert reviews and prototypes for risky parts
Complex	Evolving or disputed	Partly unknown	Agile or hybrid; short cycles, frequent feedback, small experiments
Chaotic	Unclear and changing fast	Unknown	Act to stabilise first (crisis mode), then move to a complex-zone approach

Tailoring factors

FACTOR	LEANS PREDICTIVE WHEN…	LEANS ADAPTIVE WHEN…
Requirement stability	Scope is fixed by law, design or contract	Users will discover needs as they see the product
Cost of change	Late change is very expensive (concrete, hardware)	Change is cheap (software, content, process)
Delivery options	Value only exists when everything is complete	Value can be released in usable slices
Regulation and safety	Heavy documentation and formal approvals are mandatory	Lighter governance is acceptable
Stakeholder availability	Customer reviews only at fixed gates	A product owner or users are available every week
Team experience	Team is used to phase-based delivery	Team is cross-functional and experienced with agile

Tailoring in five moves

1. **Diagnose** Assess uncertainty, size, criticality, regulation, culture and team capability.
2. **Pick a base** Choose a starting approach — predictive, agile or a hybrid pattern your organisation already knows.
3. **Adjust** Add, remove or scale practices: lighter documents for small projects, extra reviews for safety-critical ones.
4. **Agree** Confirm the approach with the sponsor, governance bodies and the team so expectations match.
5. **Inspect and adapt** Review the approach itself at retrospectives and phase gates; change what is not working.

> **WATCH OUT · BEWARE THE WORST OF BOTH WORLDS**
> Some teams call themselves hybrid but simply run sprints inside a big upfront specification followed by a single big-bang release. They carry the cost of agile ceremonies **and** the risk of late feedback. Good hybrid design is deliberate: each element is there for a reason.

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PRO TIP · TAILOR AT THREE LEVELS

Your **organisation** sets the framework (governance, gates, standard templates). Each **project** tailors that framework to its context. Each **team** then tailors its own working agreements. Write down what you tailored and why — auditors and new team members will thank you.

SCENARIO · DECIDE · CHOOSING AN APPROACH FOR A BANK

A commercial bank in Dhaka must deliver a new regulatory reporting system by a fixed central-bank deadline, with requirements defined in a published circular. In the same project the bank wants a new customer self-service portal whose features are still being debated by marketing. What approach should the PM recommend?

(A) Fully predictive for both parts, with a detailed specification signed off before any work begins.

(B) Fully agile for both parts, with no fixed milestones.

(C) A hybrid: predictive planning and formal sign-off for the regulatory component, agile sprints with frequent feedback for the portal, and a shared integration and release plan.

(D) Delay the portal until the regulatory system is complete, then decide.

Circle your choice. Answer and feedback: module answer key.

Approach selection checklist

☐ I have rated requirement clarity and technical certainty for each major deliverable.

☐ I know where late change would be very expensive and where it is cheap.

☐ I have checked whether value can be released in usable increments.

☐ Mandatory regulations, approvals and contract terms are identified.

☐ A product owner or user representative is available at the cadence the approach needs.

☐ The team has the skills — or the coaching — the chosen approach requires.

☐ Phase gates and governance checkpoints are mapped onto the life cycle.

☐ The chosen approach and the reasons for tailoring are written down and agreed with the sponsor.

☐ A date is set to review whether the approach is working.

KNOWLEDGE CHECK

11. A team delivers a payroll module first, then leave management, then performance reviews — each usable on release. This is mainly:

    (A) Iterative development
    (B) Predictive development
    (C) Prototype-driven development
    (D) Incremental development

12. Requirements are disputed and the technology is new to the team. In the uncertainty matrix this project is most likely:

    (A) Clear
    (B) Complicated
    (C) Complex
    (D) Chaotic

13. Which factor most strongly favours a predictive approach?

    (A) Users want to see early versions
    (B) The cost of late change is very high
    (C) The team is cross-functional
    (D) Requirements will emerge from customer feedback

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14. What is the main danger of a poorly designed hybrid?

(A) It always costs more than a purely predictive or agile approach

(B) It may combine heavy upfront specification with late feedback

(C) It removes the need for a sponsor and for formal governance gates

(D) It forbids sprints in any phase that has a fixed regulatory deadline

15. What should a PM do after choosing and tailoring an approach?

(A) Review it periodically and adapt it if it is not working

(B) Lock it in for the whole project to give stability

(C) Ask the team to follow it without further discussion

(D) Replace it every sprint to experiment with new practices

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 4 · 26 MIN

Business Case, Benefits & OKRs

Justify the investment, define the benefits and connect the project to strategy

- Outline the parts of a lean business case
- Calculate and interpret payback, ROI, NPV and benefit–cost ratio
- Write a benefits register entry and link a project to OKRs

Before an organisation commits people and money, someone must answer a simple question: **why this, why now, and is it worth it?** The **business case** answers that question. It is not a one-time formality — it is the yardstick you return to whenever a major decision or change arises.

Anatomy of a lean business case

1. **Problem or opportunity** What is wrong or possible today? Use evidence: data, complaints, market signals.
2. **Options** At least three, including 'do nothing' and 'do the minimum', so the decision is a real choice.
3. **Benefits** What measurable improvements will each option bring, and when?
4. **Costs and funding** One-time and ongoing costs, and where the money comes from.
5. **Risks and assumptions** What could make the case fail, and what are we assuming to be true?
6. **Recommendation** The preferred option, the decision criteria and the next step requested.

Common investment measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Payback period	`Initial investment ÷ annual net cash inflow (for even inflows)`	How long until the money spent is recovered. Shorter is safer, but it ignores what happens after payback.
Return on investment (ROI)	`(Total benefits − Total costs) ÷ Total costs × 100%`	Gain per unit of cost. Simple, but ignores the timing of money.
Net present value (NPV)	`Σ [Cash flow in year t ÷ (1 + r)^t] − Initial investment`	Value today of future cash flows, discounted at rate r. Positive NPV means the investment beats the required return.
Benefit–cost ratio (BCR)	`Present value of benefits ÷ Present value of costs`	Greater than 1 means benefits exceed costs in today's money.

EXAMPLE · SAME PROJECT, DIFFERENT VERDICTS

A garment exporter can spend **Tk 50 lakh** on an automated cutting-room system that saves **Tk 20 lakh a year for 3 years**. Payback = 50 ÷ 20 = **2.5 years**. Simple ROI = (60 − 50) ÷ 50 = **20%**. But at a 10% discount rate the savings are worth 18.18 + 16.53 + 15.03 = **Tk 49.74 lakh** today, so NPV ≈ **−Tk 0.26 lakh**. Lesson: the timing of money can turn an apparently good case into a marginal one — present more than one measure.

PRO TIP · DO NOT DECIDE ON MONEY ALONE

Some valuable projects have weak financial returns: regulatory compliance, safety, sustainability and staff wellbeing. Score them on **strategic fit, risk reduction and obligation** as well as finance. Equally, a great NPV does not rescue a project that does not fit the strategy.

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Benefits management ensures that the gains promised in the business case actually happen. Most benefits appear **after** the project closes, when operations use the new capability. That is why each benefit needs a named **owner** in the business — usually not the project manager — plus a baseline, a target and a date to measure.

TEMPLATE **Benefits register**

Record each expected benefit so it can be tracked from business case to realisation.

BENEFIT	MEASURE (KPI)	BASELINE	TARGET AND DATE	BENEFIT OWNER	HOW AND WHEN MEASURED
Faster customer onboarding	*Median minutes to open an account*	*18 min*	*6 min by June 2027*	*Head of Retail Operations*	*Monthly report from onboarding system*
Lower support load	*Assisted tickets per 1,000 sign-ups*	*120*	*70 by Sept 2027*	*Customer Service Manager*	*Quarterly ticket analysis*

OKRs (Objectives and Key Results) are a popular way to connect strategy to work. An **objective** is a short, inspiring, qualitative statement of where you want to go. **Key results** are two to five measurable outcomes that show you got there. Projects and initiatives are the bets you place to move the key results.

OKRs, KPIs and project success criteria

OKR

- Sets direction for a period (often a quarter or year)
- Ambitious, may not be fully reached
- Measures outcomes, not tasks
- Owned by a business team

KPI

- Monitors the health of ongoing operations
- Has normal ranges and alerts
- Continues indefinitely
- Example: system uptime, cost per order

Project success criteria

- Agreed at the start, recorded in the charter
- Define when this project is a success
- Mix of delivery and outcome measures
- Should trace to OKRs and the business case

EXAMPLE · OKR IN ACTION

Objective: Make digital account opening effortless for first-time customers in rural districts. **KR1:** Raise application completion from 42% to 70%. **KR2:** Cut median onboarding time from 18 to 6 minutes. **KR3:** Reduce agent-assisted tickets per 1,000 sign-ups by 40%. **Project bet:** redesign the e-KYC flow with offline-first photo capture.

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WATCH OUT · OUTPUTS DISGUISED AS KEY RESULTS

"Launch the new app" is a task, not a key result. Ask: *if we do this, what number changes?* Write that number as the key result.

SCENARIO · DECIDE · TWO PROJECTS, ONE BUDGET SLOT

The portfolio board can fund one project this year. Project A (a merchant loyalty scheme) shows a high ROI but does not support any current company objective. Project B (an e-KYC redesign) shows a modest positive NPV and directly moves two of this year's key results. As the PM preparing the recommendation, what is BEST?

(A) Recommend A because ROI is the most objective measure.

(B) Recommend B, showing its strategic alignment, financial measures and risks, and note A as a candidate for a future cycle.

(C) Recommend funding both by cutting each budget in half.

(D) Avoid making any recommendation; it is the board's job.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

16. A project costs Tk 30 lakh and returns Tk 10 lakh a year in even net inflows. What is the payback period?

(A) 2 years
(B) 3 years
(C) 10 years
(D) 30 years

17. Which measure accounts for the time value of money?

(A) Simple return on investment
(B) Payback period
(C) Percent complete
(D) Net present value

18. Who should normally own a business benefit such as 'lower support costs'?

(A) A business manager who will run the new capability
(B) The project manager who delivered the capability
(C) The vendor that supplied and supports the system
(D) The quality team that verified the deliverables

19. Which is a well-written key result?

(A) Launch the redesigned app to all customers by June
(B) Increase completion rate from 42% to 70% by June
(C) Hold weekly product meetings with the sales team
(D) Hire two senior developers before the end of March

20. Why should a business case include a 'do nothing' option?

(A) Because it is usually the cheapest and lowest-risk choice
(B) Because regulators require it in every jurisdiction
(C) To keep the document short by limiting the options
(D) To expose the cost and risk of inaction for comparison

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 5 · 25 MIN

The Project Charter & the Modern Project Manager

Authorise the project properly and step into the role with the right mindset

- Explain the purpose and contents of a project charter
- Draft a one-page charter for a real initiative
- Describe the responsibilities and mindset of a modern project manager

The **project charter** is the birth certificate of a project. Issued by the sponsor, it formally authorises the project to exist, names the project manager and gives that person authority to use organisational resources. Without one, you are leading work nobody has officially agreed to.

Business case, charter and management plan

Business case

- Answers: should we invest?
- Options, benefits, costs, risks
- Owned by the sponsor
- Revisited at major decisions

Project charter

- Answers: what is authorised and who leads it?
- Purpose, objectives, high-level scope, milestones, budget, authority
- Signed by the sponsor
- Short — often one to three pages

Project management plan

- Answers: how will we deliver and control it?
- Baselines, subsidiary plans, approach, governance
- Owned by the PM with the team
- Living document, updated through change control

TEMPLATE

One-page project charter

A lean charter the sponsor can read in five minutes and sign. Fill one row per section.

SECTION	GUIDING QUESTION	YOUR PROJECT'S ANSWER
Purpose	*Why does this project exist, and why now?*	*Field staff register flood-affected households on paper; data reaches head office 3 weeks late, delaying aid.*
Success criteria	*How will we know it worked?*	*95% of registrations synced within 48 hours; aid disbursed within 10 days of registration by Dec 2027.*

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Sections to include in your charter

- Purpose and link to strategy or OKRs
- Measurable objectives and success criteria
- High-level scope — what is in and what is explicitly out
- Key deliverables and summary milestones
- Summary budget and funding source
- Key stakeholders, sponsor and named project manager with authority level
- High-level risks, assumptions and constraints
- Approval requirements: who signs off what, and when

CASE STUDY · CHARTER FOR AN NGO DIGITAL REGISTRATION ROLLOUT

A national NGO is replacing paper-based beneficiary registration with a mobile app across 40 unions in flood-prone districts of Sylhet division. The donor has approved funding of USD 420,000 over 18 months and wants quarterly progress reports. Field officers have mixed smartphone skills; connectivity is poor in many areas. The country director is the sponsor; a regional programme coordinator has been asked to manage the project. Data protection rules for beneficiary information must be followed.

1. Who should issue and sign the charter, and why?
2. Write two measurable success criteria.
3. Name two items you would place explicitly out of scope.
4. Which life cycle would you propose, and why?

Model answer: module answer key.

The modern project manager is less a task-master and more a **value-focused integrator and leader**. You connect strategy to delivery, people to purpose and plans to reality. On agile teams some traditional PM duties are shared by a product owner, a team coach and the team itself; your contribution shifts toward coordination across teams, stakeholders, risk, budget and governance.

Six stances of the modern project manager

M1

Value navigator

Keeps every decision tied to outcomes, benefits and the business case.

M2

Integrator

Joins scope, schedule, cost, risk, quality and people into one coherent plan.

M3

Servant leader

Removes obstacles, protects focus and grows the team's capability.

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M4
Adaptive planner
Plans in the right level of detail and updates plans as learning arrives.

M5
Trusted communicator
Shares honest status, early bad news and clear options with stakeholders.

M6
Ethical steward
Acts with honesty, fairness, respect and responsibility for people and planet.

Who does what across approaches

RESPONSIBILITY	PREDICTIVE PROJECT	AGILE TEAM	HYBRID PROJECT
Setting priorities	PM with sponsor via baselines	Product owner, continuously	Product owner within PM-managed milestones
Detailed planning	PM with team, up front	Team, each iteration	Both, at different levels
Removing impediments	PM	Team coach / Scrum Master and team	PM and team coach together
Stakeholder and budget governance	PM	Often a PM, product owner or delivery lead outside the team	PM

INSIGHT · AUTHORITY VS INFLUENCE

A charter gives you **formal authority**, but most of the time you will lead people who do not report to you. Your real power comes from credibility, relationships and clarity of purpose — themes developed in the Lead stage.

SCENARIO · DECIDE · NO CHARTER, BUT 'JUST START'

A director calls you in: "The board loves this idea. Start the vendor selection tomorrow — paperwork can come later." There is no charter, no confirmed budget and no named sponsor. What should you do FIRST?

(A) Start vendor meetings tomorrow to show enthusiasm.

(B) Refuse to do anything until a full project management plan is approved.

(C) Draft a one-page charter with the director within a day or two, confirm who the sponsor is and get it signed before engaging vendors.

(D) Email the whole company announcing the project.

Circle your choice. Answer and feedback: module answer key.

Key terms

Value
The worth, importance or usefulness of a result to stakeholders, after considering cost, risk and time.

Business case
A documented justification for an investment that compares options, benefits, costs and risks and recommends a way forward.

Project charter
A document issued by the sponsor that formally authorises a project and gives the project manager authority to apply resources.

Benefit
A measurable improvement that results from an outcome and is valued by the organisation or society.

OKR
Objectives and Key Results: a qualitative objective paired with a few measurable outcomes that show progress toward it.

Life cycle
The series of phases a project passes through from start to completion.

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Tailoring
Deliberately adapting the approach, processes and documents to fit the context of a project, team and organisation.

Hybrid approach
A deliberate combination of predictive and adaptive practices within one project.

KNOWLEDGE CHECK

21. Who normally issues and signs the project charter?

(A) The sponsor or initiating authority
(B) The project manager, once appointed
(C) The lead vendor or delivery partner
(D) The quality manager for the programme

22. Which item belongs in a charter rather than in the detailed management plan?

(A) A full resource-levelled schedule
(B) High-level scope with key exclusions
(C) Day-by-day test scripts for each feature
(D) The detailed sprint backlog for release one

23. On an agile team, who usually owns day-to-day prioritisation of the backlog?

(A) The project manager
(B) The sponsor
(C) The product owner
(D) The finance team

24. A PM leads specialists who report to other managers. Their main source of power is likely to be:

(A) Formal authority from the organisation chart
(B) The threat of escalation to their line managers
(C) Influence built on credibility and relationships
(D) Control over their salaries and ratings

25. Why should a charter state what is explicitly out of scope?

(A) To prevent misunderstandings and reduce scope creep
(B) To make the charter more complete for auditors
(C) Because sponsors prefer to approve smaller scopes
(D) To avoid having to list every deliverable in detail

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MODULE 1 · FRAME

Answer key

Knowledge checks

1. C A core-system migration is temporary, unique and creates a change. The others are ongoing, repetitive operations.

2. B Related efforts coordinated for combined benefits form a program. A portfolio is the wider investment mix across many goals.

3. B Progressive elaboration means plans and requirements become more detailed as information and learning increase.

4. A Steer is about evidence-based control: measurement, risk and course correction. Shape covers stakeholders and scope; Close covers handover and lessons.

5. C Stable product teams keep knowledge and deliver value continuously. Project management skills are still applied, but in shorter loops.

6. D An outcome is the change in behaviour or capability that results from using an output. The app itself is the output.

7. C Tailoring means fitting the approach to the context instead of applying every process by default.

8. D Systems thinking means seeing interactions and ripple effects across parts of the project and organisation.

9. A That domain covers the delivery approach, cadence and phases that suit the product and context.

10. D Principles are guidance for behaviour and decisions; they help most when the situation is new or ambiguous.

11. D Adding usable pieces of value one after another is incremental delivery.

12. C Evolving requirements with partly unknown technology place the work in the complex zone, which favours agile or hybrid approaches.

13. B When late change is very expensive — such as poured concrete or manufactured hardware — careful upfront planning pays off.

14. B A hybrid without deliberate design can carry agile overhead while still getting feedback only at the end.

15. A Tailoring is continuous: inspect the approach at retrospectives and gates and adjust where needed.

16. B Payback = 30 ÷ 10 = 3 years.

17. D NPV discounts future cash flows to today's value. Simple ROI and payback ignore timing.

18. A Benefits usually appear after the project ends, so they are owned by the business manager responsible for the operation.

19. B A key result is a measurable outcome. The others are tasks or outputs.

20. D The 'do nothing' baseline reveals what happens without investment, so decision-makers can compare options fairly.

21. A The sponsor (or equivalent authority) issues the charter, which authorises the project and the PM.

22. B The charter carries high-level scope and exclusions; the detail belongs in later plans and backlogs.

23. C The product owner orders the backlog to maximise value; the PM supports governance and coordination.

24. C Most PMs lead through influence; formal authority is limited, especially in matrix organisations.

25. A Clear exclusions set expectations early and give you a reference point when extra requests arrive.

Scenarios & case studies

> **Scenario · Project or operations?**
> **B** is best. Best. You respect the distinction: operations are sustained with process controls, while a defined improvement (for example automating payroll) is a genuine project with its own end and benefits.

> **Scenario · Delivered — but will it be used?**
> **C** is best. Best. You keep value at the centre, make the trade-off visible and give the decision-maker clear options. This reflects stewardship, stakeholder engagement and value-focused principles.

> **Scenario · Choosing an approach for a bank**
> **C** is best. Best. Each part gets the approach that matches its uncertainty, and an integrated plan manages the dependencies between them.

> **Scenario · Two projects, one budget slot**
> **B** is best. Best. You give the board a balanced view — strategy, finance and risk — and keep A visible for later consideration.

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Case study · Charter for an NGO digital registration rollout

Sponsor: the country director signs, because the charter grants authority over NGO resources and links the project to the donor agreement. **Success criteria:** e.g. 95% of registrations synced within 48 hours of capture; average registration time under 8 minutes per household by month 12; zero reportable data-protection breaches. **Out of scope:** e.g. building a cash-transfer payment platform; replacing the head-office finance system; devices for partner organisations. **Life cycle:** hybrid — an agile build of the app with field pilots in two unions, then an incremental rollout in waves of about ten unions, with predictive elements for procurement, donor reporting and data-protection approval. This tests the app offline in real conditions before scaling.

Scenario · No charter, but 'just start'

C is best. Best. You respond quickly but secure authority, budget and clarity of purpose — the minimum needed to start responsibly.

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02

MODULE 2 · SHAPE

Stakeholders, Scope & Requirements

Shape the work: understand the people, discover the real needs and keep scope under control

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Identify and analyse stakeholders using the power–interest grid and salience model
- Plan engagement and choose elicitation techniques that uncover real needs
- Write INVEST user stories with Given/When/Then acceptance criteria and agree ready and done
- Decompose scope with a WBS and dictionary, and manage a product backlog
- Prioritise with MoSCoW, value vs effort, Kano and WSJF and handle change without chaos

LESSONS

1. Finding & Analysing Stakeholders · 24 min
2. Engagement Planning & Requirements Elicitation · 26 min
3. User Stories, Acceptance Criteria & Readiness · 26 min
4. Scope Structures: WBS, Dictionary & Backlog · 24 min
5. Prioritisation, Scope Creep & Change Control · 28 min

MODULE 2 · SHAPE · LESSON 1 · 24 MIN

Finding & Analysing Stakeholders

Identify everyone who matters, then understand their power, interest and urgency

- Identify stakeholders systematically, including the less obvious ones
- Analyse stakeholders with the power–interest grid and the salience model
- Build and maintain a stakeholder register

Projects rarely fail because of technology alone. They fail because someone important was missed, misunderstood or ignored until it was too late. A **stakeholder** is any person, group or organisation that can affect, be affected by, or believe they are affected by the project. Notice the last part: **perception counts**.

Identify stakeholders in four sweeps

1. **Start with the charter** List the sponsor, customers, users, team, suppliers and anyone named in contracts or the business case.
2. **Follow the value chain** Who supplies inputs, who uses outputs, who runs the result, who pays, who regulates?
3. **Ask around** Interview early stakeholders: 'Who else should I speak to? Who might object?'
4. **Look outward** Regulators, unions, community groups, media, partner NGOs, future users and those who lose something from the change.

WATCH OUT · THE QUIET STAKEHOLDERS

Some of the most influential stakeholders never attend your meetings: the IT security officer who can block go-live, the union representative, the regulator's desk officer, the night-shift supervisor whose process you are changing. Find them early.

Stakeholder identification checklist

☐ Sponsor, steering committee and funding bodies listed
☐ Direct users and their sub-groups listed separately
☐ People who operate, maintain or support the result after go-live included
☐ Regulators, auditors, data-protection and security functions checked
☐ Suppliers, contractors and partner organisations listed
☐ Groups who lose something from the change identified
☐ Community, unions, media or public interest groups considered
☐ Each stakeholder's interest, power and current engagement recorded with evidence
☐ Register reviewed with at least two experienced colleagues
☐ Date set to re-check the register

The **power–interest grid** plots each stakeholder by how much power they have over the project and how interested they are in it. It gives a quick first view of how much attention each group needs.

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Power–interest grid: four quadrants

QUADRANT	POWER / INTEREST	TYPICAL STAKEHOLDERS	STRATEGY
Manage closely	High power, high interest	Sponsor, key customer, steering committee	Involve in decisions, consult often, no surprises
Keep satisfied	High power, low interest	Finance director, regulator, CIO	Brief concisely, consult on what touches them, avoid overloading
Keep informed	Low power, high interest	End users, field staff, community groups	Regular, clear updates and channels for feedback
Monitor	Low power, low interest	Peripheral departments, general public	Light-touch updates; watch for changes in position

The **salience model** adds a third dimension. It looks at **power** (ability to impose their will), **legitimacy** (how appropriate their involvement is) and **urgency** (how time-sensitive and critical their claims are). Stakeholders who have all three are **definitive** and need immediate attention.

Salience model — the seven classes

P
Dormant
Power only. Could matter a lot if they become interested.

L
Discretionary
Legitimacy only. Fair claim but little pressure; good-citizen engagement.

U
Demanding
Urgency only. Noisy but without power or legitimacy; manage, do not ignore.

PL
Dominant
Power and legitimacy. Formal influence — often senior managers or regulators.

PU
Dangerous
Power and urgency without legitimacy. Can disrupt — handle with care.

LU
Dependent
Legitimacy and urgency, but depend on others' power — e.g. affected communities.

PLU
Definitive
All three. Highest priority for engagement.

INSIGHT · GRIDS ARE SNAPSHOTS
Power and interest change. A regulator with little interest today becomes 'manage closely' the moment a complaint lands. Re-analyse at each phase gate, after major changes and whenever a stakeholder surprises you.

TEMPLATE **Stakeholder register**

One living table that captures who your stakeholders are and how you will engage them.

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STAKEHOLDER	ROLE / ORGANISATION	INTEREST AND EXPECTATIONS	POWER / INTEREST	CURRENT → DESIRED ENGAGEMENT	ENGAGEMENT APPROACH	OWNER
Head of Nursing	*Hospital operations*	*Ward workflows stay safe during move*	*High / High*	*Neutral → Supportive*	*Monthly design walk-through; nurse champions on each ward*	*PM*
City fire authority	*Regulator*	*Compliance with fire code*	*High / Low*	*Unaware → Supportive*	*Early pre-application meeting; formal submissions at design gate*	*Facilities lead*

SCENARIO · DECIDE · THE FORGOTTEN DEPARTMENT

Two months into building a new warehouse management system for a Gazipur garment exporter, you learn the customs compliance team was never consulted. They say the system must produce bonded-warehouse reports in a specific format, or the factory may face penalties. What should you do FIRST?

(A) Tell them the scope is already approved and they should raise a change request next year.

(B) Add them to the stakeholder register, meet them quickly to understand the requirement and its urgency, and assess the impact through the change process.

(C) Ask developers to add the report quietly to avoid delaying the project.

(D) Escalate to the sponsor and ask who is to blame for the omission.

Circle your choice. Answer and feedback: module answer key.

Stakeholder analysis — practical tips

- Analyse groups, then split them when members differ (e.g. 'users' → senior nurses, junior nurses, ward clerks).
- Record **evidence** for each rating — what did they say or do?
- Keep sensitive analysis private; share engagement plans, not personal judgements.
- Note relationships: who influences whom?

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KNOWLEDGE CHECK

1. Which stakeholders belong in the 'keep satisfied' quadrant?

   (A) Low power, high interest
   (B) Low power, low interest
   (C) High power, high interest
   (D) High power, low interest

2. In the salience model, a stakeholder with power, legitimacy and urgency is:

   (A) Dormant
   (B) Discretionary
   (C) Definitive
   (D) Dependent

3. Why is 'believe they are affected' part of the stakeholder definition?

   (A) Only perceptions matter; actual effects are secondary
   (B) It ensures that competitors and the media are always included in the analysis
   (C) Perceptions can drive support or opposition as strongly as facts
   (D) It is not part of the definition; only real impact counts

4. When should stakeholder analysis be updated?

   (A) Only at project start, when the register is built
   (B) At gates, after major changes and when positions shift
   (C) Only at closure, as part of the lessons-learned review
   (D) Never; it is a one-time exercise at initiation

5. A flood-affected community demands faster aid but lacks formal power. In the salience model they are most likely:

   (A) Dependent
   (B) Dangerous
   (C) Dormant
   (D) Definitive

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MODULE 2 · SHAPE · LESSON 2 · 26 MIN

Engagement Planning & Requirements Elicitation

Move stakeholders from where they are to where you need them — and draw out what they really need

- Assess current and desired stakeholder engagement levels
- Create a practical engagement plan
- Choose suitable elicitation techniques for different situations

Analysis tells you **who** matters. Engagement is what you **do** about it. The goal is not to make everyone happy — that is impossible — but to secure the involvement, support and decisions the project needs at the right time.

Engagement assessment matrix (C = current, D = desired)

STAKEHOLDER	UNAWARE	RESISTANT	NEUTRAL	SUPPORTIVE	LEADING
Head of Nursing			C	D	
Ward clerks	C		D		
Hospital CEO (sponsor)				C	D
Junior doctors' association		C	D		

DEFINITION · ENGAGEMENT LEVELS

Unaware – does not know about the project. **Resistant** – aware and opposed. **Neutral** – aware, neither for nor against. **Supportive** – aware and in favour. **Leading** – actively championing success. The gap between C and D drives your engagement actions.

PRO TIP · DESIRED IS NOT ALWAYS 'LEADING'

Not everyone needs to champion your project. For most groups 'Supportive' or even 'Neutral' is enough. Spend energy where the gap matters most to success.

A light engagement plan

STAKEHOLDER	GAP (C → D)	KEY MESSAGE	METHOD AND FREQUENCY	OWNER
Junior doctors' association	Resistant → Neutral	Shift handovers will be faster and safer, not monitored more	Two co-design workshops, then monthly Q&A	Clinical lead
Ward clerks	Unaware → Neutral	What changes for you and when; training before go-live	Team briefings, short videos in Bangla, floor walkers at launch	Change manager
Hospital CEO	Supportive → Leading	Visible sponsorship drives adoption	Fortnightly 20-min briefing; CEO opens training days	PM

Elicitation means drawing out requirements — not just collecting what people say they want, but discovering what they actually need. People often describe solutions ('add a button') rather than problems ('I cannot find yesterday's orders'). Your job is to get to the problem.

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Elicitation techniques: when to use what

TECHNIQUE	BEST FOR	WATCH OUT FOR
Interviews	Depth, sensitive topics, senior stakeholders	Time-consuming; hear one view at a time
Facilitated workshops	Agreement across groups, fast decisions	Dominant voices; needs a skilled facilitator
Observation / job shadowing	Tacit knowledge people cannot describe	People behave differently when watched
Surveys and questionnaires	Many, dispersed respondents	Shallow answers; low response rates
Prototypes and wireframes	Clarifying user interfaces and flows	Users may think the prototype is nearly finished
Document and data analysis	Existing rules, reports, regulations, logs	Documents may be out of date
Focus groups	Attitudes and expectations of a user segment	Groupthink; not a substitute for observing real behaviour
Design thinking / journey mapping	Understanding experience end to end	Can drift without a clear problem statement

A practical elicitation cycle

01 Prepare — Goals, questions, participants → **02 Elicit** — Interview, observe, workshop → **03 Document** — Notes, models, stories → **04 Confirm** — Play back to stakeholders → **05 Refine** — Prioritise and detail just in time

EXAMPLE · OBSERVATION BEATS ASKING

A team building a garment-factory quality app asked supervisors how they record defects. Answer: 'on the defect sheet.' Observation on the floor showed that supervisors actually used WhatsApp photos and filled in the sheet at the end of the shift. The final app was built around photo capture — something no interview had revealed.

SCENARIO · DECIDE · TWO LOUD VOICES

In a requirements workshop for a new loan-origination system, two senior credit managers dominate the discussion. Branch officers — the main future users — are silent. What should the facilitator do?

(A) Let the senior managers lead; they have more authority.

(B) Switch to structured techniques — silent brainstorming on cards, round-robin input and small breakout groups — and schedule follow-up interviews or observation with branch officers.

(C) End the workshop and send a survey instead.

(D) Ask the managers to leave the room.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Pick one stakeholder on your current or recent project whose engagement was lower than it needed to be. What was their current level, what should it have been, and what single action would have closed the gap?

KNOWLEDGE CHECK

6. In an engagement assessment matrix, 'C' and 'D' stand for:
   - (A) Customer and Delivery
   - (B) Critical and Dependent
   - (C) Current and Desired
   - (D) Committed and Detached

7. Which technique is best to uncover tacit knowledge that users cannot easily describe?
   - (A) Observation / job shadowing
   - (B) Structured online survey
   - (C) Analysis of existing documents
   - (D) Email questionnaires

8. A stakeholder is aware of the project and opposes it. Their engagement level is:
   - (A) Resistant
   - (B) Unaware
   - (C) Neutral
   - (D) Supportive

9. What is a key risk of using prototypes in elicitation?
   - (A) They are always too expensive for small projects
   - (B) They cannot demonstrate user flows or navigation paths
   - (C) They replace the need for later user testing
   - (D) Users may assume the product is nearly finished

10. Which statement about desired engagement is most accurate?
    - (A) Desired levels depend on what the project needs from each person
    - (B) Every stakeholder should ideally reach the 'Leading' level
    - (C) Desired levels are set once at initiation and never change
    - (D) Only the sponsor and the most senior users need a desired level

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MODULE 2 · SHAPE · LESSON 3 · 26 MIN

User Stories, Acceptance Criteria & Readiness

Write requirements that teams can build and test — and agree when work is ready and done

- Write user stories that meet the INVEST criteria
- Express acceptance criteria in Given/When/Then form
- Apply a definition of ready and a definition of done

On adaptive projects, requirements are commonly captured as **user stories**: short statements of a need, told from the user's point of view. A story is a **promise of a conversation**, not a full specification — detail is added just in time through discussion, examples and acceptance criteria.

EXAMPLE · USER STORY FORMAT

As a rural agent, **I want to** capture a customer's ID photo while offline, **so that** I can complete onboarding where there is no mobile signal.

INVEST — the six qualities of a good story

I
Independent
Can be built and released without waiting on another story, as far as possible.

N
Negotiable
Details are open for discussion; it is not a frozen contract.

V
Valuable
Delivers something a user or customer actually cares about.

E
Estimable
The team understands it well enough to size it.

S
Small
Fits comfortably within one iteration — ideally a few days of work.

T
Testable
Clear criteria tell everyone when it is working.

Acceptance criteria set the conditions a story must satisfy to be accepted by the product owner. A popular format comes from behaviour-driven development: **Given** a starting context, **When** an action occurs, **Then** an observable result follows.

EXAMPLE · GIVEN / WHEN / THEN

Scenario: photo saved offline. Given the agent's phone has no network connection, When the agent captures and confirms an ID photo, Then the photo is stored encrypted on the device And the application shows 'Saved — will sync later'.
Scenario: automatic sync. Given a stored photo and the network returns, When the app is open, Then the photo uploads within 2 minutes And the onboarding status changes to 'Submitted'.

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Splitting large stories (epics)

Split by...

- Workflow step (capture → verify → approve)
- Business rule (individual vs joint account)
- Data type or channel (photo vs NID scan)
- Happy path first, exceptions later
- Performance level (works first, fast later)

Avoid splitting by...

- Technical layer only (database story, UI story)
- Team member or skill
- Phases like 'design story' and 'test story'
- Anything that delivers no user-visible value alone

TEMPLATE

User story map

Lay out the user's journey left to right, then stack stories under each step by priority to plan thin, valuable releases.

USER ACTIVITY (BACKBONE)	USER STEP	RELEASE 1 STORIES (WALKING SKELETON)	RELEASE 2 STORIES	LATER
Open account	*Capture identity*	*Capture ID photo offline*	*Scan NID barcode automatically*	*Face match with liveness check*
Open account	*Verify customer*	*Agent reviews data before submit*	*Auto-check against blacklist*	*Risk-score explanations*

Definition of ready vs definition of done

Definition of ready (team agreement, optional)

- Story is written and understood by the team
- Acceptance criteria are agreed
- Dependencies identified or removed
- Sized and small enough for one iteration
- Designs or data available if needed

Definition of done (quality commitment)

- Code reviewed and merged
- Automated tests written and passing
- Acceptance criteria met and demonstrated
- Security and accessibility checks passed
- Documentation and release notes updated
- Deployed to a production-like environment

WATCH OUT · DO NOT TURN 'READY' INTO A GATE

A definition of ready helps avoid starting work that is unclear, but used rigidly it becomes a mini-waterfall where nothing starts until everything is specified. Keep it short and use it as a conversation aid.

INSIGHT · DONE MEANS USABLE

If a story is 'done except testing', it is not done. A shared, visible definition of done protects quality and makes progress honest — a theme developed further in the Deliver stage.

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SCENARIO · DECIDE · A STORY THAT IS TOO BIG

During refinement, the product owner presents: "As a customer, I want a complete loan application journey so I can get credit." The team estimates it would take three sprints. What should the PM or team coach suggest?

(A) Accept it as is and spread it across three sprints.

(B) Split it into thin vertical slices — e.g. submit basic details, upload documents, see decision status — each valuable and testable on its own.

(C) Split it into database, API and user-interface stories.

(D) Ask the product owner to write a 40-page specification first.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

11. In INVEST, the 'N' stands for:

(A) Necessary
(B) Negotiable
(C) Non-functional
(D) New

12. Which acceptance criterion is best written?

(A) The search page should load quickly for all users, especially at busy times
(B) The search should be user-friendly, intuitive and easy for every customer to use
(C) Search works properly and always returns the right results to the user who asked
(D) Given 500 concurrent users, When a user searches, Then results show within 2 seconds

13. What is the main purpose of a definition of done?

(A) To set a shared quality standard every increment must meet
(B) To decide which stories may enter the next sprint
(C) To estimate stories consistently across the whole team
(D) To replace acceptance criteria with one shared list

14. Which is a good way to split a large story?

(A) By technical layer, such as database, API and interface
(B) By workflow step that still delivers user value
(C) By team member, so each person owns one part
(D) Into separate 'design', 'build' and 'test' stories

15. A user story is best described as:

(A) A placeholder for a conversation about a user need
(B) A complete specification of a feature for developers
(C) A test case written from the user's point of view
(D) A short charter that authorises a piece of work

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MODULE 2 · SHAPE · LESSON 4 · 24 MIN

Scope Structures: WBS, Dictionary & Backlog

Break the work down — with a work breakdown structure, a product backlog, or both

- Build a deliverable-oriented WBS that follows the 100% rule
- Write a WBS dictionary entry
- Compare the WBS with a product backlog and know when to use each

Whatever the approach, scope must be made visible. Predictive projects usually decompose scope into a **work breakdown structure (WBS)**. Adaptive projects keep an ordered **product backlog**. Hybrid projects often use both: a WBS for the overall project and backlogs inside the parts that evolve.

> DEFINITION · WORK BREAKDOWN STRUCTURE
> A **WBS** is a hierarchical decomposition of the total scope of work into deliverables and smaller components. The lowest level is the **work package**, which can be estimated, scheduled, assigned and controlled.

WBS example — new hospital outpatient wing (top level first)

1. Outpatient wing project	`Total scope`
Level 2: major deliverables	`1.1 Design` `1.2 Structure` `1.3 Building services` `1.4 Fit-out and equipment` `1.5 Commissioning and handover` `1.6 Project management`
Level 3: components (example for 1.3)	`1.3.1 Electrical` `1.3.2 HVAC` `1.3.3 Medical gas` `1.3.4 Fire safety systems`
Work packages (example for 1.3.3)	`Pipework installed` `Outlets fitted` `Pressure tested and certified`

Rules for a strong WBS

- **100% rule:** the children of each element add up to exactly 100% of the parent's work — nothing missing, nothing extra.
- Organise around **deliverables (nouns)**, not activities (verbs). Activities come later in the schedule.
- Include project management work itself as a branch.
- Decompose until work packages can be estimated and owned — a common heuristic is roughly 8 to 80 hours of effort, adjusted to project size.
- Use a numbering scheme (1.3.3) that links to schedule, cost and risk records.
- Build it with the team; people who do the work spot what is missing.

The **WBS dictionary** gives each element the detail the diagram cannot hold. Together, the approved scope statement, the WBS and the WBS dictionary form the **scope baseline** on predictive projects.

TEMPLATE **WBS dictionary entry**

Describe each work package so anyone can understand what it includes, who owns it and how it will be accepted.

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WBS ID AND NAME	DESCRIPTION OF WORK	OWNER	ACCEPTANCE CRITERIA	KEY MILESTONES	ESTIMATED COST / EFFORT	ASSUMPTIONS AND CONSTRAINTS
1.3.3 Medical gas — pressure tested and certified	*Pressure test oxygen, vacuum and medical air lines; obtain certification*	*MEP subcontractor lead*	*All lines pass test at specified pressure; certificate issued by approved inspector*	*Test complete by week 42*	*Tk 6.5 lakh / 120 hours*	*Inspector available within 5 days' notice*

WBS vs product backlog

WBS

- Hierarchy of deliverables
- Aims to capture 100% of known scope early
- Changes go through formal change control
- Basis for estimates, schedule and cost baseline
- Best for stable, well-understood scope

Product backlog

- Ordered list of items (features, stories, fixes)
- Emerges and evolves as learning occurs
- Refined continuously by the product owner and team
- Top items detailed, lower items rough
- Best for evolving products and uncertain needs

PRO TIP · USING BOTH IN A HYBRID

On a hybrid hospital project, the WBS covers the building, equipment and commissioning. One branch — '1.4.5 Patient flow software' — is delivered by an agile team from its own product backlog. The WBS element carries a budget and a milestone; the backlog decides what goes inside it.

SCENARIO · DECIDE · ACTIVITIES IN THE WBS

A new team member drafts a WBS whose second level reads: 'Plan', 'Design', 'Build', 'Test', 'Deploy'. The level-3 items under each are tasks such as 'Hold meeting' and 'Write code'. What feedback is BEST?

(A) Accept it; any structure is fine as long as it is complete.

(B) Suggest reorganising around deliverables (e.g. 'Mobile app', 'Admin portal', 'Data migration', 'Training', 'Project management') and moving activities into the schedule.

(C) Delete the WBS and use only a Gantt chart.

(D) Add more levels until every task is under one hour.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Stakeholder
Any person, group or organisation that can affect, be affected by, or believe they are affected by a project.

Salience model
A stakeholder classification based on power, legitimacy and urgency; those with all three are 'definitive'.

User story
A short description of a need from the user's point of view, used as a placeholder for conversation and refined with acceptance criteria.

Definition of done
A shared quality standard that every increment of work must meet before it is considered complete.

Product backlog
An ordered, evolving list of everything that might be needed in a product, owned by the product owner.

MoSCoW
A prioritisation method that sorts requirements into Must have, Should have, Could have and Won't have this time.

Power–interest grid
A two-by-two tool that groups stakeholders by their power over and interest in the project to guide engagement strategy.

Elicitation
Drawing out stakeholders' real needs through techniques such as interviews, observation, workshops and prototypes.

Acceptance criteria
Conditions a deliverable or story must satisfy to be accepted, often written as Given/When/Then.

Work breakdown structure (WBS)
A hierarchical, deliverable-oriented decomposition of the total scope of a project.

Scope creep
Uncontrolled expansion of scope without adjusting time, cost or resources and without proper approval.

KNOWLEDGE CHECK

16. What does the 100% rule state?

(A) Every task must be 100% complete before the next starts
(B) Child elements must sum to exactly the work of their parent
(C) All stakeholders must approve the WBS before work begins
(D) The whole budget must be allocated to work packages

17. The lowest level of a WBS is called the:

(A) Activity
(B) Control account
(C) Milestone
(D) Work package

18. On a predictive project, the scope baseline consists of:

(A) The charter, budget and schedule baseline
(B) The product backlog and the sprint backlog
(C) The risk register, issue log and assumption log
(D) Approved scope statement, WBS and WBS dictionary

19. Which statement about the product backlog is accurate?

(A) It is ordered and refined continuously as learning occurs
(B) It is fixed once the sponsor has approved its contents
(C) It contains only technical tasks defined by developers
(D) It is owned and ordered by the sponsor for each release

20. WBS elements should mainly be named as:

(A) Verbs describing the tasks to be done
(B) Deliverables or outcomes (nouns)
(C) The names of responsible team members
(D) Target dates for each piece of work

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MODULE 2 · SHAPE · LESSON 5 · 28 MIN

Prioritisation, Scope Creep & Change Control

Decide what matters most, and handle change without chaos

- Prioritise work using MoSCoW, value vs effort, Kano and WSJF
- Recognise scope creep and gold plating
- Compare integrated change control with backlog refinement

Every project has more ideas than capacity. Prioritisation is how you protect value: doing the most important things first and consciously saying 'not now' to the rest. Without it, teams spread themselves thin and deliver many half-finished things.

MoSCoW prioritisation

CATEGORY	MEANING	EXAMPLE — MOBILE WALLET SAVINGS FEATURE
Must have	Without it the release fails or is illegal/unsafe	Deposits into savings with correct interest calculation
Should have	Important but a workaround exists	Savings goals with progress bar
Could have	Desirable if time allows	Animated celebration when goal is reached
Won't have (this time)	Agreed to exclude from this release	Joint savings with family members

PRO TIP · KEEP THE MUSTS HONEST

A common guideline is to keep Must-haves to roughly 60% of planned effort, leaving room for Shoulds and Coulds to absorb surprises. If everything is a Must, nothing is prioritised.

Value vs effort matrix

High value

- Low effort → **Quick wins**: do first
- High effort → **Big bets**: plan carefully, slice thinly

Low value

- Low effort → **Fill-ins**: do if spare capacity
- High effort → **Money pits**: avoid or drop

Kano model — how features affect satisfaction

B

Basic (must-be)

Expected; absence causes anger, presence is barely noticed. E.g. a banking app that does not lose money.

P

Performance

More is better, roughly in proportion. E.g. faster transfers.

D

Delighters

Unexpected features that create delight; absence is not missed. E.g. smart savings nudges.

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I

Indifferent

Users do not care either way — candidates to drop.

R

Reverse

Some users actively dislike it — e.g. heavy gamification for business users.

Weighted shortest job first (WSJF) — in brief

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost of delay	`User/business value + Time criticality + Risk reduction or opportunity enablement`	Relative scores (often on a Fibonacci-like scale) estimating the economic pain of waiting.
WSJF	`Cost of delay ÷ Job size (or duration)`	Higher scores go first: the most value lost per unit of time, delivered fastest. Popularised in scaled agile frameworks.
Worked example	`Feature X: CoD 21, size 8 → 2.6. Feature Y: CoD 13, size 3 → 4.3`	Y goes first, even though X has the higher cost of delay, because it is much smaller.

Scope creep vs gold plating vs healthy change

Scope creep

- Uncontrolled growth in scope
- Requests absorbed without impact analysis
- Usually from stakeholders
- Leads to delays, overruns, burnout

Gold plating

- Team adds extras nobody asked for
- Well-meant but unapproved
- Adds cost and risk
- Hides real progress

Healthy change

- Visible, analysed and decided
- Linked to value or necessity
- Trade-offs made explicitly
- Baselines or backlog updated transparently

Integrated change control (predictive)

01 **Request** — Change request logged →

02 **Analyse** — Impact on scope, time, cost, quality, risk, benefits →

03 **Decide** — PM, sponsor or change control board approves, rejects or defers →

04 **Update** — Baselines, plans and documents revised →

05 **Communicate** — Stakeholders and team informed; change implemented

Two ways to absorb change

Integrated change control

- Used when scope, schedule and cost are baselined
- Formal request and impact analysis
- Decision by authorised body (e.g. CCB)
- Protects contracts and commitments
- Can feel slow; tailor thresholds to keep it fast

Backlog refinement

- Used for evolving product scope
- New items added to the backlog by the product owner
- Ordered against everything else; something else may move down
- Change is expected and cheap if made before work starts
- Iteration goal protected during the iteration

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INSIGHT · HYBRID CHANGE THRESHOLDS

Many hybrid projects let the product owner reorder the backlog freely **within** an agreed budget and milestone, but send anything that moves a milestone, increases total budget or affects a contract to formal change control. Write the thresholds down in the plan.

SCENARIO · DECIDE · WHICH PATH FOR THIS CHANGE?

On a hybrid hospital project, the product owner may reorder the patient-flow software backlog freely within its agreed budget and milestone. The hospital's insurance partner now asks for a claims-integration feature that the team estimates would push the software milestone back by three weeks and add 6% to the software budget. What should the project manager do?

(A) Let the product owner add it to the backlog; backlog changes are the product owner's call.

(B) Raise it through integrated change control with an impact analysis, so the sponsor or change board can decide on the milestone and budget change.

(C) Reject it because the scope baseline is fixed.

(D) Ask the team to absorb it with overtime so the milestone does not move.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · SCOPE PRESSURE AT A DHAKA FINTECH LAUNCH

A Dhaka fintech start-up is launching a merchant QR-payments app in time for Eid, when transaction volumes peak. The team works in two-week sprints with a fixed launch date eleven weeks away. Investors are pushing for a loyalty-points feature 'for launch'. A sales director has promised one big supermarket chain a custom settlement report. Meanwhile developers have been polishing animations that were never requested. The product owner is overwhelmed and the backlog has 180 items marked 'high priority'.

1. What scope problems can you identify?

2. How would you re-prioritise the backlog for an Eid launch?

3. How should the custom report request be handled?

4. What would you agree with the team about the animations?

Model answer: module answer key.

KNOWLEDGE CHECK

21. In MoSCoW, 'W' stands for:

(A) Want to have
(B) Worth having
(C) Won't have this time
(D) Will have later for sure

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22. A feature is expected by every user; its absence causes strong dissatisfaction. In Kano terms it is:

(A) A delighter (attractive) feature
(B) A performance (one-dimensional) feature
(C) An indifferent feature
(D) A basic (must-be) feature

23. Feature A has a cost of delay of 20 and size 10; Feature B has a cost of delay of 9 and size 3. By WSJF, which goes first?

(A) A, because its cost of delay is higher
(B) B, because its WSJF score is higher
(C) Both have equal priority
(D) Neither — WSJF cannot compare them

24. A developer adds an extra dashboard nobody requested. This is:

(A) Scope creep
(B) Change control
(C) Gold plating
(D) Progressive elaboration

25. On an agile team, a stakeholder asks for a new feature mid-project. What normally happens?

(A) It is added to the current sprint immediately by the team
(B) It must be rejected because the scope is already agreed
(C) The product owner adds it to the backlog and orders it by value
(D) The PM rewrites the charter and asks the sponsor to re-approve it

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MODULE 2 · SHAPE

Answer key

Knowledge checks

1. D High-power stakeholders with low interest should be kept satisfied without overloading them.

2. C Definitive stakeholders combine all three attributes and need the highest attention.

3. C People who think they are affected act on that belief; ignoring them can create resistance.

4. B Stakeholder power and interest change; analysis is a living activity.

5. A Legitimacy plus urgency without power makes a stakeholder dependent on others to advance their claims.

6. C C shows current engagement; D shows desired engagement. The gap drives your actions.

7. A Watching people work reveals workarounds and habits they would not think to mention.

8. A Aware and opposed is the definition of resistant.

9. D A realistic prototype can create false expectations about progress — explain clearly what it is.

10. A Desired engagement is set per stakeholder according to what success requires.

11. B Negotiable — details are refined through conversation rather than fixed in advance.

12. D It is specific, observable and testable, using Given/When/Then.

13. A The definition of done is the quality commitment applied to all work; acceptance criteria are story-specific.

14. B Vertical slices along workflow steps keep each piece valuable and testable.

15. A Stories capture the need briefly; detail is added just in time through conversation and criteria.

16. B The 100% rule ensures the decomposition includes all — and only — the work of the parent element.

17. D Work packages are the lowest WBS level, which can be estimated, scheduled and controlled.

18. D These three approved components together form the scope baseline.

19. A The product backlog evolves; the product owner orders it and the team helps refine it.

20. B A deliverable-oriented WBS makes completeness and ownership easy to check.

21. C Won't have (this time) records agreed exclusions for the current release.

22. D Basic features are taken for granted; missing them causes anger, having them barely raises satisfaction.

23. B A scores 20 ÷ 10 = 2; B scores 9 ÷ 3 = 3. Higher WSJF goes first.

24. C Gold plating is the team adding unrequested extras; scope creep usually comes from unmanaged stakeholder requests.

25. C Backlog refinement absorbs change: the product owner orders new items by value while the current iteration's goal is protected.

Scenarios & case studies

> **Scenario · The forgotten department**
> **B** is best. Best. You correct the identification gap, understand the real need and route any scope impact through proper change control.

> **Scenario · Two loud voices**
> **B** is best. Best. Structured facilitation balances voices in the room, and follow-up elicitation reaches users who may not speak up in front of seniors.

> **Scenario · A story that is too big**
> **B** is best. Best. Vertical slices keep stories small, testable and valuable, and let the team release and learn earlier.

> **Scenario · Activities in the WBS**
> **B** is best. Best. A deliverable-oriented WBS supports the 100% rule, clear ownership and acceptance, while activities belong in the schedule.

> **Scenario · Which path for this change?**
> **B** is best. Best. Changes that move milestones or budgets beyond agreed thresholds belong in formal change control, even on the agile part of a hybrid project.

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Case study · Scope pressure at a Dhaka fintech launch
Problems: scope creep (loyalty points, custom report), gold plating (animations) and a backlog with no real prioritisation. **Re-prioritise:** run a MoSCoW session tied to the launch goal — payments, settlement and security as Musts; use value vs effort or WSJF to order the rest; cap Musts at about 60% of the remaining capacity based on velocity. Loyalty points likely become Should or post-launch, with the trade-off shown to investors using data. **Custom report:** treat it as a change — assess value (one large merchant), effort and risk to the date; the product owner decides with the sponsor; if accepted, something of equal size moves out, and the decision is communicated to sales. **Animations:** discuss gold plating openly in the retrospective, agree that work must trace to backlog items, and channel ideas through the product owner.

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03

MODULE 3 · PLAN

Estimation, Scheduling & Budgeting

Plan the work: honest estimates, realistic schedules and budgets that hold up under uncertainty

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Choose and apply analogous, parametric, bottom-up, three-point and relative estimating techniques
- Forecast agile releases from velocity ranges and plan iterations by capacity
- Build a network diagram, find the critical path and calculate total and free float
- Compress schedules and optimise resources with clear trade-offs
- Build a cost baseline with the right reserves and interpret Monte Carlo forecasts

LESSONS

1. Estimation Techniques & Why Estimates Are Ranges · 28 min
2. Agile Estimation, Velocity & Release Planning · 26 min
3. Dependencies, Network Diagrams & the Critical Path · 32 min
4. Schedule Compression & Resource Optimisation · 26 min
5. Budgets, Reserves & Monte Carlo Forecasting · 30 min

MODULE 3 · PLAN · LESSON 1 · 28 MIN

Estimation Techniques & Why Estimates Are Ranges

Analogous, parametric, bottom-up and three-point estimating — with honest uncertainty

- Choose between analogous, parametric, bottom-up and three-point estimating
- Calculate triangular and PERT estimates and a standard deviation
- Communicate estimates as ranges with confidence levels

An estimate is a **forecast**, not a promise. It is a best judgement about duration, effort or cost, given what we know today. Because the future is uncertain, every honest estimate carries a range. Pressure to give a single precise number early is one of the most common sources of project failure.

Four core estimating techniques

TECHNIQUE	HOW IT WORKS	STRENGTHS	LIMITS	TYPICAL USE
Analogous (top-down)	Base the estimate on a similar past project, adjusted for differences	Fast, cheap, needs little detail	Only as good as the similarity	Early business cases, ROM estimates
Parametric	Multiply a unit rate by a quantity (e.g. cost per m², hours per screen)	Scalable and repeatable	Needs reliable historical rates	Construction, rollouts, repeatable work
Bottom-up	Estimate each work package or activity and add them up	Most detailed and defensible	Slow; needs a full WBS	Baselines, contracts, detailed planning
Three-point	Use optimistic, most likely and pessimistic values	Shows uncertainty explicitly	Values can still be biased	Risky or unfamiliar activities

EXAMPLE · PARAMETRIC IN PRACTICE

Floor tiling for a hospital wing covers **1,200 m²**. Historical data shows one crew lays **25 m² per day**. Effort = 1,200 ÷ 25 = **48 crew-days**. With three crews working in parallel, duration ≈ **16 working days**, before allowing for curing, access and inspections.

Three-point estimating formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Triangular mean	`E = (O + M + P) ÷ 3`	Simple average of optimistic (O), most likely (M) and pessimistic (P). Gives equal weight to the extremes.
PERT (beta) mean	`E = (O + 4M + P) ÷ 6`	Weights the most likely value four times. The classic Program Evaluation and Review Technique estimate.
Standard deviation (PERT)	`σ ≈ (P − O) ÷ 6`	A rough measure of spread. A wider gap between O and P means more uncertainty.
Approximate confidence	`E ± 1σ ≈ 68%; E ± 2σ ≈ 95%`	Assumes a roughly normal distribution — a useful approximation, not a guarantee.

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EXAMPLE · WORKED THREE-POINT EXAMPLE

Integrating a payment gateway: O = 4 days, M = 6 days, P = 14 days. Triangular = (4 + 6 + 14) ÷ 3 = **8 days**. PERT = (4 + 24 + 14) ÷ 6 = **7 days**. σ = (14 − 4) ÷ 6 ≈ **1.67 days**. So roughly 68% confidence of 5.3–8.7 days and 95% confidence of 3.7–10.3 days. Report: '**about 7 days, very likely between 4 and 11**'.

INSIGHT · ADDING UP PATHS

For a chain of independent activities, add the PERT means to get the path mean, and add the **variances** (σ^2) — not the standard deviations — then take the square root for the path σ. Uncertainty grows more slowly than the sum of individual spreads.

Estimate accuracy narrows as you learn (commonly quoted ranges)

ESTIMATE CLASS	WHEN	TYPICAL RANGE
Rough order of magnitude (ROM)	Idea and business case	about −25% to +75%
Budget estimate	After high-level design	about −10% to +25%
Definitive estimate	Detailed design / work packages known	about −5% to +10%

WATCH OUT · BIASES THAT DISTORT ESTIMATES

Anchoring — the first number spoken (often the boss's) pulls estimates toward it. **Optimism bias** — people imagine the smooth path. **Padding** — hidden buffers that get consumed anyway (work expands to fill the time available, and people delay starting when they feel time is plentiful). Counter them with historical data, independent estimates and visible, shared reserves rather than hidden padding.

TEMPLATE **Three-point estimation sheet**

Estimate uncertain activities and keep a record of the basis for each number.

WBS ID / ACTIVITY	OPTIMISTIC (O)	MOST LIKELY (M)	PESSIMISTIC (P)	PERT (O+4M+P)/6	Σ (P−O)/6	BASIS AND ASSUMPTIONS
2.3 Payment gateway integration	*4 d*	*6 d*	*14 d*	*7.0 d*	*1.67 d*	*Vendor sandbox available; similar to 2025 wallet project*

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SCENARIO · DECIDE · 'JUST GIVE ME ONE NUMBER'

During a steering meeting, the sponsor of a new ERP rollout asks: "How long will this take? I need one date for the board." Requirements workshops have not started yet. What is the BEST response?

(A) Give an optimistic date to keep the sponsor happy, then adjust later.

(B) Refuse to estimate until all requirements are documented.

(C) Give a range based on comparable rollouts (e.g. 9–15 months), explain the main uncertainties and commit to a narrower forecast after the requirements phase.

(D) Pick the midpoint of your gut feeling and present it as firm.

Circle your choice. Answer and feedback: module answer key.

Habits of good estimators

- Estimate with the people who will do the work.
- Use historical data from your own organisation wherever possible.
- Write down assumptions next to every estimate.
- Give ranges and confidence levels, not single points.
- Re-estimate as you learn, and track actuals to improve next time.

KNOWLEDGE CHECK

1. Which technique is most appropriate for a quick estimate at the business-case stage, using a similar past project?

   (A) Bottom-up
   (B) Analogous
   (C) Parametric
   (D) Monte Carlo

2. O = 2, M = 5, P = 14. What is the PERT estimate?

   (A) 5
   (B) 6
   (C) 7
   (D) 21

3. Using the same values (O = 2, P = 14), what is the PERT standard deviation?

   (A) 1
   (B) 2
   (C) 3
   (D) 12

4. Which estimating technique uses a statistical relationship such as cost per square metre?

   (A) Analogous
   (B) Parametric
   (C) Bottom-up
   (D) Expert judgement only

5. Why should estimates be communicated as ranges?

   (A) They reflect real uncertainty and help decision-makers judge risk
   (B) Ranges look more professional and credible than single figures in reports
   (C) Sponsors prefer vague answers that cannot be held against the team
   (D) Single-point estimates are not permitted by most methodologies

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MODULE 3 · PLAN · LESSON 2 · 26 MIN

Agile Estimation, Velocity & Release Planning

Story points, planning poker and forecasting with real delivery data

- Size work with story points and relative estimation techniques
- Run a planning poker session
- Forecast releases with velocity ranges and plan iterations by capacity

Agile teams often estimate **relatively** rather than in hours: is this item bigger or smaller than that one we already know? People are better at comparing than at predicting absolute durations, and relative sizes stay stable even as the team gets faster.

DEFINITION · STORY POINTS

A **story point** is a unitless, team-specific measure of the overall size of a backlog item — combining effort, complexity and uncertainty. Many teams use a modified Fibonacci scale (1, 2, 3, 5, 8, 13, 20…) so that larger items get deliberately coarse numbers.

Planning poker in five steps

01 Explain — Product owner describes the item and its acceptance criteria

→

02 Clarify — Team asks questions

→

03 Vote privately — Each estimator picks a card

→

04 Reveal and discuss — Highest and lowest explain their reasoning

→

05 Re-vote — Repeat until the team converges

INSIGHT · THE VALUE IS IN THE CONVERSATION

When one developer votes 3 and another votes 13, someone knows something the others do not — a hidden integration, a reusable component, a compliance step. Planning poker surfaces that knowledge before work starts.

Other relative sizing techniques

TECHNIQUE	HOW IT WORKS	GOOD FOR
T-shirt sizing	Items labelled XS, S, M, L, XL	Early roadmaps, large epics, non-technical audiences
Affinity (wall) estimation	Team silently arranges cards from small to large, then groups them	Sizing dozens of items quickly
Bucket system	Items placed into buckets labelled with point values	Large backlogs, many participants
Count-based (no points)	Items split to similar small size, then simply counted	Mature teams using throughput forecasting

Velocity and forecasting

MEASURE	FORMULA	WHAT IT TELLS YOU
Velocity	`Sum of points of items that met the definition of done in an iteration`	A team's observed delivery rate. Partly finished items count zero.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Forecast iterations	`Remaining backlog points ÷ velocity`	Use a low and high velocity to produce a range, not one number.
Capacity-adjusted velocity	`Average velocity × (available person-days ÷ usual person-days)`	Adjusts for holidays, leave and training in the coming iteration.

EXAMPLE · FORECASTING A RELEASE

Last three sprints: **28, 32 and 30 points** (average 30). Remaining release backlog: **240 points**. Optimistic: 240 ÷ 32 = 7.5 → **8 sprints**. Pessimistic: 240 ÷ 28 ≈ 8.6 → **9 sprints**. With two-week sprints, forecast **16–18 weeks**. Next sprint covers Eid holidays with 70% of usual availability, so plan about 30 × 0.7 = **21 points**.

Release planning vs iteration planning

Release planning

- Horizon: several iterations to a few months
- Sets a release goal and target date or scope range
- Uses velocity ranges and a prioritised backlog
- Revisited every iteration as data arrives
- Involves product owner, team and key stakeholders

Iteration planning

- Horizon: one iteration (1–4 weeks)
- Sets an iteration goal
- Team selects items it can complete to the definition of done
- Breaks items into tasks where useful
- Based on recent velocity and actual capacity

TEMPLATE **Release forecast worksheet**

Turn backlog size and velocity history into an honest, range-based release forecast.

RELEASE / GOAL	REMAINING POINTS	VELOCITY RANGE (LOW–HIGH)	ITERATIONS NEEDED (RANGE)	FORECAST DATE RANGE	KEY RISKS AND ASSUMPTIONS
R2 – Merchant QR payments	*240*	*28–32*	*8–9*	*16–18 weeks from 2 Feb*	*Payment-switch certification takes ≤ 3 weeks*

WATCH OUT · VELOCITY IS NOT A PERFORMANCE SCORE

Story points are calibrated by each team, so comparing velocity between teams is meaningless. Setting targets like 'increase velocity by 20%' simply inflates estimates. Use velocity only for the team's own forecasting.

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SCENARIO · DECIDE · A VELOCITY TARGET

A department head reviewing two Scrum teams says: "Team Padma delivers 45 points per sprint, Team Jamuna only 25. Jamuna must reach 45 next quarter." What should the PM do?

(A) Agree and set the target for Team Jamuna.

(B) Explain that points are not comparable across teams, and propose outcome measures — features delivered to users, lead time, defect rates, customer results — while offering to help both teams improve flow.

(C) Ask Team Jamuna to double their estimates quietly.

(D) Merge the two teams so velocities average out.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

6. Velocity counts points from:
   - (A) Items that met the definition of done
   - (B) All items started in the iteration
   - (C) Items estimated in sprint planning
   - (D) Items assigned to senior developers

7. Remaining backlog is 180 points; velocity ranges from 25 to 30. How many iterations are forecast?
   - (A) 5–6
   - (B) 6–8
   - (C) 7–9
   - (D) 8–10

8. Why do many teams use a Fibonacci-like scale for story points?
   - (A) It is required by the Scrum Guide for every Scrum team
   - (B) Wider gaps reflect the greater uncertainty of larger items
   - (C) It converts directly into hours for capacity planning
   - (D) It makes velocity grow faster as the team matures

9. What is the main benefit of discussion in planning poker?
   - (A) It produces an exact hour estimate for each task
   - (B) It lets the manager review and approve the final estimates
   - (C) It replaces the need for written acceptance criteria
   - (D) It surfaces hidden knowledge and assumptions early

10. Average velocity is 40, but next iteration only 75% of usual capacity is available. A sensible plan is about:
    - (A) 40 points
    - (B) 30 points
    - (C) 53 points
    - (D) 10 points

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MODULE 3 · PLAN · LESSON 3 · 32 MIN

Dependencies, Network Diagrams & the Critical Path

Sequence the work, find the critical path and calculate float

- Identify dependency types and their nature
- Draw a precedence network and perform forward and backward passes
- Calculate total float and free float and explain the critical path

Once you know the activities, you must decide their **order**. A network diagram shows which activities depend on others. From it you can find the **critical path** — the longest chain of dependent activities, which sets the shortest possible project duration.

Four logical relationships

TYPE	MEANING	EXAMPLE
Finish-to-start (FS)	B cannot start until A finishes — the most common	Install software after the server is delivered
Start-to-start (SS)	B cannot start until A starts	Proofreading can start once writing has started
Finish-to-finish (FF)	B cannot finish until A finishes	Testing cannot finish until development finishes
Start-to-finish (SF)	B cannot finish until A starts — rare	The old shift cannot end until the new shift starts

Nature of dependencies

KIND	DESCRIPTION	EXAMPLE
Mandatory (hard logic)	Required by physics, law or contract	Foundation before walls
Discretionary (soft logic)	Chosen by best practice or preference — can be changed	Finish design before starting build
External	Depends on parties outside the project	Regulator approval, vendor delivery
Internal	Within the project team's control	Testing after coding by the same team

> DEFINITION · LEADS AND LAGS
> A **lag** is a required wait between activities (concrete must cure 7 days before loading: FS + 7). A **lead** lets a successor start early, overlapping its predecessor (start training material 3 days before the system build finishes: FS – 3).

Worked example — clinic booking system go-live (durations in days)

ACTIVITY	DESCRIPTION	DURATION	PREDECESSORS
A	Confirm requirements	3	—
B	Build booking module	4	A

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ACTIVITY	DESCRIPTION	DURATION	PREDECESSORS
C	Procure tablets	2	A
D	Integrate with patient records	5	B
E	Configure tablets	3	C
F	Pilot and go-live	2	D, E

Forward and backward pass

1. **Forward pass** Start the first activity at ES = 0. For each activity, EF = ES + duration. A successor's ES is the **largest** EF of its predecessors.
2. **Project duration** The largest EF at the end of the network is the shortest possible project duration — here 14 days.
3. **Backward pass** Set the last activity's LF = 14. For each activity, LS = LF − duration. A predecessor's LF is the **smallest** LS of its successors.
4. **Float** Total float = LS − ES. Free float = earliest ES of successors − EF.
5. **Critical path** Activities with zero total float form the critical path: A → B → D → F = 3 + 4 + 5 + 2 = 14 days.

Results of the passes

ACTIVITY	DUR.	ES	EF	LS	LF	TOTAL FLOAT	FREE FLOAT
A	3	0	3	0	3	0	0
B	4	3	7	3	7	0	0
C	2	3	5	7	9	4	0
D	5	7	12	7	12	0	0
E	3	5	8	9	12	4	4
F	2	12	14	12	14	0	0

Scheduling formulas (start-at-zero convention)

MEASURE	FORMULA	WHAT IT TELLS YOU
Early finish	`EF = ES + Duration`	Earliest an activity can finish.
Late start	`LS = LF − Duration`	Latest an activity can start without delaying the project.
Total float	`TF = LS − ES = LF − EF`	How long an activity can slip without delaying the project end.
Free float	`FF = (earliest ES of successors) − EF`	How long an activity can slip without delaying any successor's early start.
Critical path	`Longest path through the network (TF = 0 when the end date equals the calculated finish)`	The chain that determines the project duration.

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INSIGHT · TOTAL FLOAT IS SHARED; FREE FLOAT IS YOURS ALONE

C and E both show 4 days of total float, but they **share** it — they sit on the same path A → C → E → F (10 days). C has **zero free float**: if C slips 2 days, E starts 2 days later and E's remaining float shrinks to 2. E has 4 days of free float because nothing waits on it until F starts on day 12.

PRO TIP · DAY-1 TOOLS

Many scheduling tools start counting at day 1, so EF = ES + duration − 1. The dates look different, but float values and the critical path are the same. Know which convention an exam or tool uses.

WATCH OUT · WATCH THE NEAR-CRITICAL PATH

A path with very little float is almost critical. If its activities are risky, it may become the critical path overnight. Also watch for **negative float**: it means an imposed deadline is earlier than the network can achieve — a signal to re-plan, compress or renegotiate.

SCENARIO · DECIDE · USING FLOAT

In the clinic example, the procurement officer asks to delay activity C (procure tablets) by 3 days to combine it with another order and get a discount. What is the impact, and what should you do?

(A) Refuse — any delay is unacceptable.

(B) Approve it without telling anyone; there is float.

(C) Approve it after confirming the impact: end date unchanged, but E shifts by 3 days and the path A–C–E–F keeps only 1 day of float. Inform the configuration lead and update the schedule and risk log.

(D) Approve it and add 3 days to the project end date to be safe.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

11. Which dependency is the most common in project schedules?

(A) Start-to-finish
(B) Finish-to-finish
(C) Start-to-start
(D) Finish-to-start

12. An activity has ES = 5, EF = 9, LS = 8, LF = 12. Its total float is:

(A) 0 days
(B) 3 days
(C) 4 days
(D) 7 days

13. In a forward pass, when an activity has two predecessors finishing on day 7 and day 10, its ES is:

(A) 7
(B) 8.5
(C) 10
(D) 17

14. 'Finish design before starting construction' chosen as best practice, though overlap is possible, is a:

(A) Discretionary dependency
(B) Mandatory (hard logic) dependency
(C) External dependency
(D) Lag

15. Negative total float indicates that:

(A) The project is ahead of schedule on its critical path
(B) Resources are idle and can be moved to other work
(C) The required finish date is earlier than the network allows
(D) The critical path has changed to a different set of activities

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MODULE 3 · PLAN · LESSON 4 · 26 MIN

Schedule Compression & Resource Optimisation

Crashing, fast-tracking, levelling and smoothing — trade-offs made visible

- Choose between crashing and fast-tracking and calculate crash cost slope
- Compress a schedule step by step along the critical path
- Distinguish resource levelling from resource smoothing

Sooner or later someone will ask for the project to finish earlier. Compression is possible, but never free: you pay with **money**, **risk** or **scope**. Your job is to make those trade-offs clear before anyone commits.

Crashing vs fast-tracking

Crashing

- Add resources to critical activities to shorten them
- Overtime, extra crews, faster equipment, express delivery
- Increases cost
- Works best on effort-driven activities
- Diminishing returns — more people do not always mean less time

Fast-tracking

- Overlap activities normally done in sequence
- Change discretionary FS links to SS or add leads
- Little direct cost
- Increases risk of rework
- Only possible where dependencies are discretionary

Crash cost slope

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost slope	`(Crash cost − Normal cost) ÷ (Normal duration − Crash duration)`	Extra cost per day saved. Crash the critical activity with the lowest slope first.
Example	`Activity D: normal 5 d at Tk 100,000; crash 3 d at Tk 160,000`	(160,000 − 100,000) ÷ (5 − 3) = Tk 30,000 per day saved.

Compressing the clinic schedule from 14 to 11 days

CRITICAL ACTIVITY	NORMAL → CRASH DURATION	COST SLOPE (TK/DAY)	DECISION
A Confirm requirements	3 → 2	40,000	Not chosen — most expensive (it would shorten both paths)
B Build booking module	4 → 3	25,000	Crash 1 day first (cheapest): 14 → 13
D Integrate with records	5 → 3	30,000	Crash 2 days next: 13 → 11
F Pilot and go-live	2 → 2	—	Cannot be crashed

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EXAMPLE · RESULT AND CHECK

Total extra cost = 25,000 + 2 × 30,000 = **Tk 85,000** for 3 days saved. Always re-check other paths after each step: A → C → E → F is 10 days, so it is still shorter than 11 and the critical path is unchanged. If you had needed 9 days, crashing A would only get you to 10, because B, D and F are already at their limits — a sign that further compression needs fast-tracking, scope or logic changes.

WATCH OUT · PEOPLE ARE NOT INTERCHANGEABLE

Adding people to knowledge work — especially late — often slows it down at first: newcomers need onboarding, and communication overhead grows. This observation, famously made about late software projects, means crashing works best on well-defined, divisible tasks.

Schedules also have to be **resource-feasible**. A network may say two activities run in parallel, but if both need the same certified electrician, they cannot. Resource optimisation adjusts the schedule to match real availability.

Resource levelling vs resource smoothing

Resource levelling

- Resolves over-allocation or limited availability
- Activities are delayed until resources are free
- Can change the critical path
- May extend the end date
- Used when resource limits are hard

Resource smoothing

- Evens out peaks and valleys in demand
- Moves activities only within their float
- Does not change the critical path
- End date unchanged
- May not remove all over-allocation

EXAMPLE · LEVELLING VS SMOOTHING ON SITE

Levelling: one licensed medical-gas engineer is needed by two parallel activities. One must wait, so the finish moves out by 4 days. **Smoothing:** painting needs 12 painters in week 6 but only 4 in week 7. Because painting the corridor has 5 days of float, you shift part of it into week 7 and use 8 painters each week — the end date does not move.

SCENARIO · DECIDE · TWO WEEKS EARLIER, PLEASE

Your sponsor asks you to bring forward the launch of a garment buyer's order-tracking portal by two weeks to match a trade fair. The critical path runs through design, build and user acceptance testing. What should you do FIRST?

(A) Immediately approve overtime for the whole team.

(B) Analyse the critical path and present options — fast-track design and build where logic is discretionary, crash specific critical activities, or reduce the scope released for the fair — with cost, risk and quality impacts for each.

(C) Tell the sponsor it is impossible.

(D) Cut user acceptance testing to save two weeks.

Circle your choice. Answer and feedback: module answer key.

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Schedule health checklist

☐ Every activity has at least one predecessor and one successor (except start and finish).

☐ Durations are estimated by the people doing the work, with assumptions recorded.

☐ Hard date constraints are used only when truly required.

☐ Critical and near-critical paths are identified and reviewed.

☐ Leads and lags are justified and documented.

☐ No resource is assigned above available capacity after levelling.

☐ Key external dependencies (vendors, regulators) have owners and dates.

☐ Milestones match the charter and contract commitments.

☐ A schedule baseline is saved and changes go through change control.

KNOWLEDGE CHECK

16. Which technique overlaps activities that were planned in sequence?

(A) Crashing
(B) Resource levelling
(C) Fast-tracking
(D) Smoothing

17. Crash cost Tk 90,000 for 4 days; normal cost Tk 60,000 for 7 days. The cost slope is:

(A) Tk 7,500/day
(B) Tk 10,000/day
(C) Tk 15,000/day
(D) Tk 30,000/day

18. When crashing, which activities should you shorten first?

(A) Any non-critical activity with float
(B) Critical activities with the lowest cost slope
(C) The longest activity, regardless of its path
(D) Activities with the most float available

19. Which technique may extend the project end date?

(A) Resource smoothing
(B) Fast-tracking
(C) Resource levelling
(D) Using available float

20. Fast-tracking is only possible where dependencies are:

(A) Mandatory
(B) Start-to-finish
(C) External
(D) Discretionary

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MODULE 3 · PLAN · LESSON 5 · 30 MIN

Budgets, Reserves & Monte Carlo Forecasting

Build a cost baseline, protect it with the right reserves and forecast with simulation

- Aggregate estimates into a cost baseline and project budget
- Distinguish contingency reserve from management reserve
- Explain how Monte Carlo simulation produces probabilistic forecasts

Cost planning turns estimates into a **time-phased budget** that the organisation can fund and you can control. A sound budget does three things: it covers the planned work, it protects against identified risks, and it shows when money will be needed.

Cost categories worth separating

CATEGORY	EXAMPLES	WHY IT MATTERS
Direct vs indirect	Contractor labour, materials vs office rent, shared IT	Indirect costs are often allocated by formula; agree the rule early
Fixed vs variable	Software licence fee vs cloud usage per transaction	Variable costs grow with scope and volume
One-time vs recurring	Implementation vs annual support and hosting	Business cases fail when recurring costs are forgotten

How a project budget is built (top first)

Project budget	`Cost baseline + management reserve`
Management reserve	`For unknown-unknowns` `Outside the baseline; needs management approval`
Cost baseline	`Approved, time-phased budget` `Control accounts + contingency reserves`
Contingency reserves	`For identified risks (known-unknowns)` `Inside the baseline`
Work package and activity estimates	`Bottom-up costs of the planned work`

Worked example — outpatient wing (Tk crore)

LINE	AMOUNT	RUNNING TOTAL
Sum of work package estimates	4.20	4.20
+ Contingency reserve (identified risks, e.g. soil conditions, import duty changes)	0.40	4.60 = cost baseline
+ Management reserve (unforeseen events)	0.30	4.90 = project budget

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Contingency reserve vs management reserve

Contingency reserve

- Covers identified risks and their responses
- Included in the cost baseline
- Usually controlled by the PM within agreed rules
- Sized from risk analysis (e.g. expected value or simulation)
- Unused amounts can be released as risks pass

Management reserve

- Covers unforeseen, unidentified risks
- Not in the cost baseline but in the total budget
- Released only with management or sponsor approval
- Often set as a policy percentage
- Using it changes the cost baseline via change control

INSIGHT · THE S-CURVE AND FUNDING

Plotting the cumulative cost baseline over time usually produces an **S-curve**: slow spending at the start, a steep middle and a flattening end. Compare it with funding tranches from the finance team or donor; if planned spend exceeds available funding in any period, re-sequence the work — this is called **funding limit reconciliation**.

Single-number estimates hide risk. **Monte Carlo simulation** runs a schedule or cost model thousands of times, each time drawing random values from the ranges you provide (for example the O–M–P range of each activity). The result is a distribution of possible outcomes and the probability of meeting any date or budget.

Monte Carlo forecasting — how it works

01 Model — Schedule network or cost breakdown →

02 Ranges — Three-point estimates, risk events, correlations →

03 Simulate — Thousands of runs with random draws →

04 Read results — Histogram and S-curve of outcomes →

05 Decide — Commit at an agreed confidence, e.g. P80

EXAMPLE · READING A SIMULATION

The deterministic plan says the wing will be ready in **13 months**. Simulation shows only about a 20% chance of achieving that. **P50 = 14.5 months**, **P80 = 16 months**, **P90 = 17 months**. The sponsor chooses to commit publicly to 16 months (P80) and to treat 13 months as an internal stretch target. On agile work, a similar simulation can sample past weekly throughput to forecast when a set of backlog items will be finished.

PRO TIP · WHY SIMULATION BEATS ADDING UP

Where parallel paths merge, the merge point waits for the latest one — so the chance of finishing on time is lower than for any single path. This **merge bias** is invisible in a critical path calculation but shows up clearly in a Monte Carlo forecast.

SCENARIO · DECIDE · 'SPARE' MONEY

Halfway through the outpatient wing project, a department head asks you to use Tk 25 lakh of the contingency reserve to add a staff café, saying the money is 'unused anyway'. What should you do?

(A) Agree; the reserve has not been needed so far.

(B) Explain that contingency is reserved for identified risks; treat the café as a change request with its own impact analysis and funding decision by the sponsor.

(C) Use the management reserve instead without telling the sponsor.

(D) Reject the idea outright without passing it on.

Circle your choice. Answer and feedback: module answer key.

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CASE STUDY · BUDGETING A PRIVATE HOSPITAL WING IN CHATTOGRAM

A private hospital in Chattogram is adding a 60-bed wing. The bottom-up estimate of all work packages is Tk 38 crore. The risk register identifies piling problems on reclaimed land, possible import duty increases on medical equipment and monsoon delays. Expected values of these risks total Tk 3.2 crore. The board's policy sets management reserve at 5% of the cost baseline. Funding arrives in four quarterly tranches of Tk 11 crore each. A Monte Carlo cost simulation shows P50 = Tk 41 crore and P80 = Tk 43.5 crore.

1. Calculate the cost baseline and total project budget.

2. Who controls each reserve?

3. What does the funding schedule mean for planning?

4. At what confidence level would you recommend the board approve the budget, and why?

Model answer: module answer key.

REFLECT

Recall a project where the budget or deadline was set before anyone estimated the work. How would a range-based estimate or a simple Monte Carlo forecast have changed the conversation with the sponsor?

Key terms

Parametric estimating
Estimating by multiplying a quantity by a unit rate derived from historical data, e.g. cost per square metre.

Story point
A unitless, team-relative measure of the size of a backlog item, combining effort, complexity and uncertainty.

Critical path
The longest sequence of dependent activities through a schedule network; it determines the shortest possible project duration.

Free float
The amount of time an activity can be delayed without delaying the early start of any successor.

PERT estimate
A weighted three-point estimate calculated as (Optimistic + 4 × Most likely + Pessimistic) ÷ 6.

Velocity
The amount of work, usually in story points, a team completes to the definition of done in one iteration.

Total float
The amount of time an activity can be delayed without delaying the project finish date.

Crashing
Shortening the schedule by adding resources to critical activities, usually at extra cost.

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Fast-tracking
Shortening the schedule by overlapping activities normally done in sequence, at a higher risk of rework.

Management reserve
Budget outside the cost baseline for unforeseen work; its use requires management approval.

Contingency reserve
Budget or time within the baseline set aside for identified risks.

Cost baseline
The approved, time-phased budget, including contingency but excluding management reserve, used to measure cost performance.

KNOWLEDGE CHECK

21. Work packages total Tk 2.0 crore, contingency is Tk 0.2 crore and management reserve is Tk 0.1 crore. The cost baseline is:

(A) Tk 2.0 crore
(B) Tk 2.2 crore
(C) Tk 2.3 crore
(D) Tk 2.1 crore

22. Management reserve is intended for:

(A) Unforeseen work within scope (unknown-unknowns)
(B) Identified risks with planned responses (known-unknowns)
(C) New features requested by users after sign-off
(D) Team bonuses and recognition at project close

23. A Monte Carlo result says 'P80 = 16 months'. This means:

(A) The project will take exactly 16 months to complete
(B) There is only a 16% chance of finishing on time
(C) About 80% of simulated runs finished within 16 months
(D) The project will be 80% complete at month 16

24. Why might a Monte Carlo forecast show a later date than the critical path calculation?

(A) Simulation always adds a fixed buffer to every activity
(B) The critical path calculation is always wrong for large projects
(C) It ignores dependencies and treats every activity as independent
(D) It captures uncertainty and merge bias where paths converge

25. Comparing the time-phased cost baseline with available funding by period is called:

(A) Funding limit reconciliation
(B) Schedule crashing
(C) Resource smoothing analysis
(D) Benefits realisation planning

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MODULE 3 · PLAN

Answer key

Knowledge checks

1. B Analogous estimating uses comparable past projects and needs little detail — ideal early on.

2. B (2 + 4×5 + 14) ÷ 6 = 36 ÷ 6 = 6.

3. B (14 − 2) ÷ 6 = 2.

4. B Parametric estimating multiplies a quantity by a unit rate derived from historical data.

5. A Ranges show the uncertainty that exists, so decisions about reserves and commitments are informed.

6. A Only completed items that meet the definition of done count; partly finished work counts zero.

7. B 180 ÷ 30 = 6; 180 ÷ 25 = 7.2, rounded up to 8. So 6–8 iterations.

8. B The widening gaps reflect growing uncertainty for larger items.

9. D Different votes reveal different understanding; discussing them improves shared understanding and the estimate.

10. B 40 × 0.75 = 30 points.

11. D Finish-to-start is by far the most common relationship.

12. B TF = LS − ES = 8 − 5 = 3 (or LF − EF = 12 − 9 = 3).

13. C A successor can only start when all predecessors finish, so ES takes the largest EF: 10.

14. A It is preferred logic, not required by physics or law — so it is discretionary and can be changed (e.g. when fast-tracking).

15. C Negative float means a constraint date cannot be met with the current logic and durations.

16. C Fast-tracking performs activities in parallel or with overlap, increasing rework risk.

17. B (90,000 − 60,000) ÷ (7 − 4) = 30,000 ÷ 3 = Tk 10,000 per day.

18. B Only critical activities shorten the project; start with the cheapest per day saved.

19. C Levelling delays activities to respect resource limits, which can change the critical path and push the end date.

20. D Discretionary logic can be relaxed; mandatory logic cannot.

21. B Cost baseline = estimates + contingency = 2.2. Management reserve is added on top to reach the total budget of 2.3.

22. A Management reserve covers unidentified risks and requires management approval to use.

23. C P80 is the value at or below which 80% of simulated outcomes fall.

24. D Uncertainty on several converging paths makes on-time completion less likely than a single-path calculation suggests.

25. A Funding limit reconciliation adjusts the timing of work so planned spending fits within funding limits.

Scenarios & case studies

> **Scenario · 'Just give me one number'**
> **C** is best. Best. An analogous ROM range is honest, useful for the board and paired with a plan to reduce uncertainty.

> **Scenario · A velocity target**
> **B** is best. Best. You correct the misuse of velocity and redirect attention to measures that reflect value and flow.

> **Scenario · Using float**
> **C** is best. Best. You use float deliberately, understand that it is shared along the path, and communicate the reduced buffer.

> **Scenario · Two weeks earlier, please**
> **B** is best. Best. You base the response on the schedule model and give the sponsor clear trade-offs to decide on.

> **Scenario · 'Spare' money**
> **B** is best. Best. New scope goes through change control, and reserves are used only for their purpose.

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> **Case study · Budgeting a private hospital wing in Chattogram**
> **Cost baseline** = 38 + 3.2 = **Tk 41.2 crore**. **Management reserve** = 5% × 41.2 ≈ Tk 2.06 crore, so **project budget ≈ Tk 43.26 crore**. **Control:** the PM manages contingency within agreed rules for the identified risks; the board or sponsor approves any use of management reserve, which then updates the baseline through change control. **Funding:** total funding is Tk 44 crore, but only Tk 11 crore per quarter — the time-phased S-curve must be checked against each tranche; expensive equipment purchases may need to be re-sequenced or financed. **Confidence:** the baseline (Tk 41.2 crore) sits near P50, so there is roughly a coin-flip chance of staying within it. The total budget (≈ Tk 43.26 crore) is just under P80 (Tk 43.5 crore). Recommend the board approve funding close to P80 and treat the gap as a visible, governed risk — not hide it.

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04

MODULE 4 · DELIVER

Agile, Hybrid & Quality Execution

Deliver the work: run Scrum and Kanban well, design hybrids on purpose and build quality in

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Run Scrum with clear accountabilities, events, artifacts and commitments
- Improve flow with Kanban practices, WIP limits and flow metrics including the CFD
- Compare scaling frameworks neutrally and design a deliberate hybrid delivery model
- Plan, assure and control quality and argue for prevention using cost of quality
- Choose suitable contract types and run effective daily execution habits

LESSONS

1. Scrum in Practice · 28 min
2. Kanban, WIP Limits & Flow Metrics · 28 min
3. Scaling Frameworks & Hybrid Delivery Design · 28 min
4. Quality Management & the Cost of Quality · 27 min
5. Procurement, Contracts & Daily Execution Habits · 30 min

MODULE 4 · DELIVER · LESSON 1 · 28 MIN

Scrum in Practice

Accountabilities, events, artifacts and commitments — explained plainly

- Describe the three Scrum accountabilities and how they work together
- Explain the purpose and timeboxes of the Scrum events
- Link each Scrum artifact to its commitment

Scrum is a lightweight framework for teams solving complex problems. It rests on **empiricism** — knowledge comes from experience, so you make work **transparent**, **inspect** it often and **adapt** quickly — and on lean thinking: reduce waste and focus on essentials. A Scrum Team is small, typically ten people or fewer, cross-functional and self-managing.

Three accountabilities

PO

Product Owner

Maximises the value of the product. Owns and orders the product backlog, sets the product goal and is one person, not a committee — though they may represent many stakeholders.

SM

Scrum Master

Accountable for the team's effectiveness. Coaches the team and organisation in Scrum, removes impediments and helps everyone understand empirical working.

DEV

Developers

Everyone who creates the increment — engineers, designers, testers, analysts. They plan the sprint, hold each other accountable and meet the definition of done.

Five events (maximum timeboxes for a one-month sprint; shorter sprints usually need less)

EVENT	PURPOSE	TIMEBOX
The Sprint	Container for all other events; a fixed period in which an increment is created	One month or less
Sprint Planning	Decide why the sprint is valuable (sprint goal), what can be done and how	Up to 8 hours
Daily Scrum	Developers inspect progress toward the sprint goal and adapt the plan for the next day	15 minutes
Sprint Review	Inspect the outcome with stakeholders and adapt the product backlog	Up to 4 hours
Sprint Retrospective	Inspect how the team worked and plan improvements in quality and effectiveness	Up to 3 hours

Artifacts and their commitments

ARTIFACT	WHAT IT IS	COMMITMENT THAT GIVES IT FOCUS
Product Backlog	Ordered, emerging list of what is needed to improve the product	Product Goal — the long-term objective for the product
Sprint Backlog	Sprint goal + selected items + the plan to deliver them	Sprint Goal — the single objective for the sprint
Increment	A usable step toward the product goal that meets the definition of done	Definition of Done — the quality standard for the increment

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One sprint at a glance

01 **Sprint Planning**
Goal and plan

→

02 **Daily Scrums**
Daily inspection and adaptation

→

03 **Building the increment**
Refinement continues alongside

→

04 **Sprint Review**
Stakeholders inspect results

→

05 **Retrospective**
Team improves how it works

DEFINITION · BACKLOG REFINEMENT

Refinement is an ongoing activity, not a formal event: the product owner and developers break down, clarify and size upcoming items so they are ready for future sprint planning. Many teams spend a modest share of each sprint on it.

INSIGHT · FIVE VALUES MAKE IT WORK

Scrum depends on **commitment, focus, openness, respect and courage**. Without them, the events become empty rituals — a stand-up where nobody admits a problem inspects nothing.

WATCH OUT · COMMON ANTI-PATTERNS

The Daily Scrum turned into a status report for a manager. A product owner with no real authority to decide. A sprint review that is a one-way demo with no stakeholder feedback. A 'Scrum Master' who is simply the old project manager assigning tasks. Each one removes the inspection or adaptation Scrum relies on.

TEMPLATE

Sprint backlog board

Make the sprint goal and the team's plan visible and update it every day.

SPRINT GOAL	BACKLOG ITEM	SIZE (PTS)	TASKS / PLAN	WHO IS ON IT	STATUS	IMPEDIMENTS
Agents can onboard customers offline	*Capture ID photo offline*	*5*	*Camera screen; local encryption; sync queue; tests*	*Rafi, Nadia*	*In progress*	*Need test devices with Android 10*
Agents can onboard customers offline	*Sync status on dashboard*	*3*	*API endpoint; status badge; tests*	*Tanvir*	*To do*	—

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PRO TIP · WHERE DOES THE PROJECT MANAGER FIT?

Scrum does not define a project manager. In practice, PMs on Scrum projects often coordinate across several teams, manage budgets, contracts, risks and external stakeholders, and protect the team's ability to work. Do not assign tasks to developers or run their Daily Scrum.

SCENARIO · DECIDE · AN URGENT REQUEST MID-SPRINT

On day 4 of a two-week sprint, the head of sales asks a developer directly to add a promotional banner for a campaign starting next week. The developer asks you what to do. What is BEST?

(A) Tell the developer to add it quickly; it is small.

(B) Direct the request to the product owner, who decides whether it is more valuable than current work and, with the developers, whether it fits without endangering the sprint goal.

(C) Cancel the sprint and start a new one including the banner.

(D) Tell the head of sales to wait until the next sprint without discussing it.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

1. Who is accountable for ordering the product backlog?
   - (A) Product Owner
   - (B) Scrum Master
   - (C) Developers
   - (D) Project sponsor
2. The commitment for the Increment is the:
   - (A) Definition of Done
   - (B) Product Goal
   - (C) Sprint Goal
   - (D) Acceptance criteria
3. What is the maximum timebox for a Daily Scrum?
   - (A) 5 minutes
   - (B) 15 minutes
   - (C) 30 minutes
   - (D) 1 hour
4. Who can cancel a sprint?
   - (A) Scrum Master
   - (B) Product Owner
   - (C) Any stakeholder
   - (D) Developers
5. Which event focuses on improving the way the team works?
   - (A) Sprint Review
   - (B) Sprint Retrospective
   - (C) Sprint Planning
   - (D) Product backlog refinement

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MODULE 4 · DELIVER · LESSON 2 · 28 MIN

Kanban, WIP Limits & Flow Metrics

Visualise work, limit work in progress and read flow like a professional

- Apply the core Kanban practices to a team's workflow
- Set WIP limits and explain why they speed delivery up
- Calculate and interpret lead time, cycle time, throughput and Little's Law
- Read a cumulative flow diagram

Kanban is a method for improving the flow of work. It does not prescribe roles or events: you **start with what you do now**, make it visible, and improve step by step. It works well for support teams, operations, marketing, and project teams whose work arrives continuously.

Six core Kanban practices

K1

Visualise the work

Show every work item and workflow stage on a board so everyone sees the same picture.

K2

Limit work in progress

Cap how many items can be in each stage at once.

K3

Manage flow

Watch how work moves, spot blockages and act to keep items moving.

K4

Make policies explicit

Write down the rules: what 'ready for test' means, how urgent items are handled.

K5

Implement feedback loops

Regular replenishment, daily stand-ups at the board, delivery and service reviews.

K6

Improve collaboratively

Use data and small experiments to evolve the process together.

Example board with WIP limits

STAGE	READY	DEVELOP	TEST	DEPLOY	DONE
WIP limit	5	4	3	2	—
Items now	5	4	3	1	27 this month

DEFINITION · WIP LIMIT

A **work-in-progress limit** caps the number of items allowed in a stage. When a column is full, people **stop starting and start finishing** — they help move existing items forward instead of pulling new ones. Fewer items in progress means less task-switching, faster feedback and shorter waits.

Flow metrics

METRIC	DEFINITION	USE IT TO…
WIP	Number of items started but not finished	Keep load manageable; predict cycle time
Throughput	Number of items finished per unit of time	Forecast how much can be delivered

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METRIC	DEFINITION	USE IT TO…
Cycle time	Elapsed time from when work starts on an item to when it is finished	Measure team responsiveness and predictability
Lead time	Elapsed time from when the request is made (or committed to) until it is delivered	Measure what the customer experiences
Work item age	Time elapsed since an in-progress item was started	Spot items getting stuck before they become late

WATCH OUT · AGREE YOUR START AND END POINTS
Organisations define lead and cycle time slightly differently. Write down exactly which board columns start and stop each clock, or your metrics will be argued about rather than used.

Flow formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Little's Law	`Average cycle time = Average WIP ÷ Average throughput`	Holds for a reasonably stable system. Example: 12 items in progress, 6 finished per week → about 2 weeks average cycle time.
Reduce WIP to speed up	`Halve WIP at the same throughput → halve cycle time`	The fastest lever most teams have is starting less work.
Flow efficiency	`Active work time ÷ Total cycle time × 100%`	Often surprisingly low in knowledge work — items spend much of their time waiting, not being worked on.

A **cumulative flow diagram (CFD)** stacks the number of items in each stage over time. Each coloured band is a stage; the top edge of 'Done' rises as work is delivered. Reading it takes practice, but a few patterns tell you most of what you need.

How to read a CFD

- **Vertical** distance across the in-progress bands on a date = WIP at that moment.
- **Horizontal** distance from where an item enters progress to where it reaches Done ≈ approximate average cycle time.
- A band that **widens** = work is piling up in that stage — a bottleneck downstream of it or slow work in it.
- A **flat** Done line = nothing is being delivered — investigate immediately.
- Bands rising **smoothly and in parallel** = stable, predictable flow.
- A sudden drop in the top (backlog) line = scope removed; a jump = scope added.

EXAMPLE · KANBAN IN A BANK'S IT SUPPORT TEAM
A Dhaka bank's internal IT team had 70 tickets in progress and average cycle time of 19 days. They visualised every ticket, set WIP limits per person and stage, and made a policy for urgent incidents (a separate expedite lane, limit 1). Within three months WIP fell to about 25 and cycle time to roughly 7 days — with the same people.

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SCENARIO · DECIDE · A FULL TEST COLUMN

Your board shows Test at its WIP limit of 3, with two items waiting in 'Develop – done'. Two developers say they have nothing to do because they cannot pull new work. What should the team do?

(A) Raise the Test WIP limit to 6 so developers can keep moving work forward.

(B) Have the idle developers help with testing — pairing with testers, automating checks or fixing issues found — to finish items in Test.

(C) Let the developers start new items anyway and park them.

(D) Send the developers home early.

Circle your choice. Answer and feedback: module answer key.

PRO TIP · SCRUM AND KANBAN TOGETHER

Many Scrum teams use a Kanban board, WIP limits and flow metrics inside their sprints. The combination keeps Scrum's cadence and goal focus while improving flow within the sprint.

REFLECT

Look at your own work right now. How many things have you started but not finished? What would happen to your delivery speed if you limited yourself to three?

KNOWLEDGE CHECK

6. Average WIP is 20 items and average throughput is 10 items per week. By Little's Law, average cycle time is about:
   (A) 0.5 weeks (B) 2 weeks
   (C) 10 weeks (D) 200 weeks
7. On a CFD, a band that keeps getting wider indicates:
   (A) Work accumulating in that stage (B) Faster delivery through that stage
   (C) Scope being removed from the backlog (D) Lower WIP across the whole system
8. Lead time differs from cycle time because lead time:
   (A) Starts when the customer request is made or accepted (B) Starts only when the team begins active work
   (C) Excludes the waiting time before work is started (D) Measures the number of items finished per week
9. What is the main purpose of a WIP limit?
   (A) To keep everyone busy (B) To measure velocity
   (C) To prevent stakeholders from adding work (D) To reduce multitasking and improve flow
10. Which metric shows how long an unfinished item has been in progress?
    (A) Work item age (B) Throughput
    (C) Sprint velocity (D) Lead time

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MODULE 4 · DELIVER · LESSON 3 · 28 MIN

Scaling Frameworks & Hybrid Delivery Design

When many teams must work as one — and how to blend predictive and agile on purpose

- Summarise the main ideas of SAFe, LeSS, Scrum@Scale and Nexus neutrally
- Decide when scaling is needed and when it is not
- Design a deliberate hybrid delivery model for a real project

When a product needs more people than one small team, coordination becomes the main challenge: shared backlogs, dependencies, integration and aligned priorities. Several frameworks offer patterns for this. None is universally best — each reflects different assumptions about structure and control.

Scaling frameworks at a glance

FRAMEWORK	CORE IDEA	TYPICAL STRUCTURE	OFTEN CHOSEN WHEN	WATCH OUT FOR
SAFe (Scaled Agile Framework)	Comprehensive, configurable model spanning team, program and portfolio levels	Agile release trains of roughly 50–125 people planning together every Planning Interval (often 8–12 weeks)	Large enterprises need alignment with portfolio governance and budgeting	Can become heavy and role-rich if adopted wholesale
LeSS (Large-Scale Scrum)	Scrum applied to many teams with as little extra structure as possible	One product, one product owner, one product backlog, several teams in a common sprint	Organisations willing to simplify structure and descale	Requires deep organisational change
Scrum@Scale	Scale-free extension of Scrum through networks of teams	Scrum of Scrums for delivery; a MetaScrum for backlog priorities; executive teams to remove impediments	Organisations wanting to grow Scrum gradually	Relies on strong Scrum at team level
Nexus	Lightweight exoskeleton around 3–9 Scrum teams	A Nexus Integration Team focuses on integration and dependencies	A single product with a few teams and integration pain	Limited guidance beyond about nine teams

WATCH OUT · SCALE LAST, NOT FIRST

Many scaling problems are really team-level problems multiplied. Before adopting a framework, reduce dependencies (feature teams, modular architecture), shrink batch sizes and fix flow at team level. Sometimes the best scaling move is fewer people on the product.

Practices common to most scaled approaches

- A shared cadence so teams plan, integrate and review together
- Joint planning events with visible cross-team dependencies
- Frequent (ideally continuous) integration of all teams' work
- A single, clearly owned view of priorities across teams
- Communities of practice to share skills and standards

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Most organisations in 2026 run **hybrid** delivery. Good hybrids are designed on purpose: each part of the work gets the approach that fits its uncertainty, and the joins between parts are managed explicitly.

Five common hybrid patterns

H1

Predictive wrapper, agile core

Gates, budget and milestones are predictive; product development inside runs in sprints.

H2

Agile discovery, predictive build

Prototype and test ideas iteratively, then build the chosen solution with a fixed plan (common in construction and hardware).

H3

Parallel tracks

Agile software teams alongside predictive hardware, construction or procurement, joined by integration milestones.

H4

Predictive build, incremental rollout

The solution is completed predictively, then deployed in waves — site by site or region by region.

H5

Project plus flow

A project delivers a new capability; a Kanban team handles support and small changes in parallel.

Designing a hybrid in six steps

1. **Split the work** Divide scope into components (software, hardware, training, process, procurement).
2. **Rate uncertainty** Place each component on the uncertainty matrix from the Frame stage.
3. **Assign approaches** Predictive for stable, high-cost-of-change parts; agile or flow for uncertain parts.
4. **Design the joins** Integration milestones, interface agreements, shared cadence for reviews.
5. **Align governance** Map phase gates and reporting to both styles — milestones and release forecasts.
6. **Review and adapt** Check at each gate and retrospective whether the design still fits.

PRO TIP · MAKE THE JOINS VISIBLE

Keep a simple **dependency board**: one row per team or track, columns for upcoming iterations or months, and strings or arrows linking each dependency to the work that needs it. Review it at every joint planning session. A dependency without a named owner and date is a risk, not a plan.

CASE STUDY · ERP FOR A KNITWEAR EXPORTER IN NARAYANGANJ

A knitwear exporter with three factories and 9,000 workers is implementing a cloud ERP for merchandising, production planning, inventory and finance. The vendor's standard modules cover finance and inventory well, but merchandising and production planning need heavy configuration and a custom mobile app for line supervisors. Buyers in Europe require full traceability data from next autumn's season. Finance staff are experienced and process-driven; line supervisors have low digital literacy. The board wants a single go-live date for simplicity.

1. Which components suit predictive delivery and which suit agile?

2. Is a single big-bang go-live wise? What would you propose?

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3. How would you manage the joins between components?

4. What governance information would you give the board?

Model answer: module answer key.

SCENARIO · DECIDE · THE PMO WANTS A GANTT CHART

Your project uses an agile team for the customer app and predictive contracts for data-centre hardware. The PMO insists the agile team provide a detailed task-level Gantt chart for the next six months. What should you do?

(A) Force the team to produce the Gantt chart; the PMO sets the rules.

(B) Refuse — agile teams do not do planning.

(C) Meet the PMO's real need: provide a milestone roadmap integrated with the hardware schedule, release forecasts as ranges and burn-up charts, and agree this format in the governance plan.

(D) Produce the Gantt chart yourself without involving the team.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

11. Which framework is built around one product backlog and one product owner for several teams, with minimal extra roles?

(A) SAFe (B) Scrum@Scale
(C) LeSS (D) PRINCE2

12. In Nexus, which group focuses on integration across teams?

(A) The Executive MetaScrum team (B) The release train engineer
(C) The PMO (D) The Nexus Integration Team

13. What should usually come before adopting a scaling framework?

(A) Hiring more teams to increase delivery capacity (B) Reducing dependencies and fixing flow at team level
(C) Buying a scaling tool licence for all teams (D) Writing a long process manual for every role

14. A project prototypes a new clinic layout iteratively with staff, then builds it with a fixed construction plan. This is the pattern:

(A) Project delivery plus continuous flow (B) Parallel agile and predictive tracks
(C) Pure agile from discovery to build (D) Agile discovery, predictive build

15. What is the key feature of a well-designed hybrid?

(A) It uses every practice from every method on the project (B) Each element is chosen deliberately and joins are managed
(C) It avoids governance so teams can move faster (D) It always has one go-live at the end of the project

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MODULE 4 · DELIVER · LESSON 4 · 27 MIN

Quality Management & the Cost of Quality

Plan quality in, assure the process, control the results

- Distinguish quality planning, assurance and control
- Classify costs of quality and argue for prevention
- Use core quality tools and a definition of done to protect quality

Quality means the result **meets its requirements and is fit for use**. It is not the same as luxury. A low-cost budget phone can be high quality; an expensive one that crashes is not. Modern quality management prevents problems early instead of inspecting them out at the end.

DEFINITION · QUALITY VS GRADE

Grade is a category for products with the same use but different features (basic vs premium hotel room). **Quality** is how well the result meets its requirements. Low grade may be acceptable; low quality never is.

Three parts of quality management

Plan quality

- Identify standards and requirements
- Set measurable quality metrics
- Define acceptance criteria and definition of done
- Choose reviews, tests and tools

Manage quality (assurance)

- Is the process sound?
- Audits, process reviews, retrospectives
- Design for quality, checklists, training
- Improve the way work is done

Control quality

- Is the product right?
- Inspection, testing, measurement
- Compare results with criteria
- Verify deliverables before acceptance

Cost of quality (CoQ)

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost of quality	`CoQ = Cost of conformance + Cost of non-conformance`	Total cost spent to achieve quality plus the cost of failing to achieve it.
Cost of conformance	`Prevention + Appraisal`	Money spent to avoid failures (training, design reviews) and to find them (testing, inspection).
Cost of non-conformance	`Internal failure + External failure`	Failures found before delivery (rework, scrap) and after delivery (warranty, recalls, lost customers).

Cost of quality — examples

CATEGORY	SOFTWARE PROJECT	CONSTRUCTION / FACTORY
Prevention	Coding standards, pair programming, training	Method statements, worker training, supplier qualification
Appraisal	Code reviews, automated tests, security scans	Material testing, inspections, quality audits
Internal failure	Bugs fixed before release, rework	Rework of faulty concrete, rejected garments, scrap

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CATEGORY	SOFTWARE PROJECT	CONSTRUCTION / FACTORY
External failure	Production outages, data breaches, refunds	Buyer claims, product recalls, lost orders, reputation damage

INSIGHT · SHIFT LEFT

The later a defect is found, the more it costs to fix — sometimes dramatically. Investing in prevention and early appraisal (clear criteria, design reviews, automated tests that run on every change) is almost always cheaper than external failure.

Core quality tools

TOOL	WHAT IT SHOWS	USE WHEN
Cause-and-effect (fishbone) diagram	Possible causes grouped by category	Exploring the root causes of a problem
Pareto chart	Causes ranked by frequency or impact	Choosing which few causes to fix first
Control chart	Process results over time against control limits	Checking whether a process is stable
Check sheet	Tally of occurrences by type	Collecting defect data on the floor
Histogram	Distribution of a measurement	Seeing the spread and shape of results
Scatter diagram	Relationship between two variables	Testing whether one factor drives another
Flowchart	Steps and decision points in a process	Finding where errors or delays enter

EXAMPLE · PARETO ON THE SEWING FLOOR

A Gazipur garment factory recorded 1,000 defects in a month. A Pareto chart showed that **broken stitches, skipped stitches and uneven hems** caused about 78% of them. Instead of retraining everyone on everything, the team fixed needle maintenance and machine tension on the worst lines first — and halved defects in six weeks.

Agile quality practices

- A visible **definition of done** applied to every increment
- Acceptance criteria agreed before work starts
- Test automation and continuous integration on every change
- Code review or pairing as standard
- Retrospectives that tackle root causes, not just symptoms
- Small batches, so problems surface early

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Release quality checklist

☐ All acceptance criteria demonstrated to the product owner or customer

☐ Definition of done met for every item in the release

☐ Automated regression tests passing

☐ Security and data-protection checks completed

☐ Performance tested under expected peak load

☐ Accessibility checked for key user journeys

☐ Known defects reviewed and accepted or fixed

☐ User guides, release notes and training materials updated

☐ Rollback plan prepared and tested

☐ Support and operations teams briefed and ready

SCENARIO · DECIDE · SKIP THE TESTS?

With two days left in a sprint, developers suggest skipping the regression test suite so they can finish all planned items. The definition of done requires regression tests to pass. What should you encourage?

(A) Skip the tests this once and catch up next sprint.

(B) Keep the definition of done; finish fewer items fully, and discuss with the product owner which items to move back to the backlog.

(C) Mark the items done but add a note that tests are pending.

(D) Extend the sprint by two days.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

16. Which is an example of an appraisal cost?

(A) Staff training in quality methods
(B) Warranty claims from customers
(C) Customer refunds after complaints
(D) Product testing and inspection

17. A buyer rejects a shipment because of defects found at their warehouse. This is:

(A) Prevention and planning cost
(B) External failure cost
(C) Appraisal cost
(D) Internal failure cost

18. Which tool helps choose the few causes responsible for most problems?

(A) Flowchart
(B) Scatter diagram
(C) Pareto chart
(D) Histogram

19. Quality assurance mainly focuses on:

(A) Inspecting finished deliverables for defects
(B) Whether the processes used are effective
(C) Setting the quality budget for the project
(D) Approving or rejecting change requests

20. A basic but reliable product is best described as:

(A) High quality, low grade
(B) Low quality, low grade
(C) High grade, low quality
(D) Neither grade nor quality applies

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MODULE 4 · DELIVER · LESSON 5 · 30 MIN

Procurement, Contracts & Daily Execution Habits

Buy well, contract smartly and run delivery day by day

- Outline the procurement process from make-or-buy to closure
- Compare fixed-price, time-and-materials and cost-reimbursable contracts
- Describe agile-friendly contract options
- Build daily execution habits that keep delivery visible and moving

Few projects do everything in-house. Equipment, software, cloud services, contractors and consultants are bought from outside. **Procurement** is how you decide what to buy, choose the right supplier and contract, and manage the relationship until the work is accepted and closed.

Procurement in six steps

01 Make or buy — Compare in-house vs external on cost, skill, risk, control

→

02 Plan and specify — Statement of work, evaluation criteria, contract type

→

03 Solicit — Request for information, quotation or proposal

→

04 Select and award — Evaluate fairly against published criteria; negotiate

→

05 Manage — Performance, payments, changes, disputes

→

06 Close — Verify deliverables, final payment, lessons, records

DEFINITION · STATEMENT OF WORK (SOW)

The **SOW** describes what the supplier must deliver: scope, deliverables, standards, acceptance criteria, timeline and reporting. A vague SOW is the single biggest cause of supplier disputes.

Contract types and who carries the cost risk

TYPE	HOW PAYMENT WORKS	COST RISK MAINLY ON	BEST WHEN
Firm fixed price (FFP)	Agreed price for defined scope	Seller	Scope is clear and stable
Fixed price incentive fee (FPIF)	Fixed price with bonus or penalty for performance against targets, up to a ceiling	Shared, mostly seller	Clear scope; buyer wants to reward performance
Fixed price with economic price adjustment	Fixed price adjusted for agreed indices (e.g. steel, currency)	Shared	Multi-year contracts with volatile input costs
Time and materials (T&M)	Rates per hour/day plus materials	Buyer	Scope unclear, short engagements, staff augmentation
Cost plus fixed fee (CPFF)	Actual costs plus a fixed fee	Buyer	High uncertainty, research-like work
Cost plus incentive fee (CPIF)	Actual costs plus a fee that varies with performance; overruns and underruns shared	Shared, mostly buyer	Uncertain scope with measurable targets

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TYPE	HOW PAYMENT WORKS	COST RISK MAINLY ON	BEST WHEN
Cost plus award fee (CPAF)	Actual costs plus an award decided by the buyer's evaluation	Buyer	Performance is hard to measure objectively

WATCH OUT · FIXED PRICE IS NOT RISK-FREE FOR THE BUYER

A seller squeezed by an unrealistic fixed price may cut quality, raise change requests aggressively or walk away. The cheapest bid is often not the lowest total cost.

Agile-friendly contracting options

- **Capped time and materials** — pay for time used, up to an agreed ceiling.
- **Fixed price per iteration or increment** — buyer can stop after any increment if enough value is achieved.
- **Target cost with shared pain and gain** — overruns and savings are split, aligning both sides.
- **Scope-swap clauses** — buyer can exchange not-yet-started backlog items for new items of equal size without renegotiation.
- **Early termination by agreement** — buyer may end early with a modest fee when the remaining value is low.
- **Framework agreements with call-offs** — pre-agreed terms, work ordered in small packages as needs become clear.

SCENARIO · DECIDE · THE VENDOR'S CHANGE REQUEST

Your firm-fixed-price contract with a software vendor covers a hospital billing system. Users now want a new insurance-claim interface not in the statement of work. The vendor says it will cost extra. The finance manager insists the vendor should absorb it because 'the price is fixed'. What should you do?

(A) Agree with the finance manager and tell the vendor to deliver it at no cost.

(B) Check the SOW to confirm the interface is new scope, then process it through the contract's change procedure — impact, price and schedule — for an authorised decision.

(C) Ask the vendor's developers informally to add it without telling their manager.

(D) Cancel the contract and find a new vendor.

Circle your choice. Answer and feedback: module answer key.

Execution is won or lost in daily habits. Plans and contracts set direction, but delivery depends on what the team sees, discusses and unblocks every day.

Six daily execution habits

E1

Short stand-up at the board

15 minutes, focused on the goal and blockers, not on reporting to a manager.

E2

Keep information radiators current

Visible boards, burn-up charts, risk and impediment lists anyone can read without asking.

E3

Hunt impediments

Capture every blocker, give it an owner and a date, and escalate fast if it threatens a goal.

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E4

Check WIP and flow

Look for stuck or aging items before starting anything new.

E5

Touch base with a stakeholder

One short, purposeful conversation a day prevents many surprises.

E6

Close the loop

Update the plan, decisions and risks so tomorrow starts from the truth.

Impediment escalation ladder

LEVEL	WHO ACTS	TYPICAL TIMEFRAME	EXAMPLE
1 — Team	Developers themselves	Same day	Pair to resolve a tricky merge conflict
2 — Team coach or PM	Scrum Master, team lead or PM	1–2 days	Get test devices or environment access
3 — Product owner or functional manager	PO, line manager, vendor manager	Within the iteration	Clarify conflicting requirements, free a shared specialist
4 — Sponsor or steering group	Sponsor, steering committee	When a milestone or goal is threatened	Resolve funding, cross-department priority clashes

PRO TIP · MAKE RADIATORS HONEST

An information radiator only works if it shows bad news as clearly as good news. A board that is always green is either a miracle or a mirage. Encourage red early — it is the cheapest time to fix things.

Key terms

Product Owner
The accountable person in Scrum who maximises product value and owns the ordering of the product backlog.

Scrum Master
The accountable person in Scrum for team effectiveness, coaching the team and organisation and removing impediments.

Sprint Goal
The single objective for a sprint that gives the developers focus and flexibility in what they deliver.

Increment
A usable step toward the product goal that meets the definition of done.

WIP limit
A cap on the number of work items allowed in a workflow stage at one time.

Cycle time
The elapsed time from when work starts on an item until it is finished.

Lead time
The elapsed time from when an item is requested or committed to until it is delivered to the customer.

Throughput
The number of work items finished per unit of time.

Cumulative flow diagram
A stacked area chart showing the number of items in each workflow stage over time.

Cost of quality
The total cost of conformance (prevention and appraisal) and non-conformance (internal and external failure).

Firm fixed price contract
A contract in which the seller delivers defined scope for an agreed price and carries most of the cost risk.

Time and materials contract
A contract that pays agreed rates for time and materials used, with most cost risk on the buyer.

KNOWLEDGE CHECK

21. Under which contract type does the buyer carry the most cost risk?

A) Firm fixed price
B) Fixed price incentive fee
C) Fixed price with economic price adjustment
D) Cost plus fixed fee

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22. Which document defines what a supplier must deliver?

(A) Risk management plan
(B) Sprint backlog
(C) Statement of work
(D) Benefits register

23. Which clause lets an agile buyer exchange unstarted backlog items for new ones of equal size?

(A) Liquidated damages
(B) Force majeure
(C) Scope-swap clause
(D) Retention clause

24. A blocker threatens a release milestone and the team and PM cannot resolve it. What is the appropriate next step?

(A) Wait and raise it with the team at the next retrospective
(B) Keep it from the board until a fix is found
(C) Escalate promptly to the right decision-maker with options
(D) Remove the blocked item from the definition of done

25. What makes an information radiator effective?

(A) It is updated monthly with written commentary
(B) Only managers can view it, so issues stay discreet
(C) It shows the true status, problems included, to everyone
(D) It is always green, so stakeholders stay confident

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MODULE 4 · DELIVER

Answer key

Knowledge checks

1. A The product owner orders the backlog to maximise value.

2. A Each artifact has a commitment; for the increment it is the definition of done.

3. B The Daily Scrum is timeboxed to 15 minutes, regardless of sprint length.

4. B Only the product owner can cancel a sprint, typically if the sprint goal becomes obsolete.

5. B The retrospective inspects people, interactions, processes, tools and the definition of done to plan improvements.

6. B 20 ÷ 10 = 2 weeks.

7. A A widening band means more items are entering a stage than leaving it.

8. A Lead time reflects the customer's wait from request to delivery; cycle time starts when work begins.

9. D Limiting WIP reduces task switching and queues, which shortens cycle time.

10. A Work item age tracks items still in progress, so you can act before they become late.

11. C LeSS keeps Scrum's simplicity: one product, one backlog, one product owner, several teams.

12. D The Nexus Integration Team helps the Scrum teams integrate their work and resolve dependencies.

13. B Many scaling problems are multiplied team-level problems; fix them first.

14. D Iterative discovery followed by predictive execution suits work where building is expensive to change.

15. B A deliberate design with managed joins avoids the 'worst of both worlds'.

16. D Appraisal costs are spent finding defects: testing, inspections and audits.

17. B Defects found after delivery to the customer are external failure costs — usually the most expensive.

18. C A Pareto chart ranks causes so the team focuses on the vital few.

19. B Assurance looks at the process; control checks the product.

20. A It meets its requirements (high quality) with fewer features (low grade).

21. D In cost-reimbursable contracts the buyer pays actual costs, so overruns fall mainly on the buyer.

22. C The statement of work describes scope, deliverables, standards and acceptance criteria for the supplier.

23. C A scope-swap clause keeps price and timeline while allowing priorities to change.

24. C Impediments that threaten goals should move up the escalation ladder quickly with clear options.

25. C Radiators work because they make reality visible to all without needing to ask.

Scenarios & case studies

> **Scenario · An urgent request mid-sprint**
> **B** is best. Best. The product owner owns priorities; the developers own the sprint plan. Together they can adjust scope while protecting the goal.

> **Scenario · A full test column**
> **B** is best. Best. 'Stop starting, start finishing': swarming on the bottleneck improves flow and builds shared skills.

> **Case study · ERP for a knitwear exporter in Narayanganj**
> **Predictive:** finance and inventory using standard modules (stable requirements, accounting rules, data migration with clear cut-over). **Agile:** merchandising configuration and the supervisor app — uncertain needs, high user-experience risk; build in sprints with supervisors testing on the floor. **Go-live:** a big bang across three factories is risky. Propose finance and inventory first, then production modules piloted in one factory, then incremental rollout to the other two (pattern H4 combined with H3), with the traceability deadline as a fixed milestone. **Joins:** integration milestones every few sprints, an interface agreement on data structures, a shared release calendar and a cross-team integration lead. **Board:** a milestone roadmap plus release forecasts as ranges, budget burn against the baseline, top risks and readiness measures (training completion, data quality) — not task-level detail.

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> **Scenario · The PMO wants a Gantt chart**
> **C** is best. Best. You satisfy the governance need for visibility and integration while using planning artefacts that fit adaptive work.

> **Scenario · Skip the tests?**
> **B** is best. Best. Quality standards are not negotiable within the sprint — scope is. Fewer done items beat more half-done ones.

> **Scenario · The vendor's change request**
> **B** is best. Best. The contract and SOW define the baseline; genuine new scope is handled through the agreed change mechanism.

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05

MODULE 5 · STEER

Steer — Risk, Performance & Control

Spot uncertainty early, measure progress truthfully and forecast the finish with confidence.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Identify, analyse and prioritise risks using qualitative and quantitative techniques, including EMV and decision trees.
- Plan owned responses for threats and opportunities and manage issues with clear escalation.
- Calculate and interpret earned value measures and forecasts for a sponsor.
- Use agile and flow metrics without misusing them, and build dashboards that drive decisions.
- Communicate forecasts as honest ranges with confidence levels.

LESSONS

1. Risk Thinking & Identification · 25 min
2. Analysing Risk: From Heat Maps to Decision Trees · 30 min
3. Responding to Risk & Managing Issues · 25 min
4. Earned Value Management · 35 min
5. Agile Metrics, Dashboards & Honest Forecasting · 30 min

MODULE 5 · STEER · LESSON 1 · 25 MIN

Risk Thinking & Identification

See uncertainty early, name it precisely and write it down where it can be managed.

- Distinguish a risk, an issue and an opportunity using a single clear test.
- Write well-formed risk statements using cause–event–effect structure.
- Run a pre-mortem and use a risk breakdown structure and an assumptions log to widen identification.

Every project is a bet on the future. Steering — stage 5 of the **ZPMP Value Loop** — is the discipline of watching that bet, adjusting it and telling the truth about it. It begins with risk: the uncertain events that could push your outcomes off course, for better or for worse.

Risk, issue or opportunity? The ZPMP time test

R

Risk (threat)

An **uncertain** future event that would hurt at least one objective if it happened. It has a probability below 100%. You manage it *before* it happens.

O

Opportunity

An uncertain future event that would **help** an objective — finish earlier, cost less, deliver more value. Positive risk deserves the same attention as negative risk.

I

Issue

Something that **has already happened** or is certain and is affecting the project now. You resolve it, you do not ‘mitigate’ it.

A

Assumption

Something treated as true for planning without proof. Every assumption hides a risk: ‘what if this is false?’

INSIGHT · ONE QUESTION SORTS THEM ALL

Ask: **‘Has it happened yet?’** If no and it would hurt — risk. If no and it would help — opportunity. If yes — issue. A risk that materialises moves from the risk register to the issue log, and the risk entry is closed with a note.

Vague risks produce vague responses. ‘Supplier problems’ is a worry, not a risk. A useful risk statement names the cause you can see today, the uncertain event, and the effect on a specific objective.

A well-formed risk statement

MEASURE	FORMULA	WHAT IT TELLS YOU
Cause–event–effect	`Because of <cause>, <event> may occur, which would lead to <effect on objective>.`	Forces precision. The cause is a fact today; the event is uncertain; the effect names cost, time, scope, quality or benefit.
Example	`Because our only certified network engineer is due for parental leave in March, cut-over testing may slip, which would delay go-live by 3–4 weeks.`	Now the team can see who owns it, what would trigger it and which responses make sense.

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Ways to find risks (use more than one)

PM

Pre-mortem

Imagine it is a year from now and the project has failed badly. Each person silently writes why. Then share and cluster. Hindsight imagined in advance unlocks honest concerns people would not otherwise voice.

RBS

Risk breakdown structure

A hierarchy of risk sources — e.g. Technical, External, Organisational, Project management — each split into sub-categories. Walk through it to spot blind areas.

AL

Assumptions & constraints log

Review every assumption and ask what happens if it is wrong. Track who validates it and by when.

SH

Stakeholder & expert interviews

Operations staff, suppliers and finance see risks the core team cannot. Ask: 'What usually goes wrong with projects like this?'

DR

Document & data review

Lessons-learned registers, audit findings, past schedules, contracts and incident logs from similar work.

SW

SWOT & prompt lists

Strengths, weaknesses, opportunities, threats — or checklists such as PESTLE — to cover the external environment.

Running a 45-minute pre-mortem

1. **Set the scene (5 min)** Explain the project goal and say: 'It is twelve months from now. This project was a painful failure. Why?'
2. **Silent writing (8 min)** Everyone writes reasons independently — sticky notes or a shared board. Silence protects junior voices from anchoring on senior opinions.
3. **Round-robin share (12 min)** One reason per person per round until exhausted. No debate yet.
4. **Cluster and vote (10 min)** Group similar reasons and dot-vote the most worrying clusters.
5. **Convert (10 min)** Turn the top clusters into cause–event–effect risk statements, each with a proposed owner.

A starter risk breakdown structure

LEVEL 1	LEVEL 2 EXAMPLES	TYPICAL QUESTIONS
Technical	Requirements, technology maturity, integration, performance, data migration	Have we built this before? How many interfaces?
External	Suppliers, regulation, market, weather, currency, politics	What could change outside our control?
Organisational	Funding, priorities, shared resources, other projects	Could our people or budget be pulled away?
Project management	Estimating, planning, communication, control	How confident are we in our estimates?
People & change	Adoption, skills, morale, key-person dependency	Will users actually change how they work?

WATCH OUT · IDENTIFICATION IS NOT A ONE-OFF WORKSHOP

Risk lists go stale within weeks. Revisit risks at each iteration review, stage gate or major change, and give anyone on the team a simple way to raise a new risk at any time.

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SCENARIO · DECIDE · THE POLITE SILENCE

During a risk workshop for a hospital pharmacy system, the head of IT talks for 20 minutes and the rest of the team only agrees with him. You suspect clinical staff have concerns they are not raising. What is the BEST move?

(A) Switch to a silent-writing pre-mortem round so everyone contributes independently before discussion.

(B) Thank the head of IT and close the workshop since the list looks complete.

(C) Ask the clinical staff to email you their risks privately after the meeting.

(D) Escalate to the sponsor that the head of IT dominates meetings.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a project you have worked on. Write one risk that actually materialised as an issue. How would a cause–event–effect statement written at the start have changed what the team did?

Key terms

Risk
An uncertain event or condition that, if it occurs, affects at least one project objective — negatively (threat) or positively (opportunity).

Risk breakdown structure (RBS)
A hierarchical grouping of risk sources used as a prompt to identify risks systematically.

Pre-mortem
A workshop in which the team imagines the project has failed and works backwards to identify the reasons, surfacing risks early.

KNOWLEDGE CHECK

1. A key supplier has just informed you their factory is closed for two weeks. How should this be recorded?

   (A) As an issue in the issue log

   (B) As a new risk with high probability

   (C) As an assumption to validate

   (D) As an opportunity to renegotiate

2. Which risk statement is best formed?

   (A) Vendor risk: the vendor might be poor at support, which could cause problems for users

   (B) There is a risk of delays, cost overruns and quality problems across the whole project

   (C) The vendor has no local office and is new to us, so the support contract is at risk

   (D) With no local vendor office, fixes may take over 48 hours and breach the 24-hour SLA

3. What is the main purpose of silent writing in a pre-mortem?

   (A) To reduce anchoring and status effects so all contribute

   (B) To save meeting time by avoiding any group discussion

   (C) To avoid having to document risks formally afterwards

   (D) To let the PM select the risks without team influence

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4. An assumption log is useful for risk identification because…

   (A) each assumption that might be false is a potential risk
   (B) assumptions never change once they have been logged
   (C) it can replace the risk register on smaller projects
   (D) sponsors must sign off every assumption

5. A new regulation might allow your team to reuse an approved design, saving six weeks. This is…

   (A) an issue
   (B) a constraint
   (C) an opportunity
   (D) a residual risk

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MODULE 5 · STEER · LESSON 2 · 30 MIN

Analysing Risk: From Heat Maps to Decision Trees

Rank risks quickly, then put numbers on the ones that matter.

- Use a probability–impact matrix with defined scales to prioritise risks.
- Calculate expected monetary value (EMV) and use it to size contingency.
- Solve a simple decision tree and read a Monte Carlo output (P50, P80, S-curve, tornado).

Once you have a list of risks, you cannot treat them all equally. Analysis answers two questions: **which risks deserve attention first**, and **how much exposure does the project carry overall?** Qualitative analysis is fast and covers every risk. Quantitative analysis is slower and is reserved for the risks and decisions that drive outcomes.

Qualitative vs quantitative analysis

Qualitative

- Ranks individual risks using scales (e.g. 1–5)
- Fast, cheap, done on every project
- Output: prioritised risk list, heat map
- Weakness: subjective unless scales are defined

Quantitative

- Models combined effect of risks on cost/schedule
- Needs data, tools and skill
- Output: probability of hitting dates/budgets, EMV, sensitivity
- Use on large, complex or high-stakes projects

Define the scales before you score

SCORE	PROBABILITY	SCHEDULE IMPACT	COST IMPACT
1 Very low	< 10%	< 1 week	< 1% of budget
2 Low	10–30%	1–2 weeks	1–3%
3 Medium	31–50%	2–4 weeks	3–5%
4 High	51–70%	1–2 months	5–10%
5 Very high	> 70%	> 2 months	> 10%

PRO TIP · CALIBRATE, DON'T GUESS

Agree the scale definitions with the sponsor before scoring. Without them, one person's 'high' is another's 'medium' and the heat map becomes decoration. Also capture **urgency** (how soon a response is needed) and **proximity** (how soon it could occur).

Multiply probability by impact to get a risk score (1–25). Plot each risk on a 5×5 grid: the top-right zone (scores of roughly 15+) gets immediate, owned responses; the middle zone gets watched; the bottom-left zone may be accepted and monitored.

Expected monetary value

MEASURE	FORMULA	WHAT IT TELLS YOU
EMV of one risk	`EMV = Probability × Impact`	Threats carry negative impact, opportunities positive. A 30% chance of a −50,000 loss has EMV −15,000.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Project risk exposure	`Σ EMV of all quantified risks`	Netting threats and opportunities gives a rational starting point for the contingency reserve.
Decision node	`Choose the branch with the best EMV`	In a decision tree, each option's EMV = its direct cost + Σ (probability × outcome) of what follows.

Worked example: sizing contingency with EMV

RISK	PROBABILITY	IMPACT (BDT)	EMV (BDT)
R1 Customs delay on imported routers	30%	−500,000	−150,000
R2 Data migration defects need rework	10%	−1,200,000	−120,000
O1 Bulk-licence discount if order before June	20%	+300,000	+60,000
Net exposure			−210,000

A contingency reserve of about **BDT 210,000** covers the expected net exposure for these known risks. It is held by the project manager for identified risks ('known unknowns'). A **management reserve**, held by the sponsor or organisation, covers unidentified risks ('unknown unknowns') and is outside the cost baseline.

CASE STUDY · CHATTOGRAM PORT: SHOULD WE REHEARSE THE CUT-OVER?

A container terminal at Chattogram is replacing its gate and yard-planning software. The vendor proposes an optional full cut-over rehearsal on a weekend, costing **USD 40,000**. Without a rehearsal, the team estimates a **35%** chance that go-live fails and the old system must be restored, causing berth delays and penalties of about **USD 300,000**. With a rehearsal, the chance of failure drops to **10%** (the same USD 300,000 impact if it happens).

1. Draw the decision tree and calculate the EMV of each option.

2. Which option should the project choose, and by how much does it improve expected cost?

3. What non-monetary factors should the sponsor also weigh?

Model answer: module answer key.

Reading a Monte Carlo output

S

S-curve

Cumulative probability of finishing by each date (or within each cost). Read across from 80% to find the P80 date.

P50

P50 / P80

P50: half of simulated outcomes finish by this date. P80: four in five do. Commit externally to a higher-confidence value; manage internally to a stretch value.

TOR

Tornado chart

Ranks which inputs most influence the result. Spend risk effort on the longest bars.

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CI

Criticality index

How often an activity appears on the critical path across simulations — a task at 70% deserves more attention than its single-point float suggests.

EXAMPLE · TURNING THE S-CURVE INTO A SENTENCE

'Our deterministic plan says 30 November. The simulation gives only a 22% chance of that. P50 is 14 December and P80 is 9 January. We recommend committing to early January and protecting the top three tornado drivers.' That is honest forecasting — a range with a confidence level, not a single hopeful date.

SCENARIO · DECIDE · THE CONFIDENT SPONSOR

Your Monte Carlo analysis shows the target launch date has a 25% probability. The sponsor says: 'Just remove the pessimistic estimates and run it again.' What should you do FIRST?

(A) Rerun the model with the optimistic values to keep the sponsor happy.

(B) Walk the sponsor through the tornado chart, show which risks drive the spread and discuss responses that would genuinely raise confidence.

(C) Escalate to the steering committee that the sponsor is ignoring risk.

(D) Quietly add a buffer to the schedule without telling anyone.

Circle your choice. Answer and feedback: module answer key.

Key terms

Expected monetary value (EMV)

Probability multiplied by monetary impact; summed across risks or branches to compare options and size reserves.

Contingency reserve

Budget or time within the baseline set aside for identified risks; controlled by the project manager.

KNOWLEDGE CHECK

6. A risk has a 40% probability and an impact of −80,000. Its EMV is...

(A) −32,000 (B) −48,000

(C) −80,000 (D) +32,000

7. Which is the strongest reason to define probability and impact scales before scoring?

(A) It makes the heat map clearer and more colourful (B) It is required by law in most regulated industries and sectors

(C) It reduces subjective, inconsistent scoring between people (D) It removes the need for later quantitative risk analysis

8. A Monte Carlo S-curve shows the P80 finish date as 20 March. This means...

(A) the project is certain to finish on 20 March as planned (B) 80% of simulated outcomes finished on or before 20 March

(C) there is an 80% chance that the project finishes after 20 March (D) 20 March is the end date of the critical path schedule

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9. Option A costs 20,000 and leaves a 10% chance of a −100,000 loss. Option B costs nothing and leaves a 25% chance of the same loss. Which is better by EMV?

   (A) B (−25,000 vs −30,000)
   (B) A (−30,000 vs −25,000)
   (C) They are equal in EMV terms
   (D) It cannot be decided from EMV

10. Who normally controls the management reserve?

   (A) The project manager, within agreed limits
   (B) The team lead responsible for the work
   (C) The vendor, under the contract terms
   (D) Senior management or the sponsor

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MODULE 5 · STEER · LESSON 3 · 25 MIN

Responding to Risk & Managing Issues

Pick the right response, give it an owner and a trigger, and resolve issues fast.

- Select response strategies for threats and opportunities and justify the choice.
- Assign risk owners, triggers, residual and secondary risks in a working risk register.
- Run an issue log and escalate at the right level and time.

Analysis tells you which risks matter. Response planning decides **what you will actually do** — and who will do it. A response without an owner, a budget and a date is just a hope.

Response strategies

For threats

- **Escalate** – outside the project's authority; hand to the right level and confirm they accept it
- **Avoid** – change the plan to remove the threat (e.g. drop a risky feature)
- **Transfer** – shift the financial impact to a third party (insurance, fixed-price contract, warranty)
- **Mitigate** – reduce probability and/or impact (prototype, extra testing, second supplier)
- **Accept** – active (set aside contingency) or passive (deal with it if it happens)

For opportunities

- **Escalate** – beyond project authority; pass to programme/portfolio
- **Exploit** – act to make it certain (assign your best people to finish early)
- **Share** – partner with a party better placed to capture it (joint venture, gain-share)
- **Enhance** – raise probability and/or impact (add resources to a promising path)
- **Accept** – take the benefit if it comes, without active pursuit

DEFINITION · TRANSFER MOVES THE MONEY, NOT THE RISK

Buying insurance or signing a fixed-price contract shifts the financial consequence, but the event can still delay your project and hurt your reputation. Transfer almost always leaves residual risk you must still watch.

Every response needs these five parts

OWN

Risk owner

One named person who monitors the risk and makes sure the response happens. Not a department, not 'the team'.

TRG

Trigger

An observable warning sign that the risk is about to occur or has occurred — e.g. 'supplier misses the second milestone'.

ACT

Response actions

Specific tasks, with dates and cost, added to the schedule and budget.

RES

Residual risk

The exposure that remains after the response. If it is still above tolerance, plan more.

SEC

Secondary risk

A new risk created by the response itself — e.g. a second supplier introduces integration risk.

PRO TIP · FALLBACK PLANS

For high-impact risks, prepare a **fallback plan** ('Plan B') to use if the primary response fails, and decide in advance what trigger activates it. Decisions made calmly beforehand are better than decisions made in a crisis.

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TEMPLATE **Risk register**

Track every identified risk from first sighting to closure. Review the top risks at every status meeting.

ID / STATEMENT (CAUSE–EVENT–EFFECT)	P × I = SCORE	OWNER	TRIGGER	STRATEGY & ACTIONS	RESIDUAL / SECONDARY	STATUS
R07 Because only one engineer holds the PLC licence, the cut-over may slip if he is unavailable, delaying go-live 3 weeks	*3 × 4 = 12*	*Farhana (ops lead)*	*Engineer leave request or illness*	*Mitigate: certify a second engineer by 15 May (BDT 85k)*	*Residual 1 × 4 = 4; secondary: training pulls engineer off build for 1 week*	*Open – action on track*

When a risk becomes real, or a problem appears that nobody predicted, it moves to the **issue log**. Issues need speed, clarity of ownership and a clear path for escalation.

Issue lifecycle

01 Log
Describe impact, date raised, who raised it

→

02 Assess
Severity, urgency, objectives affected

→

03 Assign
One owner and a target resolution date

→

04 Resolve or escalate
Within tolerance: act. Beyond tolerance: escalate with options

→

05 Close & learn
Record the fix and feed lessons learned

Escalate with options, not just problems

ESCALATION ELEMENT	WHAT TO INCLUDE
What happened	One-sentence description and when it was discovered
Impact	Effect on scope, schedule, cost, quality, benefits — with numbers
Tolerance	Which agreed threshold it breaches (e.g. > 2 weeks or > 5% cost)
Options	2–3 realistic options with trade-offs
Recommendation	Your preferred option and why
Decision needed by	The date after which options close

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SCENARIO · DECIDE · THE TRIGGERED RISK

Your risk register lists ‘key subcontractor insolvency’ with a fallback plan to activate a pre-qualified alternative. Today a news report says the subcontractor has entered administration. What should the project manager do FIRST?

- (A) Call a team meeting to brainstorm what to do.
- (B) Ask the sponsor for permission to look for a new subcontractor.
- (C) Implement the agreed fallback plan and update the issue log and risk register.
- (D) Wait to confirm the news through official channels before doing anything.

Circle your choice. Answer and feedback: module answer key.

Key terms

Residual risk
The risk that remains after responses have been implemented.

Secondary risk
A new risk that arises as a direct result of implementing a risk response.

KNOWLEDGE CHECK

11. A team buys an insurance policy covering equipment damage during shipment. Which strategy is this?
    - (A) Transfer
    - (B) Avoid
    - (C) Mitigate
    - (D) Exploit
12. To make sure a likely early-finish opportunity actually happens, the PM assigns the most experienced engineers to the task. This is...
    - (A) Enhance
    - (B) Share
    - (C) Accept
    - (D) Exploit
13. A response to one risk creates a new integration risk. The new risk is called...
    - (A) residual risk
    - (B) an emerging issue
    - (C) a trigger
    - (D) secondary risk
14. A threat is outside the project’s authority but within the programme’s. What should the PM do?
    - (A) Accept it passively and note it in the register
    - (B) Ignore it, as it is outside the project’s authority
    - (C) Escalate it to the programme manager and confirm ownership
    - (D) Avoid it by recommending that the project is cancelled
15. Which element of an escalation is MOST often missing and most valuable to decision makers?
    - (A) The project logo and reporting template
    - (B) A list of everyone who has been informed
    - (C) Realistic options with a recommendation
    - (D) The full risk register as an attachment

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MODULE 5 · STEER · LESSON 4 · 35 MIN

Earned Value Management

Measure progress in money, read the variances, and forecast the finish honestly.

- Calculate PV, EV, AC and the core variances and indices.
- Choose the right EAC formula for the situation and compute ETC, VAC and TCPI.
- Interpret EVM results in plain language for a sponsor.

'We have spent 42% of the budget and we feel about 40% done' tells a sponsor almost nothing. **Earned value management (EVM)** combines scope, schedule and cost into one measurement system so you can say exactly how much work has been accomplished for the money spent and time elapsed — and forecast where you will end up.

The three base measures

PV

Planned value

The budgeted cost of the work **scheduled** to be done by now. The baseline line on the S-curve.

EV

Earned value

The budgeted cost of the work **actually completed** by now. EV = % complete × BAC (or the sum of the budgets of completed work packages).

AC

Actual cost

What you have **really spent** on the work completed by now.

BAC

Budget at completion

The total approved budget for the work (the cost baseline, without management reserve).

Variances and indices

MEASURE	FORMULA	WHAT IT TELLS YOU
Schedule variance	`SV = EV − PV`	Negative means behind schedule (in value terms).
Cost variance	`CV = EV − AC`	Negative means over budget for the work done.
Schedule performance index	`SPI = EV ÷ PV`	Below 1.0: progressing slower than planned. 0.9 = earning 90 paisa of planned work per taka planned.
Cost performance index	`CPI = EV ÷ AC`	Below 1.0: getting less value than spent. 0.8 = 80 paisa of value per taka spent.

INSIGHT · THE MEMORY TRICK

EV comes first in every variance and index formula. A negative variance or an index below 1.0 is always bad news; positive or above 1.0 is good news.

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Forecasts

MEASURE	FORMULA	WHAT IT TELLS YOU
EAC — typical (CPI continues)	`EAC = BAC ÷ CPI`	Use when current cost performance is expected to continue. The most common default.
EAC — atypical (variance was a one-off)	`EAC = AC + (BAC − EV)`	Use when the overrun had a cause that will not recur; remaining work runs at budget.
EAC — CPI and SPI both matter	`EAC = AC + (BAC − EV) ÷ (CPI × SPI)`	Use when a fixed date is pushing you to spend more to catch up.
EAC — new estimate	`EAC = AC + bottom-up ETC`	Use when the original plan is no longer valid; re-estimate the remaining work.
Estimate to complete	`ETC = EAC − AC`	How much more money is needed.
Variance at completion	`VAC = BAC − EAC`	Negative means expected to finish over budget.
To-complete performance index	`TCPI = (BAC − EV) ÷ (BAC − AC)`	The CPI you must achieve on remaining work to finish on BAC. Use (BAC − EV) ÷ (EAC − AC) if an EAC has been approved.

CASE STUDY · WORKED EXAMPLE: A WAREHOUSE APP IN GAZIPUR

A logistics firm is building a warehouse management app. **BAC = BDT 1,000,000**, planned evenly over 10 months. At the end of month 4, the plan said 40% of the work should be done. The team has actually completed 35% of the work and spent **BDT 420,000**.

1. Calculate PV, EV, SV, CV, SPI and CPI.
2. Calculate EAC (typical), ETC, VAC and TCPI to BAC.
3. Calculate EAC if the overrun is judged atypical, and the CPI × SPI version.
4. Explain the status to the sponsor in two sentences.

Model answer: module answer key.

Reading SPI and CPI together

SPI	CPI	PLAIN-LANGUAGE STATUS	TYPICAL RESPONSE
> 1	> 1	Ahead and under budget	Check estimates were not padded; consider releasing funds
> 1	< 1	Ahead but over budget	Maybe crashing or overtime — is the speed worth the cost?
< 1	> 1	Behind but under budget	Often under-resourced; add capacity if the date matters

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SPI	CPI	PLAIN-LANGUAGE STATUS	TYPICAL RESPONSE
< 1	< 1	Behind and over budget	Root-cause analysis, re-plan, escalate if beyond tolerance

WATCH OUT · EVM TRAPS

1) Measuring % complete by gut feel — use rules such as **0/100**, **50/50** or milestone weights. 2) SPI drifts back to 1.0 at the end of every project because all work eventually gets 'earned' — late in a project, check the critical path too. 3) EVM says nothing about quality: work earned but full of defects is not value.

TEMPLATE **EVM tracker**

Update at each reporting period to see trends rather than single snapshots.

PERIOD	PV	EV	AC	CPI	SPI	EAC (BAC ÷ CPI)
Month 4	*400,000*	*350,000*	*420,000*	*0.83*	*0.88*	*1,200,000*

SCENARIO · DECIDE · THE CPI CONVERSATION

At month 6, CPI = 0.78 and the TCPI to BAC is 1.35. The sponsor asks you to 'tighten up and finish on budget'. What is the BEST response?

(A) Agree and promise to finish on budget through extra effort.

(B) Explain that a TCPI of 1.35 is very unlikely to be achieved and present options: approve a revised EAC, reduce scope or change approach.

(C) Stop reporting CPI since it upsets the sponsor.

(D) Cut testing to save money.

Circle your choice. Answer and feedback: module answer key.

Key terms

Earned value (EV)
The budgeted value of the work actually completed to date.

Cost performance index (CPI)
EV ÷ AC — the cost efficiency of work performed; below 1.0 means over budget.

To-complete performance index (TCPI)
The cost efficiency that must be achieved on remaining work to meet a target (BAC or EAC).

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KNOWLEDGE CHECK

16. EV = 90,000, PV = 100,000, AC = 120,000. What is CPI?

(A) 0.90
(B) 0.75
(C) 1.11
(D) 1.33

17. Using the same numbers, what is SV?

(A) −30,000
(B) +10,000
(C) −20,000
(D) −10,000

18. BAC = 500,000 and CPI = 0.80. Assuming performance continues, what is the EAC?

(A) 625,000
(B) 400,000
(C) 580,000
(D) 600,000

19. A one-time cost spike came from a hardware price shock that will not recur. Which EAC formula fits best?

(A) BAC ÷ CPI
(B) AC + (BAC − EV)
(C) AC + (BAC − EV) ÷ (CPI × SPI)
(D) BAC × SPI

20. TCPI to BAC is 0.92. What does this mean?

(A) The project is 8% over budget and must recover the overrun
(B) Remaining work can run at 92% efficiency and still meet budget
(C) The project is 8% behind schedule on the remaining work
(D) The project must improve its cost efficiency by 92% to finish on budget

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MODULE 5 · STEER · LESSON 5 · 30 MIN

Agile Metrics, Dashboards & Honest Forecasting

Show progress so people can act on it — without watermelons or vanity charts.

- Read burndown, burnup, velocity and flow metrics and avoid common misuse.
- Design a status dashboard with defined RAG criteria and decision-oriented content.
- Track OKRs and communicate forecasts as ranges with confidence.

Agile teams rarely calculate EVM, but they still need to see whether they are on track. Their metrics are lighter, more frequent and more visual — and just as easy to misuse. (Scrum and Kanban mechanics were covered in **Deliver**; here we focus on steering with the data they produce.)

Burndown vs burnup

Burndown

- Plots work **remaining** against time
- Simple; good for a single sprint
- Hides scope change: if scope grows and the team delivers, the line looks flat
- Question it answers: 'Will we finish what we committed to?'

Burnup

- Plots work **completed** and **total scope** as two lines
- Makes scope growth visible
- Better for releases and stakeholder conversations
- Question it answers: 'When will the lines meet — and is the target moving?'

Velocity: useful, fragile

USE

Planning aid

A team's average completed points per sprint helps it decide how much to pull into the next sprint and roughly how many sprints a backlog needs.

X1

Never compare teams

Story points are relative to each team's own scale. Comparing velocity between teams is meaningless and invites gaming.

X2

Never make it a target

When velocity becomes a goal, estimates inflate and the number rises while delivered value does not.

ALT

Prefer flow measures

Throughput (items finished per week) and cycle time (start to finish per item) are harder to game and work well for forecasting.

INSIGHT · WHEN A MEASURE BECOMES A TARGET…

…it stops being a good measure. This idea, usually attributed to economist Charles Goodhart, explains why velocity targets, 'lines of code' goals and 'tickets closed' leaderboards backfire. Measure to learn, not to judge.

A **status dashboard** exists so stakeholders can make decisions. If a reader cannot tell within 30 seconds what is on track, what is not and what they need to do, the dashboard has failed.

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Define RAG criteria in advance

STATUS	SCHEDULE	COST	MEANING FOR READERS
Green	Forecast within 1 week of baseline	EAC within 3% of BAC	On track — no action needed
Amber	Forecast 1–4 weeks late	EAC 3–8% over	At risk — team has a recovery plan; watch
Red	Forecast > 4 weeks late or milestone missed	EAC > 8% over	Off track — decision or help needed from the reader

WATCH OUT · THE WATERMELON PROJECT

Green on the outside, red on the inside. Projects report green for months and then turn red overnight. Causes: subjective RAG, fear of bad news, and reporting activity instead of outcomes. Fix it with objective criteria, forecasts rather than snapshots, and a culture that rewards early warnings.

What a good one-page dashboard shows

1. **Headline**
   Overall status, one-sentence summary and trend arrow (improving, stable, worsening).
2. **Decisions needed**
   What you need from the reader and by when — put this near the top.
3. **Forecast**
   Finish date and cost as ranges with confidence (e.g. P50/P80).
4. **Top risks & issues**
   The three to five that matter, with owners.
5. **Outcomes**
   Leading indicators of value: adoption, OKR key results, quality.

OKRs on a project

MEASURE	FORMULA	WHAT IT TELLS YOU
Objective	`Qualitative, inspiring direction`	e.g. ‘Make clinic booking effortless for patients in Rajshahi’.
Key result	`Measurable outcome with a baseline and target`	e.g. ‘Online bookings rise from 12% to 45% by December’.
Progress score	`(current − baseline) ÷ (target − baseline)`	From 12% to 45%, a current value of 34% scores (34 − 12) ÷ 33 ≈ 0.67. For ambitious ‘stretch’ OKRs many organisations treat about 0.6–0.7 as a healthy result.

Forecasting honestly with throughput

1. **Gather history** Collect weekly throughput (items finished) for the last 10–15 weeks.
2. **Size the remaining work** Count backlog items, and add a realistic allowance for the scope growth you have historically seen.
3. **Simulate** Randomly sample past weekly throughputs many times to see how many weeks the remaining items take (a simple Monte Carlo).
4. **Report a range** ‘85% likely by 30 June; 50% likely by 9 June.’

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5. **Re-forecast often** Update weekly as new data arrives; forecasts are living, not promises carved in stone.

SCENARIO · DECIDE · THE DEMAND FOR ONE DATE

A marketing director wants ‘one firm date’ for an app launch. Your throughput forecast gives 50% by 12 May and 85% by 2 June. What is the BEST response?

(A) Recommend committing externally to 2 June, plan internally toward mid-May, and agree to update the forecast weekly.

(B) Give 12 May — it is the most likely date.

(C) Refuse to give any date because agile teams do not commit to dates.

(D) Add 50% to the 12 May date to be safe.

Circle your choice. Answer and feedback: module answer key.

Dashboard health check

☐ RAG criteria are written down and agreed with the sponsor

☐ The first thing a reader sees is overall status and decisions needed

☐ Forecasts are shown as ranges with confidence levels

☐ Trends are shown, not just this week’s snapshot

☐ Top risks and issues have named owners

☐ At least one outcome or benefit measure appears, not only activity

☐ Data refreshes automatically from the work-management tool where possible

☐ No metric is used to compare or rank individual team members

☐ Someone has tested the dashboard with a real reader in under one minute

Key terms

Burnup chart
A chart showing completed work and total scope over time, making scope change visible.

Watermelon reporting
Status that appears green externally while the underlying reality is amber or red.

KNOWLEDGE CHECK

21. Which chart BEST reveals that scope is growing faster than the team delivers?

(A) Burnup
(B) Burndown
(C) Pie chart of story types
(D) Velocity bar chart

22. Management proposes ranking teams by velocity to reward the best. What is the main problem?

(A) Velocity is too difficult to calculate consistently across sprints
(B) Points are team-relative, so comparisons mislead and invite inflation
(C) Teams with the highest velocity will burn out sooner
(D) Velocity is only a valid measure for Kanban teams, not Scrum teams

23. A key result moves from a baseline of 20 to a current 50 against a target of 80. What is its progress score?

(A) 0.38
(B) 0.50
(C) 0.63
(D) 0.75

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24. What is the MOST effective way to prevent watermelon reporting?

(A) Use objective RAG criteria and reward early bad news

(B) Ban the colour red so teams focus on solutions, not blame

(C) Report less often so problems have time to be resolved

(D) Let each PM define green in a way that suits their project

25. A forecast states ‘85% likely by 2 June’. This is an example of…

(A) a probabilistic forecast

(B) a delivery guarantee

(C) a schedule baseline

(D) a velocity target for the team

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MODULE 5 · STEER

Answer key

Knowledge checks

1. A It has already happened, so it is an issue. Risks are uncertain future events; this one is certain and current.

2. D It names a present cause, an uncertain event and a specific effect on an objective. The others are vague, or leave out the cause, the event or a measurable effect.

3. A Independent input before discussion stops the loudest or most senior voice from shaping everyone's answers.

4. A Asking 'what if this is wrong?' about each assumption converts hidden uncertainty into explicit risks.

5. C It is uncertain and would benefit the schedule — a positive risk, or opportunity.

6. A 0.40 × −80,000 = −32,000. Threats carry negative values.

7. C Shared definitions turn opinions into comparable scores.

8. B P80 is the value that 80% of iterations meet or beat.

9. A A = −20,000 + (0.10 × −100,000) = −30,000. B = 0.25 × −100,000 = −25,000. B has the less negative EMV.

10. D Management reserve sits outside the baseline and requires management approval to release.

11. A Insurance shifts the financial impact to a third party: transfer.

12. D Exploit aims to make the opportunity certain. Enhance would only increase its likelihood or size.

13. D A risk caused by a response is a secondary risk.

14. C Escalation is correct when the response belongs at a higher level; the PM confirms acceptance and records it.

15. C Leaders decide faster and better when given options, trade-offs and a clear recommendation.

16. B CPI = EV ÷ AC = 90,000 ÷ 120,000 = 0.75.

17. D SV = EV − PV = 90,000 − 100,000 = −10,000.

18. A EAC = BAC ÷ CPI = 500,000 ÷ 0.80 = 625,000.

19. B When the variance is atypical, assume the remaining work will run at budget rate.

20. B A TCPI below 1.0 means the remaining work can be done slightly less efficiently than planned and still finish on BAC.

21. A A burnup shows total scope as its own line, so growth is visible.

22. B Velocity is a team's internal planning aid, not a cross-team productivity measure.

23. B (50 − 20) ÷ (80 − 20) = 30 ÷ 60 = 0.50.

24. A Objective thresholds plus psychological safety around bad news keep status honest.

25. A It expresses a date with a confidence level based on data.

Scenarios & case studies

> **Scenario · The polite silence**
> **A** is best. Correct. Silent, independent input reduces anchoring and status effects and is the quickest way to surface hidden concerns.

> **Case study · Chattogram port: should we rehearse the cut-over?**
> Rehearse: −40,000 + (0.10 × −300,000) = **−70,000**. No rehearsal: 0 + (0.35 × −300,000) = **−105,000**. Rehearsing is better by **USD 35,000** in expected value, so choose the rehearsal. Beyond money: reputation with shipping lines, safety in the yard, staff confidence with the new system, and the fact that the rehearsal also trains people. EMV supports the decision; it does not replace judgement, especially for low-probability, catastrophic outcomes.

> **Scenario · The confident sponsor**
> **B** is best. Correct. Focus the conversation on the real drivers and actions that change the probability — scope, resources, risk responses — rather than on the numbers themselves.

> **Scenario · The triggered risk**
> **C** is best. Correct. The trigger has occurred; execute the pre-planned response, record the new issue and close or update the risk.

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Case study · Worked example: a warehouse app in Gazipur

PV = 40% × 1,000,000 = **400,000**. EV = 35% × 1,000,000 = **350,000**. AC = **420,000**. SV = 350,000 − 400,000 = **−50,000** (behind). CV = 350,000 − 420,000 = **−70,000** (over). SPI = 350/400 = **0.875**. CPI = 350/420 ≈ **0.833**.

EAC typical = 1,000,000 ÷ 0.833 = **1,200,000**. ETC = 1,200,000 − 420,000 = **780,000**. VAC = 1,000,000 − 1,200,000 = **−200,000**. TCPI = (1,000,000 − 350,000) ÷ (1,000,000 − 420,000) = 650/580 ≈ **1.12**.

EAC atypical = 420,000 + 650,000 = **1,070,000**. EAC with CPI × SPI = 420,000 + 650,000 ÷ (0.833 × 0.875 ≈ 0.729) ≈ **1,311,000**.

To the sponsor: ‘We are about 12% behind schedule and getting 83 paisa of value for every taka spent. If this continues we will finish around BDT 1.2 million; finishing on budget would need every remaining taka to deliver 1.12 taka of planned work — far better than our current 0.83 — so we need a decision on scope, approach or budget.’

Scenario · The CPI conversation

B is best. Correct. Use the data to reset expectations and give the sponsor real choices.

Scenario · The demand for one date

A is best. Correct. Match the commitment’s confidence to the cost of missing it, and keep the forecast live.

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06

MODULE 6 · LEAD

Lead — People, Teams & Communication

Grow teams that trust each other, resolve conflict well and communicate with purpose.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Apply servant and situational leadership and support teams through development stages.
- Motivate people using intrinsic drivers and lead with emotional intelligence.
- Resolve conflict with the right mode and negotiate on interests using BATNA.
- Clarify decision rights with RACI/DACI and plan communication that is read and acted upon.
- Lead remote, hybrid and cross-cultural teams and coach using GROW and SBI.

LESSONS

1. Leadership Styles & Team Development · 30 min
2. Motivation & Emotional Intelligence · 25 min
3. Conflict Resolution & Negotiation · 30 min
4. Decisions, Roles & Communication Planning · 30 min
5. Remote, Hybrid & Cross-Cultural Teams; Coaching & Feedback · 30 min

MODULE 6 · LEAD · LESSON 1 · 30 MIN

Leadership Styles & Team Development

Serve the team, adapt your style, and build the safety that high performance needs.

- Explain servant leadership and apply situational leadership to individual team members.
- Recognise Tuckman's stages and choose leader behaviours for each.
- Describe the traits of high-performing teams and practical ways to build psychological safety.

Plans do not deliver projects; people do. **Lead** — stage 6 of the ZPMP Value Loop — is where a project manager earns the trust that makes every other stage work. In modern practice, the PM is less a commander and more a **multiplier**: someone who makes a group of capable people more effective together than apart.

Servant leadership in practice

S1

Remove impediments

Clear the blockers the team cannot clear themselves — access, approvals, missing information, outside interference.

S2

Shield focus

Protect the team from constant interruptions and unprioritised requests; channel demand through the backlog or change process.

S3

Grow people

Invest in skills, give stretch opportunities and share credit publicly.

S4

Listen first

Ask before telling. The people closest to the work usually see the problem most clearly.

S5

Hold the purpose

Keep reminding everyone why the work matters and who benefits from it.

INSIGHT · SERVANT DOES NOT MEAN PASSIVE

A servant leader still sets direction, holds standards and makes hard calls. The difference is the motive — the leader's authority is used for the team's success and the project's value, not for personal status.

No single style fits every person or moment. **Situational leadership** (associated with Hersey and Blanchard) matches the leader's mix of direction and support to each person's competence and commitment on a specific task.

Situational leadership: match style to readiness

PERSON'S STATE ON THIS TASK	LEADER STYLE	WHAT IT LOOKS LIKE
New and enthusiastic, low skill	Directing	Clear instructions, close follow-up, frequent check-ins
Some skill, confidence dipping	Coaching	Still directive, plus explanation, encouragement and two-way talk
Capable but hesitant or unsure	Supporting	Less direction; ask questions, build confidence, share decisions
Skilled and committed	Delegating	Agree outcomes, then step back; be available when asked

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EXAMPLE · SAME PERSON, DIFFERENT STYLES

Tanvir is a senior back-end developer (delegate API design to him) but has never presented to a steering committee (direct and coach him on that). Style is chosen per **task**, not per person.

Tuckman's team development stages

01 Forming
Polite, unsure; leader provides clarity on goals and roles

→

02 Storming
Friction over ideas and approaches; leader coaches and facilitates conflict

→

03 Norming
Agreements form; leader supports and reinforces working norms

→

04 Performing
Self-managing and productive; leader delegates and removes blockers

→

05 Adjourning
Work ends; leader recognises contributions and supports transitions

PRO TIP · STAGES ARE NOT A ONE-WAY STREET

A new member, a reorganisation or a crisis can send a performing team back to storming. Expect it, name it and help the team re-norm — often with a fresh working agreement.

Traits of high-performing teams

PS — Psychological safety
People can ask questions, admit mistakes and challenge ideas without fear of humiliation or punishment.

CG — Clear, shared goal
Everyone can state what success looks like and why it matters.

DT — Dependability
People do what they say, on time and to standard.

RS — Clear roles & structure
Who decides what, and how work flows, is understood.

CF — Constructive conflict
Debate ideas openly, then commit — disagreement is about the work, never the person.

SO — Self-organisation
The team decides how to do its work within agreed boundaries.

Five habits that build psychological safety

1. **Frame work as learning** Say early: 'This is new; we will get things wrong and that is how we find the right answer.'
2. **Model fallibility** Admit your own mistakes and say 'I might be missing something — what do you see?'
3. **Ask genuine questions** Invite the quietest voices first; use silent brainstorming.
4. **Respond productively** Thank people who bring bad news; focus on the fix, not the blame.
5. **Run blameless reviews** After incidents, ask what in the system allowed this, not who is at fault.

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SCENARIO · DECIDE · THE QUIET EXPERT

In stand-ups, a newly joined data engineer never raises blockers. Yesterday you learned she has been stuck for three days waiting for database access. What should the project manager do FIRST?

(A) Remind the whole team at the next stand-up that blockers must be raised immediately.

(B) Report to her line manager that she is not following the process.

(C) Assign her a buddy to monitor her work.

(D) Fix the access, then speak with her privately to understand what made raising it hard and agree an easy way to flag blockers.

Circle your choice. Answer and feedback: module answer key.

Key terms

Servant leadership
A leadership approach that focuses on serving the team — removing obstacles, growing people and enabling them to deliver value.

Tuckman model
A model of team development through forming, storming, norming, performing and adjourning.

Psychological safety
A shared belief within a team that it is safe to take interpersonal risks such as speaking up, asking questions or admitting mistakes.

KNOWLEDGE CHECK

1. A team member is highly skilled and motivated on a task. Which situational style fits best?
   (A) Directing (B) Coaching
   (C) Supporting (D) Delegating
2. Team members are openly arguing about the technical approach two weeks after kick-off. Which Tuckman stage is this most likely?
   (A) Forming (B) Storming
   (C) Norming (D) Adjourning
3. Which behaviour MOST directly builds psychological safety?
   (A) Publicly naming who caused each defect to deter repeats (B) Thanking someone who reports a problem early
   (C) Keeping every meeting short and focused (D) Using a strict RACI for every decision
4. Which statement about servant leadership is accurate?
   (A) The leader avoids making decisions for the team (B) The leader uses authority to enable the team's success
   (C) It only works in Scrum teams with a dedicated coach (D) It means doing the team's work for them when needed

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MODULE 6 · LEAD · LESSON 2 · 25 MIN

Motivation & Emotional Intelligence

Understand what really drives people — including you — and lead with self-awareness.

- Distinguish intrinsic and extrinsic motivation and apply autonomy, mastery and purpose.
- Use hygiene-factor thinking to diagnose dissatisfaction.
- Apply the components of emotional intelligence to everyday project moments.

Project managers often lead people who do not report to them and whose pay they do not set. That makes understanding motivation essential: you must create conditions in which people **want** to do great work.

Two sources of motivation

Extrinsic

- Comes from outside: pay, bonuses, titles, deadlines, praise
- Works well for routine, well-defined tasks
- Can crowd out interest in creative work if overused
- Its effect fades once the reward is expected

Intrinsic

- Comes from the work itself: interest, growth, meaning
- Drives quality and creativity in complex work
- Sustains effort when no one is watching
- Grows when people have choice and see impact

Autonomy, mastery, purpose

AU

Autonomy

Let people choose **how** they do their work — methods, tools, sequence — within agreed outcomes and constraints.

MA

Mastery

Provide challenges slightly beyond current skill, plus time to learn, feedback and visible progress.

PU

Purpose

Connect tasks to the people who benefit: show the farmer using the app, the patient who books in two minutes.

DEFINITION · HYGIENE FACTORS VS MOTIVATORS

A two-factor view (associated with Frederick Herzberg) separates **hygiene factors** — pay, working conditions, policies, job security — whose absence causes dissatisfaction, from **motivators** — achievement, recognition, responsibility, growth — that create real engagement. Fixing hygiene stops complaints; it rarely creates enthusiasm on its own.

Diagnose before you motivate

WHAT YOU OBSERVE	LIKELY ROOT CAUSE	USEFUL RESPONSE
Complaints about laptops, overtime pay, seating	Hygiene problem	Fix or escalate the condition; do not answer with a pep talk
Work is done but with no initiative	Low autonomy or purpose	Delegate decisions; connect work to users
Bored senior engineer	No mastery challenge	Offer a stretch problem or mentoring role
Drop in energy after a reorganisation	Uncertainty and loss of purpose	Explain the why, clarify roles, listen

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Emotional intelligence (EI) is the ability to recognise and manage your own emotions and to read and influence the emotions of others. Popularised by Daniel Goleman, it is often grouped into five components.

Five components of emotional intelligence

SA
Self-awareness
Know your moods, triggers, strengths and limits — e.g. noticing you get curt when a vendor misses dates.

SR
Self-regulation
Pause before reacting; choose the response. 'Let me come back to you in an hour' is a powerful sentence.

MO
Motivation
Drive that comes from inner standards and the goal, staying constructive under pressure.

EM
Empathy
Understand others' perspectives and feelings — including the sponsor's fears and the user's frustrations.

SS
Social skills
Build networks, influence, manage conflict and inspire — EI turned outward into action.

A 90-second reset when emotions spike

1. **Notice** Name the feeling privately: 'I am angry because I feel blamed.'
2. **Breathe and delay** Slow breathing; delay replies to heated messages.
3. **Get curious** Ask: 'What pressure might the other person be under?'
4. **Respond to the problem** Return to facts and next steps.

SCENARIO · DECIDE · THE ANGRY EMAIL

At 11 p.m. a client director emails the whole steering group saying your team is 'incompetent' because a demo crashed. You feel furious. What is the BEST action?

(A) Wait until morning, gather facts, then call the director to listen, acknowledge the impact and share a recovery plan.

(B) Reply-all immediately, pointing out the client's own environment caused the crash.

(C) Ignore the email; it will blow over.

(D) Forward it to your team so they know how the client feels.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · UNIVERSITY LMS MIGRATION: MOTIVATION SLUMP

A private university in Sylhet is moving 600 courses to a new learning management system. Six instructional designers were seconded part-time from faculties. After two months, quality is falling: courses are copied over without redesign, and designers skip review meetings. In one-to-ones they say: 'This is admin work — nobody will notice if it is good', and two mention they still have full teaching loads.

1. Which issues here are hygiene problems and which are motivator problems?

2. Propose three actions using autonomy, mastery and purpose.

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3. What should the PM escalate, and to whom?

Model answer: module answer key.

Key terms

Intrinsic motivation
Motivation that comes from the satisfaction of the work itself — interest, growth and meaning.

Emotional intelligence
The ability to recognise, understand and manage one's own emotions and to recognise and influence others' emotions.

KNOWLEDGE CHECK

5. Team members complain that their overtime has not been paid for two months. In two-factor terms, this is...
   - (A) a motivator (achievement) issue
   - (B) a lack-of-purpose issue
   - (C) a mastery and growth issue
   - (D) a hygiene factor issue

6. Which action BEST increases autonomy?
   - (A) Giving a detailed checklist and method for every task
   - (B) Letting the team choose its own testing approach within standards
   - (C) Adding a team bonus for finishing work faster than planned
   - (D) Holding daily status calls to review each person's tasks

7. A PM notices she gets defensive whenever QA reports defects. Recognising this pattern is an example of...
   - (A) empathy
   - (B) social skill
   - (C) self-awareness
   - (D) situational leadership

8. For creative, complex knowledge work, which approach is MOST likely to sustain high-quality effort?
   - (A) Large bonuses tied to output volume
   - (B) Strict monitoring of hours and output
   - (C) Public ranking of individual performance
   - (D) Building autonomy, mastery and purpose

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 6 · LEAD · LESSON 3 · 30 MIN

Conflict Resolution & Negotiation

Use conflict as fuel, not fire — and negotiate on interests, not positions.

- Choose among the five Thomas–Kilmann conflict modes based on the situation.
- Separate interests from positions and use BATNA to negotiate with confidence.
- Facilitate a conflict conversation step by step.

Conflict is normal on projects: competing priorities, scarce resources, technical opinions, schedules and personalities. Unmanaged, it drains energy. Managed well, it surfaces better ideas. The skill lies in choosing the right **mode** and handling it early, privately and on the facts.

Thomas–Kilmann modes: when to use each

MODE	ASSERTIVE / COOPERATIVE	WHEN IT FITS	RISK IF OVERUSED
Collaborating (problem-solving)	High / high	Important issue, time available, need commitment from both sides	Slow; not worth it for trivial issues
Competing (forcing)	High / low	Emergencies, safety, legal or ethical non-negotiables	Resentment, lost ideas, low buy-in
Compromising (reconciling)	Medium / medium	Equal power, time pressure, temporary fix acceptable	Both sides partly unhappy; may mask the real problem
Accommodating (smoothing)	Low / high	Issue matters more to the other side; preserving relationship; you are wrong	Your concerns pile up unheard
Avoiding (withdrawing)	Low / low	Trivial issue, emotions need cooling, someone else is better placed	Problems grow while ignored

INSIGHT · COLLABORATE IS THE DEFAULT, NOT THE ONLY ANSWER

Situational exam questions usually favour **collaborating / problem-solving** because it addresses root causes. But a safety violation calls for forcing, and a two-minute disagreement over font colours calls for accommodating or avoiding. Match the mode to the stakes and time.

Facilitating a conflict conversation

1. **Meet early and privately** Address it before it hardens; public settings make people defend positions.
2. **Set ground rules** One person speaks at a time; focus on the problem; assume good intent.
3. **Hear each side fully** Ask each person to describe the situation and what matters to them. Summarise back.
4. **Find shared goals and facts** ‘We both want the release to be stable by 15 June.’ Separate facts from interpretations.
5. **Generate options together** Brainstorm before judging. Look for options that serve both sets of interests.
6. **Agree and follow up** Write down the agreement, owners and a check-in date.

Negotiation is structured conflict resolution. Project managers negotiate with vendors, functional managers, sponsors and team members constantly — for people, money, dates and scope.

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Positions vs interests

Position (what they say)

- ‘I need two of your testers for the whole of March.’
- ‘The price is final.’
- ‘We cannot go live before Eid.’

Interest (why they say it)

- Their own release has a regulatory deadline mid-March
- The vendor needs predictable cash flow this quarter
- Branch staff will be on leave; nobody to support users

Principled negotiation toolkit

INT

Focus on interests

Ask ‘why does that matter to you?’ until you reach the underlying need.

OPT

Invent options

Expand the pie: phased delivery, different payment terms, shared resources, scope swaps.

CRI

Use objective criteria

Market rates, benchmarks, contract terms, past data — so neither side has to back down to the other’s will.

BAT

Know your BATNA

Your Best Alternative To a Negotiated Agreement — what you will do if no deal is reached. A strong BATNA gives you the confidence to walk away from a bad deal.

ZOPA

Find the ZOPA

The zone of possible agreement lies between each side’s walk-away point. No overlap, no deal — look for new options.

EXAMPLE · BATNA IN ACTION

A Dhaka fintech is negotiating cloud-hosting renewal. The incumbent asks for a 30% increase. The PM’s team has already priced a migration to another provider at a one-time cost plus 10% higher run costs for six months. That is their BATNA. Knowing it, they can calmly propose a 5% increase with a two-year commitment — and mean it when they say they have an alternative.

CASE STUDY · 5G ROLLOUT: CIVIL WORKS VS RADIO PLANNING

A mobile operator is rolling out 5G sites across Dhaka and Narayanganj. The radio planning lead wants to change antenna positions at 40 sites after new drive-test data. The civil works contractor manager refuses: towers are already reinforced to the old design, and every change costs money and two weeks. The two have stopped speaking and copy you on increasingly hostile emails. The regulator’s coverage milestone is in 10 weeks.

1. What are the likely interests behind each position?

2. Which conflict mode would you aim for and why?

3. Outline the steps you would take this week.

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Model answer: module answer key.

SCENARIO · DECIDE · TWO SENIORS, ONE TESTER

Two functional managers each claim the same senior tester for the same two weeks. Both are senior to you and are getting heated in a meeting. What should you do FIRST?

(A) Pause the debate, meet both privately to understand the interests and constraints, then bring options to a joint conversation.

(B) Decide yourself, based on whichever project you think matters more.

(C) Escalate straight to the portfolio board.

(D) Suggest they each take the tester for one week to split it evenly.

Circle your choice. Answer and feedback: module answer key.

Key terms

Thomas–Kilmann modes
Five conflict-handling approaches — competing, collaborating, compromising, avoiding and accommodating — mapped on assertiveness and cooperativeness.

BATNA
Best Alternative To a Negotiated Agreement: the course of action a party will take if negotiations fail.

KNOWLEDGE CHECK

9. A team member is about to deploy code that bypasses a mandatory security control. Which conflict mode is appropriate?
   (A) Accommodating / smoothing
   (B) Avoiding
   (C) Competing / forcing
   (D) Compromising

10. Which mode produces a win-win outcome and long-term commitment, but takes the most time?
   (A) Collaborating
   (B) Compromising
   (C) Accommodating
   (D) Avoiding

11. A vendor's BATNA becomes weaker. What is the likely effect?
   (A) The vendor gains more negotiating power
   (B) The vendor loses negotiating power
   (C) BATNA has no effect
   (D) The contract becomes void

12. 'We need the server room by Friday' is best described as...
   (A) an interest
   (B) a BATNA
   (C) a position
   (D) an objective criterion

13. Two team members disagree mildly about a variable-naming convention. The release is tomorrow. Best mode?
   (A) Collaborate through a two-hour design workshop
   (B) Escalate the decision to the project sponsor
   (C) Force a company-wide naming standard immediately
   (D) Accommodate or defer, given the low stakes

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · LEAD · LESSON 4 · 30 MIN

Decisions, Roles & Communication Planning

Make decisions visible, roles unambiguous and communication deliberate.

- Choose a decision method (consensus, consent, fist-of-five, delegated) to suit the stakes.
- Build RACI and DACI charts that avoid common errors.
- Plan communication using channel counts, push/pull/interactive methods and status reports that work.

Many project delays are really **decision delays**: nobody knows who decides, so nobody does. Pair clear decision rights with deliberate communication and you remove a huge source of waste.

Choosing a decision method

Group methods

- **Consensus** – everyone can live with and support it; slow but strong buy-in
- **Consent** – proceed unless someone has a reasoned objection; faster than consensus
- **Majority vote** – quick, but the minority may disengage
- **Fist-of-five** – quick read of support levels

Individual methods

- **Decide with input** – one decider consults, then decides; good default for most work
- **Delegated** – the person closest to the work decides within boundaries
- **Command** – for emergencies when speed matters most

Fist-of-five

FINGERS	MEANING	ACTION
0 (fist)	I object — I cannot support this	Stop; hear the concern
1	Serious concerns	Discuss before proceeding
2	Some concerns I would like to discuss	Discuss; adjust if needed
3	I can live with it	Proceed
4	I support it	Proceed
5	I strongly support it and will champion it	Proceed

RACI: one chart, four roles

R
Responsible
Does the work. Can be several people.

A
Accountable
Owns the outcome and approves. **Exactly one per activity.**

C
Consulted
Gives input before the work or decision — two-way.

I
Informed
Kept up to date after the fact — one-way.

TEMPLATE **RACI matrix**

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Clarify who does what for key deliverables and decisions; check every row has exactly one A.

ACTIVITY / DECISION	SPONSOR	PROJECT MANAGER	PRODUCT OWNER	TECH LEAD	OPS MANAGER
Approve go-live	*A*	*R*	*C*	*C*	*C*
Prioritise backlog	*I*	*C*	*A/R*	*C*	*I*

WATCH OUT · RACI SMELLS

Too many A's on one row (nobody really owns it); no A at all; a person with R everywhere (bottleneck); a column of only I's (why are they on the chart?); and too many C's, which slows every decision.

DEFINITION · DACI FOR DECISIONS

For key decisions, many teams use **DACI**: a **Driver** who runs the decision process, one **Approver** who makes the call, **Contributors** who provide expertise and **Informed** people who hear the outcome. It makes clear that the person organising the decision is not necessarily the one making it.

Communication channels

MEASURE	FORMULA	WHAT IT TELLS YOU
Number of channels	`n × (n − 1) ÷ 2`	n is the number of people. 10 people → 45 channels; 15 people → 105.
Channels added	`new total − old total`	Growing a team from 10 to 15 adds 105 − 45 = 60 channels — complexity grows much faster than headcount.

Push, pull and interactive

Push

- Sent to specific people
- Email, status report, notification
- Good for: updates, formal notices
- Risk: sent ≠ read

Pull

- People access when they need it
- Wiki, dashboard, shared repository
- Good for: large audiences, reference info
- Risk: nobody looks

Interactive

- Real-time, two-way exchange
- Meetings, calls, workshops, chat
- Good for: decisions, conflict, complex or sensitive topics
- Risk: time-consuming

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TEMPLATE

Communication plan

Decide deliberately who needs what, when, how and from whom.

AUDIENCE	INFORMATION NEEDED	METHOD (PUSH/PULL/INTERACTIVE)	FREQUENCY	OWNER	FEEDBACK LOOP
Steering committee	*Status, forecast, decisions needed, top risks*	*Interactive meeting + pushed one-pager*	*Monthly*	*PM*	*Decisions log reviewed next meeting*
Branch staff (users)	*What changes, when, training dates*	*Push (SMS/email) + pull (FAQ page)*	*Fortnightly, weekly near go-live*	*Change lead*	*Pulse survey*

A status report people actually read

1. **Bottom line up front** Status colour, trend and one-sentence summary.
2. **Decisions or help needed** What, from whom, by when.
3. **Progress against outcomes** Milestones and value delivered, not a list of meetings held.
4. **Forecast** Dates and cost as ranges, with change since last report.
5. **Top risks and issues** Three to five, with owners and actions.
6. **Keep it to one page** Link to detail for those who want it.

SCENARIO · DECIDE · THE IGNORED REPORT

Your weekly 6-page status email has a 20% open rate, and the sponsor was surprised by a delay you reported three weeks ago. What is the BEST next step?

A Send the report twice a week so it gets noticed.

B Mark the email as high-priority.

C Note in the lessons learned that the sponsor did not read the report.

D Ask the sponsor how they prefer to receive information, then switch to a one-page, decision-first summary plus a short monthly conversation for key items.

Circle your choice. Answer and feedback: module answer key.

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Key terms

RACI
A responsibility assignment matrix showing who is Responsible, Accountable, Consulted and Informed for each activity or decision.

Fist-of-five
A quick group technique in which people show zero to five fingers to signal their level of support for a proposal.

KNOWLEDGE CHECK

14. A team grows from 6 to 9 people. How many communication channels are added?

(A) 3
(B) 15
(C) 36
(D) 21

15. Which RACI problem is MOST serious?

(A) Two people marked Responsible on one task
(B) Several people marked Informed on a decision
(C) The sponsor marked Consulted on a deliverable
(D) Three people marked Accountable for one decision

16. A sensitive change to staff roles needs to be explained to affected employees. Which method fits best?

(A) Interactive — meet them and allow questions
(B) Pull — post a detailed explanation on the team wiki
(C) Push — send a mass email to all employees
(D) None — say nothing until the go-live date

17. In a fist-of-five vote, one member shows two fingers. What should happen?

(A) Pause to hear and discuss their concerns
(B) Proceed immediately, as most members agree
(C) Ignore it, because the majority view wins
(D) Remove the member from the decision

18. In DACI, who makes the final decision?

(A) Driver
(B) Contributor
(C) Approver
(D) Informed

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MODULE 6 · LEAD · LESSON 5 · 30 MIN

Remote, Hybrid & Cross-Cultural Teams; Coaching & Feedback

Lead people you rarely see, across cultures and time zones, and help them grow.

- Design working agreements for remote and hybrid teams, including async-first practices.
- Adapt communication across cultural dimensions without stereotyping.
- Use GROW coaching and SBI feedback in one-to-ones.

In 2026 most project teams are at least partly distributed — a developer in Khulna, a product owner in Dhaka, a vendor in Kuala Lumpur and a client in Frankfurt. Distance strips away the hallway conversations that used to carry context, so leaders must design what used to happen naturally.

Remote and hybrid essentials

WA

Working agreement

A short, team-written page: core hours, response-time expectations, which channel for what, camera norms, how decisions are recorded.

ASY

Async-first

Default to written updates, recorded demos and shared documents so people in different time zones can contribute; use meetings for what truly needs real time.

OVR

Overlap windows

Protect two to three hours of shared time for collaboration; rotate inconvenient meeting times fairly.

EQ

Hybrid equity

If one person is remote, everyone joins from their own screen so remote people are not second-class participants.

CON

Connection

Deliberate informal time — virtual coffee, show-and-tell, occasional in-person meet-ups at key moments such as kick-off.

Remote team working agreement checklist

☐ Core collaboration hours defined in every member’s time zone

☐ Which tool for urgent, normal and FYI messages is agreed

☐ Expected response times for each channel are written down

☐ Decisions are recorded in one agreed place

☐ Meeting norms: agenda in advance, notes and actions shared within 24 hours

☐ Public holidays and religious observances across locations are on the shared calendar

☐ Inconvenient meeting times rotate fairly

☐ A ‘how to reach me’ and focus-time preference is published by each member

☐ Security rules for devices, networks and shared files are clear

☐ The agreement is reviewed at least every quarter or when membership changes

Culture shapes how people give feedback, disagree, make decisions and view hierarchy and time. Research frameworks such as Geert Hofstede’s dimensions and Erin Meyer’s culture map describe national **tendencies** — useful as hypotheses, never as labels for individuals.

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Cultural dimensions that show up on projects

DIMENSION	ONE END	OTHER END	WHAT THE PM CAN DO
Power distance	Hierarchy expected; juniors defer	Flat; anyone challenges anyone	Invite input explicitly and privately; use anonymous tools
Communication context	High-context: meaning in what is unsaid	Low-context: meaning in explicit words	Confirm understanding in writing; ask open questions
Feedback style	Indirect, softened	Direct, blunt	Agree team feedback norms; explain intent
Time orientation	Flexible, relationships first	Linear, schedule first	Make deadline importance and reasons explicit
Uncertainty avoidance	Prefers rules and detailed plans	Comfortable with ambiguity	Balance clarity of process with room to experiment

WATCH OUT · GENERALISE CAREFULLY

A Bangladeshi engineer educated in Germany and working for a Singapore firm will not fit a national average. Use dimensions to notice and ask — 'How do you prefer to receive feedback?' — never to assume.

GROW coaching model

G

Goal

What do you want to achieve? 'By the end of this conversation, what would be useful?'

R

Reality

What is happening now? What have you tried? What is getting in the way?

O

Options

What could you do? What else? What would someone you admire do?

W

Will / way forward

What will you do, by when? How will you know it worked? What support do you need?

SBI feedback

MEASURE	FORMULA	WHAT IT TELLS YOU
Situation	`When and where`	'In yesterday's client demo…'
Behaviour	`Observable action — not a judgement`	'…you answered the integration question with specific test data…'
Impact	`Effect on you, the team or the outcome`	'…and the client agreed to sign off the milestone on the spot.' Works for praise and for correction.

PRO TIP · COACH BY ASKING, NOT TELLING

A good coaching ratio is mostly questions. The answer someone finds themselves sticks better than the one you hand them — and it builds their capability for the next problem.

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SCENARIO · DECIDE · FEEDBACK ACROSS CULTURES

A talented analyst from a high-power-distance background never challenges the client's architect, even when the analyst's data shows the architect is wrong. What is the BEST approach?

(A) Tell the analyst in the next team meeting that they must speak up more.

(B) Present the analyst's findings yourself without crediting them.

(C) Speak privately, use SBI to recognise the strong analysis, explore what makes challenging difficult (GROW), and agree a supported way — e.g. you introduce the data and invite them to present it.

(D) Accept that this is their culture and do nothing.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of your current or most recent team. Which cultural or remote-working difference has caused the most friction? Write one clause you would add to your team's working agreement to address it.

Key terms

Working agreement
A team-created set of norms describing how members will collaborate, communicate and make decisions.

SBI feedback
A feedback structure describing the Situation, the observable Behaviour and its Impact.

GROW model
A coaching structure: Goal, Reality, Options, Will (way forward).

KNOWLEDGE CHECK

19. A hybrid team holds meetings in a room with four people while two join remotely and struggle to follow. What is the BEST fix?

(A) Have everyone join from their own device for equal participation

(B) Ask remote members to come into the office for meetings

(C) Record the meetings so remote members can catch up afterwards

(D) Hold separate meetings for the in-room and remote members

20. In SBI, which is a behaviour rather than a judgement?

(A) You were unprofessional in front of the client

(B) You do not care about the project or the client

(C) You have a negative attitude with clients

(D) You arrived 20 minutes after the client call started

21. Which GROW question belongs to the 'Reality' stage?

(A) What do you want to achieve?

(B) What will you do by Friday?

(C) What other options are there?

(D) What have you already tried?

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22. Why is ‘async-first’ useful for teams spread over several time zones?

(A) It removes the need for any live meetings across time zones

(B) It is cheaper than paying for video-conferencing tools

(C) It lets people contribute without being online at the same time

(D) It hides time-zone differences from managers, clients and partners

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 6 · LEAD

Answer key

Knowledge checks

1. D High competence and high commitment call for delegating: agree outcomes and step back.

2. B Friction over ideas and ways of working is typical of storming.

3. B Rewarding candour signals it is safe to speak up.

4. B Servant leaders still lead; they direct their influence toward enabling the team.

5. D Pay is a hygiene factor; its absence causes dissatisfaction.

6. B Autonomy is about choice in how work is done within agreed outcomes.

7. C Recognising one's own emotional triggers is self-awareness.

8. D Intrinsic drivers sustain quality in complex work; heavy extrinsic incentives can backfire.

9. C Safety, security and legal non-negotiables justify a firm, directive stance.

10. A Collaboration searches for solutions satisfying both parties' interests fully.

11. B A weaker alternative makes walking away more costly, reducing leverage.

12. C It is a stated demand. The interest is the reason behind it.

13. D Low-stakes issues under time pressure do not justify heavy collaboration; revisit later if needed.

14. D 6 people: 6×5÷2 = 15. 9 people: 9×8÷2 = 36. Added: 36 − 15 = 21.

15. D Each activity needs exactly one Accountable person; multiple A's blur ownership.

16. A Sensitive and complex messages need two-way, real-time communication.

17. A Two or fewer signals concerns worth discussing before proceeding.

18. C The Approver makes the call; the Driver runs the process.

19. A Equal participation through individual devices removes the in-room advantage.

20. D Behaviour is observable and specific; the others are interpretations.

21. D Reality explores the current situation and what has been tried.

22. C Async communication respects time zones while keeping meetings for collaboration that truly needs real time.

Scenarios & case studies

> **Scenario · The quiet expert**
> **D** is best. Correct. Remove the impediment, then build safety one-to-one. The goal is to learn what got in the way, not to blame.

> **Scenario · The angry email**
> **A** is best. Correct. Self-regulation first, then empathy and a fact-based plan — ideally in a conversation, not an email war.

> **Case study · University LMS migration: motivation slump**
> Hygiene: the unchanged teaching load (workload and fairness) — a structural issue the PM should escalate to the deans and the sponsor, asking for workload relief or a revised timeline. Motivators: lack of purpose ('nobody will notice') and no mastery. Actions: (1) **Purpose** — share student feedback and data from pilot courses; invite students to demo days. (2) **Autonomy** — let designers choose which redesign patterns to use within a quality rubric they help write. (3) **Mastery** — run short workshops on active-learning design and showcase excellent courses with credit to their designers. Recognition by the Pro-VC at a faculty meeting costs nothing and signals that the work matters.

> **Case study · 5G rollout: civil works vs radio planning**
> Interests — radio planning: meet coverage and quality targets, avoid poor customer experience; civil works: protect margin, avoid unpaid rework, keep crews productive. Shared: hit the regulator milestone. Aim to **collaborate**: the stakes are high and both groups' commitment is needed. Steps: (1) meet each lead privately to hear concerns; (2) joint session with data — which of the 40 sites truly need changes for coverage, and which are nice-to-have? (3) generate options: change only sites critical to the milestone now and batch others later; use objective criteria (contracted change rates) for payment; re-sequence crews to absorb some work; (4) put the agreed approach through change control; (5) agree a weekly sync between both leads. If no agreement is possible before a hard deadline, escalate with options to the sponsor.

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Scenario · Two seniors, one tester
A is best. Correct. Understand interests first; often a split schedule, a substitute or a re-sequence solves it. Escalate with options only if they cannot agree.

Scenario · The ignored report
D is best. Correct. Tailor the method to the audience and use interactive communication for important news.

Scenario · Feedback across cultures
C is best. Correct. Private, respectful coaching builds confidence and gives a safe path to contribute.

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07

MODULE 7 · SCALE

Scale — AI-Powered Project Management, Change & Governance

Multiply your impact with AI and automation, lead adoption and connect projects to strategy responsibly.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Use generative AI copilots for project tasks with strong prompts, verification and data-privacy discipline.
- Apply automation, leading indicators and basic predictive analytics, and manage the distinctive risks of AI projects.
- Lead organisational change using ADKAR and Kotter and respond constructively to resistance.
- Explain PMO types, portfolio and programme governance, prioritisation and stage gates.
- Integrate ESG and sustainability considerations into project decisions.

LESSONS

1. Generative AI Copilots in Project Management · 35 min
2. Automation, Data-Driven PM & AI Project Risk · 30 min
3. Leading Organisational Change · 30 min
4. Governance: PMOs, Portfolios, Programmes & Sustainability · 30 min

MODULE 7 · SCALE · LESSON 1 · 35 MIN

Generative AI Copilots in Project Management

Use an AI assistant to draft, summarise and explore — while you stay accountable.

- Identify project tasks where a generative AI copilot adds value and where it does not.
- Write effective prompts using the ZPMP CRAFT pattern.
- Verify AI output and protect confidential and personal data.

Scale — stage 7 of the ZPMP Value Loop — is about multiplying your impact beyond a single project: through technology, through people adopting change, and through governance. Generative AI copilots are now part of everyday project work. Used well, they remove hours of drafting and summarising. Used carelessly, they create confident-sounding errors and data leaks. The project manager remains **accountable for every word and number that leaves the project**, whoever — or whatever — drafted it.

Where an AI assistant helps (and where to be careful)

PM TASK	GOOD USE	WATCH OUT FOR
Charter and plan drafts	First draft of objectives, scope statements, assumptions from your notes	Invented constraints or targets; generic text that fits no real project
Risk brainstorming	Generate candidate risks by RBS category to widen thinking	Plausible but irrelevant risks; missing local context
Status summaries	Turn raw updates and tracker exports into a one-page draft	Softening bad news; mis-stated numbers
Meeting notes	Summarise transcripts into decisions, actions and owners	Wrong attribution; confidential remarks captured; consent to record
Schedule what-ifs	Explain trade-offs and suggest scenarios to test	Arithmetic and date errors — run the real numbers in your scheduling tool
Stakeholder messages	Adapt one message for different audiences and reading levels	Tone that does not fit the relationship; over-promising

The ZPMP CRAFT prompt

C

Context

Give the facts the assistant cannot know: project goal, audience, stage, constraints, relevant data (sanitised).

R

Role

Say which perspective to take: ‘Act as a sceptical risk reviewer’ or ‘as a procurement specialist’.

A

Ask

State one clear task with a verb: draft, critique, summarise, compare, list.

F

Format

Specify structure and length: ‘a table with columns…’, ‘under 150 words’, ‘bullet points for a sponsor’.

T

Tests

Add checks: ‘list your assumptions’, ‘flag anything you are unsure of’, ‘do not invent figures — say if data is missing’.

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EXAMPLE · A CRAFT PROMPT FOR RISK BRAINSTORMING

Context: We are replacing the billing system for a water utility serving 400,000 customers in Khulna; go-live is in 5 months; data migration from a 15-year-old database. **Role:** Act as an experienced risk manager. **Ask:** Suggest 15 candidate risks grouped by technical, external, organisational and people/change. **Format:** A table with cause–event–effect statement and the category. **Tests:** Mark which risks are generic versus specific to our context, and list questions we should ask to validate them.

Prompting patterns that work

1. **Iterate, don't one-shot** Treat the first answer as a draft; refine with follow-ups: 'make it shorter', 'more specific to utilities'.
2. **Ask it to critique** Paste your own draft and ask: 'What is unclear, missing or risky in this charter?'
3. **Give examples** Show one example of the style or format you want; quality jumps.
4. **Split big tasks** Outline first, then each section — easier to check and steer.
5. **Ask for alternatives** 'Give three different ways to structure this steering committee update.'

WATCH OUT · HALLUCINATION IS A FEATURE OF THE TECHNOLOGY, NOT A RARE BUG

Generative models predict plausible text. They can invent references, regulations, statistics and even quotes from your own meeting. **Never** publish AI-generated facts, figures or legal statements without checking them against a reliable source.

Verify before you use

☐ Every number has been recalculated or traced to the source data
☐ Names, dates, roles and decisions match the actual records
☐ Any law, standard or policy cited has been checked in the original document
☐ Tone is right for the audience and does not soften bad news
☐ Nothing confidential or personal has been included that should not be
☐ Output reflects our project, not a generic template
☐ A human with the right knowledge has reviewed it before it is shared
☐ AI use is disclosed where policy or the audience expects it

Data-privacy rules for AI use

APP

Approved tools only

Use the AI tools your organisation has approved under an enterprise agreement. Consumer accounts may retain or use your inputs in ways your contracts forbid.

CLS

Classify before you paste

Public and internal data may be fine; confidential, client, personal or regulated data usually is not — unless the tool is explicitly approved for that class.

MIN

Minimise and anonymise

Remove names, phone numbers, account numbers and contract values; use placeholders such as 'Vendor A'.

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SEC

No secrets ever

Never paste passwords, access keys or security configurations.

REC

Consent for recording

Tell participants before an AI note-taker records or transcribes a meeting, and follow local law and company policy.

TEMPLATE

AI-use log

Keep a lightweight record of significant AI use on the project for transparency, quality and audit.

DATE	TASK	TOOL (APPROVED?)	DATA SHARED (CLASSIFICATION)	VERIFICATION DONE	REVIEWER	WHERE THE OUTPUT WAS USED
12 Mar	*Draft risk list for billing migration*	*Enterprise AI assistant (yes)*	*Internal; anonymised*	*Checked against RBS and team workshop; 6 of 15 kept*	*Risk lead*	*Risk register v0.3*

SCENARIO · DECIDE · THE HELPFUL PASTE

A team member pastes the full vendor contract, including pricing and named client contacts, into a free public chatbot to get a summary of penalty clauses. What should the project manager do FIRST?

(A) Thank them — the summary will save time.

(B) Stop further use, report the incident through the organisation's data or security process, and then guide the team on approved tools and anonymisation.

(C) Delete the chat history and say nothing.

(D) Ban all AI tools for the team permanently.

Circle your choice. Answer and feedback: module answer key.

Key terms

Generative AI copilot

An AI assistant that generates text, summaries, code or other content from natural-language prompts to support a person's work.

Hallucination

Output from a generative model that is fluent and plausible but factually wrong or invented.

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Data minimisation

Sharing only the data strictly needed for a task, with personal or confidential details removed or masked.

KNOWLEDGE CHECK

1. In the CRAFT pattern, 'Flag anything you are unsure of and do not invent figures' belongs to...
   - (A) Context
   - (B) Role
   - (C) Tests
   - (D) Format
2. An AI assistant produces a schedule what-if showing the project finishing three weeks earlier if two tasks overlap. What should the PM do NEXT?
   - (A) Send the new, earlier date to the sponsor straight away
   - (B) Ask the AI assistant to confirm that its result is correct
   - (C) Ignore it, since AI schedule outputs are never reliable enough to use
   - (D) Model it in the schedule and check resources and dependencies
3. Which data is generally SAFEST to share with an approved AI assistant?
   - (A) Anonymised meeting notes with internal-only information
   - (B) A customer list with names and phone numbers
   - (C) Administrator passwords for the test environment
   - (D) A signed contract with pricing, in a public chatbot
4. Who is accountable for a status report drafted by an AI assistant and sent by the PM?
   - (A) The AI tool provider
   - (B) The IT department
   - (C) The project manager
   - (D) Nobody, as it was automated
5. Before an AI note-taker records a client workshop, the PM should...
   - (A) record silently to avoid disrupting the discussion
   - (B) only record the client's comments, not the team's
   - (C) inform participants and follow consent rules and policy
   - (D) turn off transcription but keep recording audio

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · SCALE · LESSON 2 · 30 MIN

Automation, Data-Driven PM & AI Project Risk

Automate the routine, decide with data, and manage the special risks of AI work.

- Design automation rules and integrations that keep one source of truth.
- Use leading and lagging indicators and explain basic predictive analytics for projects.
- Identify the distinctive risks and ethical duties of AI projects and AI use in PM.

Modern work-management platforms can do much of the administrative heavy lifting: moving cards, chasing updates, refreshing dashboards. The aim is to free people for judgement and conversation — not to build a fragile web of automations nobody understands.

Anatomy of an automation rule

01 Trigger
Something happens: status changes, a date arrives, a form is submitted

→

02 Condition
Only if… e.g. priority is High and the item is in the current release

→

03 Action
Do something: assign, notify, create a task, update a field

→

04 Log
Record that it ran so people can trace what happened

High-value automations for PM tools

AUTOMATION	EXAMPLE	BENEFIT
Intake routing	New request form → creates backlog item, tags the product owner	No requests lost in email
Overdue nudges	Task 2 days past due → reminder to assignee; 5 days → flag on dashboard	Fewer manual chases
Status roll-up	Milestone dates and RAG pulled automatically into the portfolio dashboard	One source of truth, fresher data
Definition-of-done gates	Item cannot move to Done until test and review fields are complete	Built-in quality
Integration sync	Code repository or service desk events update linked work items	Less double entry
Recurring work	Monthly risk review task created with the register attached	Habits that do not depend on memory

WATCH OUT · AUTOMATION SPRAWL

Dozens of overlapping rules created by different people lead to notification overload and mysterious changes. Keep an inventory of automations with an owner and purpose, and review it each quarter. If people start ignoring notifications, the automation is costing attention instead of saving it.

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Leading vs lagging indicators

Leading (predict)

- Requirements churn rate
- Unresolved blockers older than 3 days
- Test coverage of new code
- Training completion before go-live
- Sponsor meeting attendance

Lagging (confirm)

- Final cost vs budget
- Delivery date vs baseline
- Defects found in production
- Adoption rate after 90 days
- Customer satisfaction score

Predictive analytics uses historical data to estimate what is likely to happen. Organisations with records of past projects can train simple models — often no more than regression or classification — to flag projects likely to overrun based on features such as scope churn, number of external dependencies, team turnover and early SPI. **Reference-class forecasting** (associated with Bent Flyvbjerg) applies a similar idea without complex models: compare your estimate with the actual outcomes of a class of similar past projects and adjust for typical overruns.

INSIGHT · GARBAGE IN, CONFIDENT GARBAGE OUT

Predictions are only as good as the data behind them. If time-sheets are filled in on Friday from memory and status fields are rarely updated, a model will learn the noise. Invest in data hygiene before investing in dashboards and models.

Distinctive risks of AI projects

DAT

Data

Not enough, poor quality, unrepresentative or legally restricted data — often discovered late.

PRF

Performance uncertainty

You cannot promise accuracy upfront; plan experiments, success thresholds and stop/go decisions.

BIA

Bias & fairness

Models can reproduce historic discrimination, e.g. in credit scoring or hiring.

DRF

Drift

Real-world data changes over time and accuracy decays; monitoring and retraining must be budgeted.

EXP

Explainability

Regulators and users may need to know why a decision was made.

SEC

Security & misuse

Prompt injection, data leakage and misuse of generated content in gen-AI products.

REG

Regulation

Risk-based laws such as the EU AI Act and sector rules impose duties depending on use; check obligations in every market served.

DEFINITION · USEFUL REFERENCES FOR AI GOVERNANCE

Organisations increasingly align AI work with frameworks such as the **NIST AI Risk Management Framework** and the management-system standard **ISO/IEC 42001**. A PM does not need to be an expert, but should ask early: ‘Which AI policy and risk process applies to this project?’

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Ethical duties when the PM uses AI

- Be transparent with stakeholders about significant AI-generated content
- Do not use AI tools to covertly monitor or score individual team members
- Keep humans in the loop for decisions that affect people's jobs, pay or rights
- Check outputs for bias, especially in people-related content such as role descriptions
- Respect intellectual property and confidentiality in what you input and publish
- Remain accountable — 'the AI said so' is never a defence

SCENARIO · DECIDE · THE PRODUCTIVITY SCORE

An executive suggests using an AI add-on to score each developer's productivity from keystrokes and messages, and to share rankings in the monthly review. What is the BEST response from the PM?

(A) Implement it — data-driven management is modern practice.

(B) Raise the ethical, legal and trust concerns, suggest team-level flow metrics and outcome measures instead, and involve HR and legal before any monitoring decision.

(C) Agree but keep the rankings secret.

(D) Refuse and say nothing more.

Circle your choice. Answer and feedback: module answer key.

Key terms

Leading indicator
A measure that changes before an outcome and helps predict it, allowing early action.

Reference-class forecasting
Estimating by comparing a project with the actual outcomes of a group of similar past projects.

Model drift
The decline in an AI model's performance over time as real-world data diverges from its training data.

KNOWLEDGE CHECK

6. Which is a leading indicator for a software release?

(A) Defects found in production after release
(B) Final project cost compared with the approved budget
(C) Customer satisfaction scores after launch
(D) Number of blockers open longer than three days

7. Your team ignores most tool notifications. What is the BEST action?

(A) Add more automated reminders and escalations so nothing is missed
(B) Turn off all automation and go back to manual updates
(C) Review the automation inventory and prune low-value rules
(D) Escalate the team's behaviour to their line managers

8. An AI model for loan approvals performed well at launch but accuracy falls after a year. This is most likely...

(A) scope creep
(B) a hallucination
(C) a lagging indicator
(D) model drift

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9. Why should an AI project plan include stop/go decision points after experiments?

(A) Because achievable performance is uncertain until tested

(B) Because most AI projects fail and need early exits

(C) To satisfy the PMO's standard gate review requirements

(D) To avoid documenting detailed requirements

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MODULE 7 · SCALE · LESSON 3 · 30 MIN

Leading Organisational Change

Deliverables are not benefits until people change how they work.

- Apply ADKAR to diagnose where individuals are stuck in a change.
- Use Kotter's eight steps to plan an organisational change effort.
- Respond to resistance with empathy and targeted actions.

A new system can go live perfectly and still fail if nobody uses it properly. **Organisational change management (OCM)** is the people side of delivering value: helping individuals and groups move from the current way of working to the new one. Project management delivers the solution; change management makes sure it is **adopted**.

ADKAR: the individual journey

A
Awareness
of why the change is needed. 'Why are we doing this, and why now?'

D
Desire
to support and take part. 'What is in it for me — and what happens if we do not?'

K
Knowledge
of how to change. Training, guides, examples of the new process.

Ab
Ability
to actually perform the new way. Practice, coaching, time, the right tools.

R
Reinforcement
to make it stick. Recognition, measurement, removing the old way.

PRO TIP · FIND THE FIRST BARRIER

ADKAR is sequential. If a branch manager lacks **desire**, more training (knowledge) will not help. Survey or interview, find the earliest element that is weak, and fix that first.

ADKAR focuses on individuals. **John Kotter's** eight-step model focuses on leading change across an organisation. In our own words, the steps are:

Kotter's eight steps, summarised

1. **Create urgency** Help people see, with evidence, why the status quo is risky.
2. **Build a guiding coalition** Assemble influential, credible leaders from across the organisation.
3. **Form a vision and strategy** A simple picture of the future and how to get there.
4. **Communicate and enlist** Share the vision constantly through many channels; recruit volunteers.
5. **Remove barriers** Change structures, systems and processes that block the new way.
6. **Generate short-term wins** Plan visible, early successes and celebrate them.
7. **Sustain acceleration** Use credibility from wins to tackle bigger changes; do not declare victory early.
8. **Anchor in culture** Connect new behaviours to success; build them into hiring, promotion and routines.

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Why people resist — and what helps

ROOT OF RESISTANCE	WHAT YOU MIGHT HEAR	HELPFUL RESPONSE
Not understanding why	‘The old system works fine.’	Share data and stories on the problem; let them see it themselves
Loss of status, control or skill	‘I have been the expert for 12 years.’	Involve them as super-users or trainers; honour their expertise
Fear of job loss	‘Is this automation going to replace us?’	Be honest about impacts; explain redeployment and reskilling plans
Change fatigue	‘Another initiative?’	Sequence changes; stop something old; show quick wins
Past broken promises	‘Last time they said it would be easier.’	Under-promise; pilot first; let peers report results

INSIGHT · RESISTANCE IS INFORMATION

People who push back often know something the project does not — a broken workflow, a regulatory edge case, a peak season. Listen before you persuade.

CASE STUDY · DIGITAL ACCOUNT OPENING AT A COMMERCIAL BANK

A Bangladeshi commercial bank is introducing tablet-based digital account opening at 120 branches, replacing paper forms. The pilot at 10 branches is technically successful, but after six weeks only 30% of new accounts are opened digitally. Interviews show: many officers did not know why the change was happening beyond ‘head office says so’; senior officers worry that faster processing will be used to cut staff; several were trained once, two weeks before the tablets arrived; and branch targets still reward the number of accounts, not the channel used.

1. Diagnose the situation with ADKAR. Where is the first barrier?

2. What would you change before rolling out to the other 110 branches?

3. Which Kotter steps were skipped?

Model answer: module answer key.

TEMPLATE **Change impact assessment**

Identify who is affected, how much and what support they need before go-live.

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STAKEHOLDER GROUP	WHAT CHANGES FOR THEM	IMPACT (H/M/L)	ADKAR GAP	SUPPORT ACTIONS	CHANGE CHAMPION
Branch customer-service officers	*Tablet process replaces paper forms; new KYC checks*	*H*	*Desire (job-security fears)*	*Leadership briefing on reskilling; super-user buddies; scorecard update*	*Senior officer, Agrabad branch*

SCENARIO · DECIDE · THE VOCAL SCEPTIC

A respected senior supervisor openly criticises the new warehouse scanning process in front of her team, and adoption on her shift is lowest. What should the project manager do FIRST?

(A) Meet her privately to understand her concerns, then involve her in fixing them — possibly as a change champion.

(B) Report her to her manager for undermining the project.

(C) Exclude her shift from the rollout for now.

(D) Run another training session for her shift.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Recall a change you were asked to adopt at work. Using ADKAR, which element was weakest for you, and what would have helped most?

Key terms

Organisational change management (OCM)
Structured methods for helping people move from current to future ways of working so that project outputs are adopted and benefits realised.

ADKAR
A model of individual change: Awareness, Desire, Knowledge, Ability, Reinforcement.

KNOWLEDGE CHECK

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10. Users have been trained and understand the new process, but still prefer the old spreadsheet because their bonus depends on speed. Which ADKAR element needs attention?
    - (A) Awareness of the need for change
    - (B) Desire / Reinforcement
    - (C) Knowledge of the new process
    - (D) Ability to use the new tools
11. Which Kotter step is MOST concerned with celebrating visible early successes?
    - (A) Generate short-term wins
    - (B) Create a sense of urgency
    - (C) Anchor changes in the culture
    - (D) Build a guiding coalition
12. The BEST way to treat resistance is to...
    - (A) overcome it with stronger mandates from leadership
    - (B) ignore it until go-live, when it usually fades
    - (C) explore it as information and address root causes
    - (D) remove persistent resisters from the project team
13. Training delivered three months before people get the new system is a problem mainly for which ADKAR element?
    - (A) Awareness and Desire
    - (B) Desire and Reinforcement
    - (C) Knowledge and Ability
    - (D) Awareness and Reinforcement

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MODULE 7 · SCALE · LESSON 4 · 30 MIN

Governance: PMOs, Portfolios, Programmes & Sustainability

Connect projects to strategy, choose the right work and deliver it responsibly.

- Compare supportive, controlling and directive PMOs and explain how a PMO adds value.
- Distinguish project, programme and portfolio management and use weighted scoring and stage gates.
- Integrate environmental, social and governance (ESG) considerations into project decisions.

A single well-run project is good. An organisation that consistently picks the **right** projects and runs them well is far better. Governance is the framework of authority, decision rights and oversight that makes that consistency possible — without drowning teams in bureaucracy.

PMO types

TYPE	DEGREE OF CONTROL	WHAT IT DOES	BEST FIT
Supportive	Low	Templates, training, best practice, lessons-learned library; a consultative role	Organisations new to formal PM or with diverse, autonomous units
Controlling	Moderate	Requires compliance with methods, reporting and reviews; audits and assures	Organisations needing consistency and visibility, e.g. regulated sectors
Directive	High	Directly manages projects; PMs report to the PMO	Strategic or high-risk portfolios needing central control

INSIGHT · PMO VALUE IS MEASURED BY OUTCOMES, NOT TEMPLATES

A PMO earns its place when it improves decisions and delivery: better prioritisation, fewer failed projects, faster benefit realisation, shared resource planning. Many modern PMOs reposition as a **value management office** that supports agile teams with lightweight guardrails, coaching and portfolio flow.

Project, programme, portfolio

Project

- Temporary effort to create a specific output
- Success: deliver agreed scope, time, cost, quality and value
- Managed by a project manager

Programme

- Group of related projects and activities managed together
- Success: realise benefits that individual projects could not alone
- Managed by a programme manager; focuses on interdependencies and benefits

Portfolio

- All projects, programmes and operations chosen to meet strategy
- Success: best overall value for the organisation's goals and capacity
- Managed by portfolio leadership; selects, prioritises and balances

Weighted scoring for prioritisation

MEASURE	FORMULA	WHAT IT TELLS YOU
Weighted score	`Σ (criterion weight × score)`	Weights reflect strategy and must sum to 100%. Scores often use 1–5 scales.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Example	`Strategic fit 40% × 5 + ROI 30% × 3 + Risk (inverse) 20% × 4 + ESG 10% × 2 = 3.9`	Compare candidate projects on the same criteria; use results to inform, not replace, the discussion.

EXAMPLE · COST OF DELAY AS A LENS

In agile portfolios many organisations rank work by **cost of delay divided by duration** (often called WSJF in scaled frameworks), so small, urgent, valuable items move first. Whichever method you use, make the criteria explicit and visible.

Stage-gate governance

01
Idea / mandate
Is the problem worth exploring?

→

02
Gate 1: business case
Is it viable and aligned? Fund discovery

→

03
Gate 2: plan approved
Baseline, risks, funding for delivery

→

04
Gate 3: ready to launch
Quality, readiness, change plan

→

05
Gate 4: closure & benefits
Accept, hand over, track benefits

DEFINITION · TOLERANCES KEEP GOVERNANCE LIGHT

Boards set **tolerances** — e.g. ±10% cost, ±3 weeks — within which the PM manages without escalation. Only forecasts beyond tolerance come back to the board. This is management by exception: senior time goes where it is needed.

ESG in project decisions

E
Environmental
Energy and carbon over the life cycle, materials, waste, water and pollution, climate resilience (e.g. flood-proofing sites).

S
Social
Labour standards in the supply chain, health and safety, community impact, accessibility and inclusion of users.

G
Governance
Anti-corruption, transparency in procurement, data protection, ethical AI, honest reporting.

Practical ways to build sustainability into a project

- Add a sustainability objective or key result to the charter where relevant
- Include carbon and lifecycle cost — not just purchase price — in option appraisal
- Use supplier criteria for labour standards and environmental performance
- Choose energy-efficient hosting and right-size infrastructure
- Plan for reuse, recycling or safe disposal of equipment and materials
- Report ESG measures alongside cost and schedule at stage gates

CASE STUDY · SOLAR IRRIGATION PUMPS IN NORTHERN BANGLADESH

An NGO-funded programme will install 300 solar irrigation pumps across Rangpur and Dinajpur over three years, run as three projects: procurement and installation, a farmer payment-and-service scheme, and a maintenance

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training project. The board has limited funds and two other proposals: rooftop solar for clinics, and a mobile weather-advisory service.

1. Why is this better run as a programme than three separate projects?

2. Suggest four weighted criteria the portfolio board could use to compare the three proposals.

3. Name two environmental and two social risks the programme should manage.

Model answer: module answer key.

SCENARIO · DECIDE · GATE REVIEW PRESSURE

At Gate 3, the readiness review shows user training is only 40% complete and a critical security test is outstanding. The sponsor wants to pass the gate anyway to meet a public announcement. What should the project manager do?

(A) Pass the gate; announcements matter more than training.

(B) Cancel the gate meeting until everything is ready.

(C) Mark training as complete to keep the gate green.

(D) Present the readiness facts and risks to the gate board with options — e.g. delay, limited pilot launch, or conditional pass with dated actions — and let the authorised body decide.

Circle your choice. Answer and feedback: module answer key.

Key terms

PMO
Project (or portfolio/programme) management office: a function that standardises governance, supports or directs projects and improves delivery capability.

ESG
Environmental, social and governance considerations used to assess the wider impact and responsibility of projects and organisations.

Stage gate
A decision point between phases where governance reviews progress and decides whether to continue, change or stop.

KNOWLEDGE CHECK

14. A PMO provides templates and coaching but does not require their use. Which type is it?

(A) Directive
(B) Controlling
(C) Portfolio
(D) Supportive

15. Which is the main focus of programme management?

(A) Selecting and ranking projects against strategy
(B) Managing the scope, time and cost of one deliverable
(C) Auditing projects for compliance with standards
(D) Coordinating related projects for combined benefits

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16. A project forecast is 4% over budget; the board's tolerance is ±10%. What should the PM do?

(A) Escalate to the board immediately for a decision
(B) Pause non-urgent work until the board has reviewed it
(C) Manage within the project and report normally
(D) Request a new cost baseline from the board

17. Criteria weights are 50%, 30% and 20%. Project X scores 4, 2 and 5. Its weighted score is…

(A) 3.6
(B) 3.8
(C) 4.0
(D) 3.2

18. Comparing options by lifecycle energy cost rather than purchase price alone is an example of…

(A) social sustainability and community benefit
(B) environmental and financial sustainability
(C) gold plating of the specification
(D) scope creep driven by stakeholders

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MODULE 7 · SCALE

Answer key

Knowledge checks

1. C Tests are checks and constraints that improve reliability.

2. D AI suggestions are hypotheses; verify them in the authoritative schedule.

3. A Anonymised internal data in an approved tool is the low-risk option.

4. C Accountability stays with the human who uses and sends the output.

5. C Recording and transcription require transparency and, often, consent.

6. D Ageing blockers signal future delay before it happens.

7. C Notification overload means automations cost attention; prune them.

8. D Changing real-world data degrades model performance over time.

9. A Experiments reveal whether the needed accuracy is achievable; checkpoints prevent sunk-cost escalation.

10. B They know how but lack motivation, and the reward system reinforces the old way.

11. A Short-term wins build credibility and momentum.

12. C Resistance often reveals real issues; understanding it leads to better solutions and adoption.

13. C Knowledge fades without use; just-in-time training plus practice builds ability.

14. D Supportive PMOs provide resources with low control.

15. D Programmes manage interdependencies to deliver benefits not available separately.

16. C Within tolerance, the PM manages; management by exception keeps the board for larger deviations.

17. A 0.5×4 + 0.3×2 + 0.2×5 = 2.0 + 0.6 + 1.0 = 3.6.

18. B Lifecycle costing captures operating energy and environmental impact.

Scenarios & case studies

> **Scenario · The helpful paste**
> **B** is best. Correct. Contain and report first — security teams may need to act quickly — then fix the behaviour through clear guidance, not blame.

> **Scenario · The productivity score**
> **B** is best. Correct. Redirect toward measures that improve the system, and ensure proper governance for anything touching individuals.

> **Case study · Digital account opening at a commercial bank**
> ADKAR: **Awareness** is weak (no compelling why), **Desire** is undermined by job-loss fear and by targets that do not reward the change, **Knowledge** decayed because training came too early, **Ability** lacked practice time, **Reinforcement** is absent. The first barrier is awareness/desire, so more training alone will fail. Before rollout: leaders explain the customer and compliance reasons (faster opening, fewer errors, better KYC data); HR and leadership give an honest message on staffing and reskilling; train just-in-time with practice sessions and branch super-users; add digital-channel share to branch scorecards; publish pilot-branch wins. Kotter steps skipped: urgency, guiding coalition beyond head office, communicating the vision, removing barriers (targets) and short-term wins.

> **Scenario · The vocal sceptic**
> **A** is best. Correct. Her influence is an asset; if her concerns are addressed, she can bring the team along.

> **Case study · Solar irrigation pumps in northern Bangladesh**
> Programme: the benefit — higher, sustainable farmer incomes and lower diesel use — only appears when pumps, payment scheme and maintenance skills work together; a programme manages those dependencies and benefits. Criteria (example): impact on income/livelihoods 35%, emissions reduction 25%, cost per beneficiary 25%, delivery risk 15%. Environmental: groundwater over-extraction when water becomes cheap; end-of-life disposal of panels and batteries. Social: unequal access (larger landowners capturing benefits), and safety/theft issues at remote sites — plus fair labour practices among installers.

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Scenario · Gate review pressure

D is best. Correct. Governance works when decision makers get the facts, risks and options. The PM informs and recommends; the gate body decides.

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08

MODULE 8 · CLOSE

Close — Closing, Benefits, Ethics & Certification Readiness

Finish well, secure the value, act with integrity and prepare for the ZPMP exam and capstone.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Close phases, projects and contracts with formal acceptance, handover and archiving.
- Run retrospectives and lessons-learned processes that change behaviour, and plan benefits realisation.
- Apply responsibility, respect, fairness and honesty to real dilemmas.
- Build a PM career portfolio and use a reliable strategy for situational-judgement questions.
- Integrate all eight ZPMP Value Loop stages on one project and plan the capstone.

LESSONS

1. Closing a Phase or Project · 25 min
2. Retrospectives, Lessons Learned & Benefits Realisation · 30 min
3. Professional Responsibility & Ethics · 30 min
4. Career Portfolio & Exam Strategy · 30 min
5. The Whole Loop: Integrative Walk-Through & Capstone · 40 min

MODULE 8 · CLOSE · LESSON 1 · 25 MIN

Closing a Phase or Project

Finish properly: accepted deliverables, clean handover, settled contracts and a usable archive.

- List the activities required to close a phase, project or contract.
- Plan a handover to operations that protects benefits after go-live.
- Close a project that is terminated early with the same discipline.

Close is the last stage of the ZPMP Value Loop, and it is the one most often rushed. People are pulled onto new work, the budget code is shut and documents scatter. Yet closure is where you secure acceptance, protect the benefits and capture knowledge for the next project. A project that never formally closes keeps consuming attention and money.

The ZPMP closing sequence

01 Confirm acceptance — Formal sign-off that deliverables meet the agreed criteria →

02 Transition — Hand over to operations, support and the business owner →

03 Close contracts — Verify, settle, pay, release →

04 Close finances & release people — Final costs, return resources, recognise contributions →

05 Capture learning & archive — Lessons, final report, records stored →

06 Celebrate — Mark the end; thank people publicly

DEFINITION · VERIFIED VS ACCEPTED

Quality control checks that a deliverable meets specifications (it is **verified/correct**). Acceptance is the customer or sponsor formally agreeing it meets acceptance criteria (it is **accepted**). Both are needed; only acceptance closes the scope.

Handover to operations

ELEMENT	WHAT 'DONE' LOOKS LIKE
Operational owner	A named business owner has accepted responsibility for the product and its benefits
Support model	Service desk, escalation paths and support hours agreed; tickets routed correctly
Hypercare	An agreed period (often 2–6 weeks) of extra project support after go-live, with exit criteria
Knowledge transfer	Runbooks, configuration records, known issues list, training for support staff
Warranties & licences	Vendor warranties, licence renewals and support contracts transferred to the operational owner
Measures	Benefit and performance KPIs set up and reporting to the business owner

Closing a contract

1. **Verify deliverables** Confirm everything contracted was delivered and accepted; check against the statement of work.

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2. **Resolve open items** Settle claims, disputes and change requests — negotiation first, escalation or alternative dispute resolution if needed.
3. **Final payment** Release retention money and final invoices according to the terms.
4. **Procurement audit** Review the procurement process for lessons: what worked in selection, contract type and management.
5. **Formal closure notice** Written notice to the seller that the contract is complete; update vendor performance records.

WATCH OUT · EARLY TERMINATION STILL NEEDS CLOSURE

If a project is cancelled — strategy changed, business case collapsed — close it with the same discipline: document why, what was completed, what can be reused, how contracts end (termination clauses), and what was learned. Stopping a failing project early is a **good governance outcome**, not a failure of the PM.

Project closure checklist

☐ Final deliverables formally accepted in writing by the authorised person
☐ Operational owner, support model and hypercare exit criteria agreed
☐ All contracts closed, claims settled and final payments made
☐ Open risks and issues closed or transferred to an owner in operations
☐ Final costs reconciled and project budget codes closed
☐ Team members released with feedback to their line managers
☐ Lessons learned captured with owners for follow-up actions
☐ Benefits realisation plan handed to the business owner
☐ Project records archived according to retention policy and searchable
☐ Final report issued and sponsor sign-off on project closure obtained
☐ Team and stakeholders thanked and successes celebrated

CASE STUDY · ERP GO-LIVE AT A GARMENT FACTORY IN ASHULIA

A knitwear exporter implemented an ERP system for planning, inventory and buyer compliance. Go-live happened in October. In December the PM was moved to a new project. In February: the vendor has not received final payment because nobody confirmed acceptance; 45 support tickets have no owner; the factory's IT staff do not know the vendor warranty expires in April; and buyers are asking for compliance reports that the system can produce but no one knows how to run.

1. Which closure activities were missed?
2. What should be done now, and by whom?
3. How would you prevent this on the next project?

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Model answer: module answer key.

SCENARIO · DECIDE · THE KEEN SPONSOR

The sponsor says: 'The system is live and users seem happy. Let's skip the formal acceptance meeting — just move the team to the next project tomorrow.' What should the project manager do?

(A) Agree; users are happy so acceptance is implied.

(B) Explain the risks of skipping formal acceptance and propose a short, focused sign-off session this week alongside a staged team release.

(C) Refuse to release anyone until all documentation is perfect.

(D) Send an email stating that silence will be treated as acceptance.

Circle your choice. Answer and feedback: module answer key.

Key terms

Formal acceptance
Documented confirmation by an authorised stakeholder that deliverables meet agreed acceptance criteria.

Contract closure
Completing all contract obligations, settling claims and payments and formally ending the contractual relationship.

Hypercare
A defined period of intensified support immediately after go-live, ending when agreed stability criteria are met.

KNOWLEDGE CHECK

1. A project is cancelled halfway because strategy changed. What should the PM do?

(A) Stop all work immediately and release the team without records

(B) Perform closure: document status, close contracts, capture lessons

(C) Continue until the original end date to protect the investment

(D) Hand all materials to the PMO without notes and move to new work

2. Which activity confirms that the customer agrees deliverables meet acceptance criteria?

(A) Formal acceptance

(B) Quality control

(C) Procurement audit

(D) Hypercare

3. Why should open risks be transferred at closure?

(A) So the risk register can be deleted at handover

(B) To justify increasing the final project budget

(C) It is not necessary once the project has closed

(D) Some risks continue into operations and need an owner

4. A procurement audit at closure mainly aims to...

(A) penalise the vendor for any missed obligations

(B) reopen negotiation on the final contract price

(C) identify procurement lessons for future use

(D) approve the final payment to the supplier

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 8 · CLOSE · LESSON 2 · 30 MIN

Retrospectives, Lessons Learned & Benefits Realisation

Turn experience into changed behaviour, and make sure the promised value actually arrives.

- Facilitate retrospectives that produce owned improvement actions.
- Maintain a lessons-learned register that leads to applied change.
- Build and hand over a benefits realisation plan with owners, baselines and measures.

Most organisations have folders full of lessons learned that nobody reads. The ZPMP view is simple: **a lesson is only learned when behaviour changes**. Until then it is merely a lesson observed.

Retrospective vs lessons-learned review

Retrospective

- Frequent — end of each iteration or phase
- Owned by the team, about how the team works
- Output: 1–3 improvement actions for the next cycle
- Learning applied immediately

Lessons-learned review

- At milestones and closure (ideally also during the project)
- Broader: process, governance, vendors, stakeholders
- Output: register entries and recommendations for the organisation
- Learning must travel to other projects

Retrospective formats

SSC

Start / Stop / Continue

Quick and clear; good for new teams.

4L

4Ls

Liked, Learned, Lacked, Longed for — balances positives and gaps.

SB

Sailboat

Wind (what pushes us), anchors (what holds us back), rocks (risks ahead), island (goal).

TL

Timeline

Plot events and the team's energy over time; great after long phases or incidents.

A retrospective that changes something

1. **Set the stage** Remind everyone of the shared assumption: people did their best with what they knew at the time. Check in briefly.
2. **Gather data** Use a format to collect facts and feelings silently first.
3. **Generate insights** Cluster and ask 'why?' to find root causes, not symptoms.
4. **Decide actions** Choose at most one to three actions, each with an owner and a date.
5. **Close and follow up** Put actions in the backlog; start the next retro by reviewing them.

TEMPLATE **Retrospective board**

Capture one retrospective's output so actions are visible and tracked.

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WHAT WENT WELL	WHAT DID NOT	ROOT CAUSE	ACTION	OWNER	DUE / REVIEW
Pairing on data migration scripts	*Late test data from the bank*	*No named contact in client data team*	*Agree a data contact and a weekly test-data drop*	*Rina*	*Next sprint review*

TEMPLATE **Lessons-learned register**

Record lessons so that they change templates, checklists and decisions on future projects.

DATE / PHASE	WHAT HAPPENED	IMPACT	LESSON (DO THIS NEXT TIME)	ACTION TO EMBED IT	OWNER	STATUS
Build	*Integration testing found 60% of defects in the last 2 weeks*	*3-week delay*	*Start integration tests from sprint 2 using service stubs*	*Add to PMO delivery checklist and test strategy template*	*QA lead*	*Done – template v2.1*

PRO TIP · FROM OBSERVED TO APPLIED

Every lesson needs an **embedding action**: change a template, a checklist, a contract clause, a training module or a gate criterion. And at every project kick-off, review the register for similar projects — pull, not just push.

Projects deliver **outputs** (a system, a building, a process). Users adopting those outputs create **outcomes** (faster processing, fewer errors). Outcomes produce **benefits** that the organisation values (lower cost, more revenue, better service, lower emissions). Most benefits arrive **after** the project closes — which is why benefit ownership sits with the business, not with the PM.

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From output to benefit

01 Output
Online booking system live

→

02 Capability
Patients can book and reschedule by phone or web

→

03 Outcome
Shorter queues, fewer no-shows

→

04 Benefit
More patients seen per doctor-day; higher satisfaction

→

05 Strategic objective
Accessible, efficient district healthcare

TEMPLATE Benefits realisation plan

Define how each benefit will be measured, by whom and when, and hand it to the business owner at closure.

BENEFIT	MEASURE (KPI)	BASELINE	TARGET & DATE	BENEFIT OWNER	HOW & WHEN MEASURED	DEPENDENCIES / DIS-BENEFITS
Fewer missed appointments	*No-show rate*	*22% (Jan–Mar)*	*12% by December*	*Hospital administrator*	*Monthly report from booking system*	*SMS reminders working; dis-benefit: front-desk retraining time*

WATCH OUT · DIS-BENEFITS AND BENEFIT LEAKAGE

Some outcomes are negative for someone — extra steps for a department, a role that disappears. Record these **dis-benefits** honestly. Also watch for benefit leakage: time saved that is simply absorbed rather than redeployed to valuable work.

SCENARIO · DECIDE · THE MISSING BENEFIT

Six months after go-live, the business owner says the expected 20% reduction in processing time has not appeared. You have moved to another project. What is the BEST action?

(A) Explain that benefits are not your responsibility after closure.

(B) Reopen the project and add new features.

(C) Help the business owner analyse the measures and adoption data to find the cause, and suggest corrective actions or a follow-up change.

(D) Lower the target to match current results.

Circle your choice. Answer and feedback: module answer key.

Key terms

Retrospective
A regular team meeting to inspect how the team worked and agree concrete improvements for the next cycle.

Benefits realisation
Measuring and managing the outcomes of a project after delivery to ensure intended benefits are achieved.

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Dis-benefit
A measurable negative consequence of a change, perceived as adverse by one or more stakeholders.

KNOWLEDGE CHECK

5. Which shows that a lesson has been truly learned?
   - (A) It is stored in the organisation's archive
   - (B) A practice has changed and is used on new projects
   - (C) It was discussed in detail at the closure meeting
   - (D) The sponsor has read the final report

6. Who should normally own a benefit after the project closes?
   - (A) The project manager
   - (B) A named business owner
   - (C) The PMO benefits team
   - (D) The supplier's account manager

7. A retrospective produces 14 improvement ideas. What is the BEST next step?
   - (A) Assign all 14 ideas to the PM to implement
   - (B) Email them all to the sponsor for a decision
   - (C) Wait until the next retrospective to discuss
   - (D) Pick one to three actions with owners and dates

8. Why must a benefit have a baseline?
   - (A) To make the benefits report more complete
   - (B) Without a starting value, change cannot be measured
   - (C) Because auditors require one for every benefit
   - (D) Because the baseline sets the project budget

9. 'Patients can book appointments online' is best described as...
   - (A) a benefit
   - (B) a strategic objective
   - (C) a capability/output
   - (D) a dis-benefit

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CLOSE · LESSON 3 · 30 MIN

Professional Responsibility & Ethics

Make the right call when nobody is watching — and when someone is pressuring you.

- Explain responsibility, respect, fairness and honesty as practical values.
- Work through ethical dilemmas using a structured decision approach.
- Recognise conflicts of interest, gifts, misreporting and corruption risks.

Project managers hold information, money and influence. Sponsors, clients and teams rely on what they report and how they decide. The professional community broadly agrees on four core values — **responsibility, respect, fairness and honesty** — and these are reflected, for example, in the code of ethics published by PMI. Here is what they mean in day-to-day project life, in ZPMP's own words.

Four values in action

RSP

Responsibility

Own your decisions and their consequences. Take only work you are competent to do. Keep commitments. Report errors — including your own — promptly and fix them.

RES

Respect

Treat people with dignity. Listen to other views. Honour other cultures and norms. Never abuse power or speak about others behind their backs to harm them.

FAI

Fairness

Decide impartially. Disclose and manage conflicts of interest. Give equal access to information and opportunity in procurement and staffing.

HON

Honesty

Seek and tell the truth. Report status accurately, even when it is unwelcome. Do not mislead by omission, selective data or silence.

INSIGHT · ASPIRATIONAL AND MANDATORY

Ethics codes usually combine **aspirational** standards (what we strive for — e.g. seeking diverse perspectives) and **mandatory** ones (what we must do — e.g. disclosing conflicts of interest, not taking bribes). Breaching a mandatory standard can carry professional and legal consequences.

A five-step ethical decision check

1. **Recognise** Is there an ethical issue? Who could be harmed or treated unfairly?
2. **Get the facts** What do you actually know? What laws, policies, contracts and codes apply?
3. **Consider options** List possible actions, including 'raise it', 'decline' and 'seek advice'.
4. **Test them** Would I be comfortable if this appeared in the news, or if my mentor saw it? Is it legal? Fair to all? Honest?
5. **Act and reflect** Act, document your reasoning, and raise concerns through proper channels if needed.

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Common dilemmas and the professional response

SITUATION	RISK	PROFESSIONAL RESPONSE
A bidder's relative works on your evaluation panel	Conflict of interest	Disclose immediately; the relative steps out of evaluation
A vendor sends an expensive gift during tender evaluation	Undue influence	Decline politely; follow gift policy; report if required
Sponsor asks you to show the project as green before the board meeting	Misrepresentation	Explain the real status; offer to present the recovery plan alongside
An official asks for a 'speed money' payment to clear equipment	Bribery / corruption	Refuse; escalate to legal/compliance; record the request
A new team member brings a former employer's confidential designs	IP and confidentiality breach	Do not use them; inform legal; remind of obligations
You realise your own estimate had a major error	Hidden mistake	Disclose quickly with corrected figures and impact

WATCH OUT · LOCAL CUSTOM IS NOT A LICENCE

Gift-giving and hospitality norms differ across cultures, and respecting them matters. But where a custom conflicts with law, your organisation's policy or the integrity of a decision — for example, gifts during a live tender — the law and policy prevail. When unsure, disclose and ask.

SCENARIO · DECIDE · THE GREEN REPORT

Two days before a board meeting, your sponsor asks you to change the project's status from red to amber 'to avoid panic', promising to fix things next month. What should you do?

(A) Change it to amber — the sponsor is accountable for the project.

(B) Report the sponsor to the ethics hotline immediately.

(C) Keep the status red, and offer to help the sponsor present a credible recovery plan and decisions needed alongside it.

(D) Skip the board meeting to avoid presenting.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · ETHICS AT A CITY FLYOVER PROJECT

You manage the traffic-management and utilities package of a city flyover. A subcontractor offers to pay for your team's end-of-year dinner at a five-star hotel. The same week, a site engineer tells you quietly that concrete test results for two piers were 'adjusted' to meet the deadline. Your sponsor is under heavy political pressure to open before the monsoon.

1. Identify each ethical issue and the value(s) involved.

2. What should you do about the test results, and in what order?

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3. How should you respond to the dinner offer?

Model answer: module answer key.

REFLECT

Describe a time you felt pressure to bend the truth or a rule at work. Which of the four values was under strain, and what would you do differently now?

Key terms

Conflict of interest
A situation in which personal interests could improperly influence, or appear to influence, professional judgement.

Honesty (professional)
Understanding and communicating the truth accurately, including unwelcome news, and not misleading by omission.

KNOWLEDGE CHECK

10. You discover your company's bid evaluation includes a firm owned by your cousin. What should you do FIRST?
    - (A) Evaluate the bid fairly and say nothing about it
    - (B) Disclose the relationship and step aside from it
    - (C) Mark the firm down to avoid any suspicion of favour
    - (D) Ask your cousin to withdraw the bid quietly before scoring

11. A sponsor asks you to omit a major risk from a report to investors. Which value is primarily at stake?
    - (A) Honesty
    - (B) Respect
    - (C) Fairness
    - (D) Responsibility only for the sponsor

12. A team member makes derogatory jokes about a colleague's religion. Which action BEST reflects the value of respect?
    - (A) Ignore it to avoid creating conflict in the team
    - (B) Address it promptly and involve HR as policy requires
    - (C) Laugh along so you are seen to fit in with the team
    - (D) Wait for a formal complaint before taking any action

13. In which case does local custom NOT justify accepting a gift?
    - (A) A small festival sweet box shared with the whole team after delivery
    - (B) A thank-you card from a user
    - (C) Tea offered at a site meeting
    - (D) A costly gift from a bidder during an active tender

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MODULE 8 · CLOSE · LESSON 4 · 30 MIN

Career Portfolio & Exam Strategy

Show what you can do, keep growing, and think like a professional under exam conditions.

- Build a project-management career portfolio with evidence of outcomes.
- Apply a reliable method to situational-judgement questions.
- Recognise agile and predictive context cues and typical best-practice answers.

A certificate opens a door; a **portfolio** walks you through it. Employers want proof that you can deliver outcomes, lead people and handle uncertainty. Build that proof deliberately, project by project.

What goes in a PM portfolio

CS

Case summaries

Three to five short project stories: context, your role, actions, results with numbers.

ART

Sanitised artefacts

A charter, risk register, dashboard or RACI — with confidential details removed and permission obtained.

MET

Metrics

Budget size, team size, delivery performance, benefits achieved, adoption rates.

REF

References & feedback

Short quotes from sponsors or team members, with their consent.

LRN

Continuous learning

Courses, certifications, communities, talks and what you applied from them.

RFL

Reflection

What you would do differently — evidence of a learning mindset.

TEMPLATE **Portfolio case entry (CAR)**

Write each project as Context, Action, Result so achievements are easy to scan.

PROJECT & DATES	CONTEXT (PROBLEM, SIZE)	MY ROLE	ACTIONS I TOOK	RESULTS (MEASURED)	EVIDENCE / ARTEFACT
Clinic booking app, 2025	*4 clinics, 20k patients/yr, 22% no-shows*	*PM, team of 7*	*Ran discovery with patients, hybrid delivery, SMS reminders, change champions*	*No-shows down to 13% in 6 months; on budget*	*Sanitised dashboard + sponsor quote*

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PRO TIP · TRANSLATE TASKS INTO OUTCOMES
Not ‘Responsible for weekly status reports’ but ‘Introduced a one-page decision-first report; sponsor decisions came 2 weeks faster on average.’ Numbers, verbs and outcomes.

Many PM certifications — including ZPMP’s final exam — use **situational-judgement** questions: a short scenario and four plausible actions. Usually more than one option is reasonable; you must find the **best** one for the context.

The ZPMP 5-R method for situational questions

1. **Read the last line first** Spot the qualifier: FIRST, NEXT, BEST, MOST likely, LEAST, EXCEPT.
2. **Recognise the context** Agile, predictive or hybrid? Who is involved? What stage of the Value Loop?
3. **Reduce** Eliminate options that are unethical, passive, blame-focused or bypass the process.
4. **Rank** Between the remaining two, prefer the action that addresses the root cause, gathers facts or involves the right people directly.
5. **Re-check** Does your pick answer the actual question — e.g. FIRST rather than eventually?

Context cues

Agile cues

- Product owner, sprint, backlog, iteration review
- Self-organising team, servant leader
- Best answers: collaborate with the product owner on priorities, let the team decide how, inspect and adapt, frequent feedback

Predictive cues

- Baseline, change control board, WBS, signed contract
- Phase gates, formal approvals
- Best answers: assess impact, follow the change management plan, update baselines only after approval

Universal cues

- Ethics, safety, conflict, stakeholder concern
- Best answers: understand first, talk directly and privately, act within your authority, escalate with options when needed

Answer patterns to be suspicious of

PATTERN	WHY IT IS USUALLY WEAKER
Escalate to the sponsor immediately	Skips analysis and actions within the PM’s own authority
Ignore it / wait and see	Passive; problems grow
Fire, punish or publicly blame someone	Damages safety; rarely solves root causes
Implement the change straight away because the client asked	Bypasses impact analysis and change control
Add more people immediately	May not fix the cause; can slow a late project further
Absolute words: always, never, all	Real practice is contextual

EXAMPLE · APPLYING 5-R
‘A key stakeholder complains that a feature does not meet their needs during a sprint review. What should the PM do FIRST?’ Context: agile. Eliminate ‘tell them the scope is fixed’ (rigid) and ‘escalate to the sponsor’ (premature). Between ‘add the change to the current sprint’ and ‘work with the product owner to capture it in the backlog for prioritisation’, the second respects the product owner’s role and sprint focus — the best answer.

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SCENARIO · DECIDE · EXAM-STYLE PRACTICE

During execution of a predictive construction project, the client asks the site engineer to add an extra loading bay. The engineer agrees and starts the work. What should the project manager do FIRST?

(A) Let the work continue since the client requested it.

(B) Discipline the site engineer.

(C) Update the scope baseline to include the bay.

(D) Stop the work on the extra bay and evaluate the impact of the request through the change control process.

Circle your choice. Answer and feedback: module answer key.

Exam-day habits

- Pace yourself: note the time per question and flag hard ones to return to
- Answer every question — an unanswered question cannot score
- Do not overthink: choose the best answer from the options given, not the one you wish were there
- Watch for qualifiers: FIRST and NEXT are about sequence, BEST is about overall quality
- Use breaks to reset attention

Key terms

Situational-judgement question
An exam item presenting a realistic scenario and asking for the best action among plausible alternatives.

CAR format
Context, Action, Result — a concise structure for describing professional achievements.

KNOWLEDGE CHECK

14. A question asks what the PM should do FIRST after a team member reports a new risk. Which option is most likely best?

(A) Escalate it to the sponsor straight away
(B) Assess and record it in the risk register
(C) Implement a response immediately to be safe
(D) Inform all stakeholders by email

15. Which option pattern is usually weakest in situational questions?

(A) Ignore the issue and wait for the next status meeting
(B) Meet the stakeholder privately to understand their concern
(C) Review the relevant plan and baseline before acting
(D) Collaborate with the product owner on the backlog

16. Which cue suggests a predictive context?

(A) Sprint review with stakeholders
(B) Change control board approval
(C) Self-organising, cross-functional team
(D) Product backlog refinement

17. Which portfolio entry is strongest?

(A) Led 9 people to deliver a booking app on budget; no-shows fell from 22% to 13%
(B) Managed a large and complex digital project for a well-known healthcare client
(C) Responsible for weekly meetings, status reports and stakeholder communications
(D) Very hard-working, results-driven project manager with excellent people skills

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 8 · CLOSE · LESSON 5 · 40 MIN

The Whole Loop: Integrative Walk-Through & Capstone

Watch all eight ZPMP stages work together on one project — then run your own.

- Trace a mini-project through all eight stages of the ZPMP Value Loop.
- Explain how decisions in one stage affect the others.
- Plan your ZPMP capstone and its deliverables.

You have studied each stage on its own. Real projects do not separate so neatly: a risk spotted in Steer changes the plan; a stakeholder concern in Lead reshapes scope; a lesson in Close improves the next Frame. Let us follow one mini-project through the whole **ZPMP Value Loop**.

EXAMPLE · THE MINI-PROJECT

Cumilla District Hospital — SMS and web appointment booking. Patients queue from dawn for outpatient tickets; 22% of booked follow-ups are missed. A 6-month project with a budget of BDT 4.5 million and a team of 7 (PM, product owner from hospital administration, 3 developers, 1 tester, 1 change and training lead) will deliver online and SMS booking with reminders for four outpatient departments.

The ZPMP Value Loop at a glance

1 **Frame**
Why are we doing this and what value will it create?

2 **Shape**
Who is involved and what exactly will we deliver?

3 **Plan**
How, when and for how much — with honest uncertainty.

4 **Deliver**
Build, assure quality and manage suppliers.

5 **Steer**
Measure, control, manage risk and forecast.

6 **Lead**
Grow the team, resolve conflict, communicate.

7 **Scale**
Drive adoption, use AI responsibly, govern well.

8 **Close**
Accept, hand over, realise benefits and learn.

Walking the loop

STAGE	KEY QUESTION	WHAT THE TEAM DID	KEY ARTEFACT
Frame	Is this worth doing?	Business case: fewer queues and no-shows, more patients per doctor-day; hybrid life cycle chosen	Business case, charter
Shape	Who cares and what exactly?	Mapped patients, doctors, front desk, IT, health directorate; prioritised backlog with booking, reminders, rescheduling	Stakeholder register, backlog, change control
Plan	How long and how much?	Fixed hardware and SMS-gateway contract; agile build in 2-week sprints; Monte Carlo gave P80 at 6.5 months	Roadmap, budget, risk-adjusted schedule

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STAGE	KEY QUESTION	WHAT THE TEAM DID	KEY ARTEFACT
Deliver	Are we building it right?	Scrum for software, Kanban for training logistics; usability tests with patients	Increments, test results, vendor contract
Steer	Are we on track?	Risk: low smartphone use among older patients → SMS-first design; burnup and EVM on fixed-price parts	Risk register, dashboard
Lead	Is the team healthy and aligned?	Conflict between doctors and front desk on slot rules resolved collaboratively; weekly sponsor huddle	Communication plan, RACI
Scale	Will people adopt it?	ADKAR survey showed desire gap at front desk; champions appointed; AI assistant used to draft Bangla and English FAQs, verified by staff	Change impact assessment, AI-use log
Close	Did we deliver value and learn?	Formal acceptance, 4-week hypercare, benefits plan to hospital administrator, retrospective	Closure checklist, benefits plan, lessons register

INSIGHT · THE LOOP REALLY LOOPS

In month 2 a Steer finding (older patients rarely use the web) went back to Shape (SMS-first backlog), which changed Plan (SMS gateway cost) and Lead (new message to doctors). In Close, the lesson ‘test with real patients in week 2’ went into the PMO’s template for the next hospital — feeding the next Frame.

SCENARIO · DECIDE · A MID-PROJECT CURVEBALL

In month 4, the health directorate announces that all hospital systems must integrate with a new national patient ID within 6 months. The product owner wants to add it now. The P80 forecast already sits at the deadline. What should the PM do FIRST?

(A) Add the integration to the current sprint to show responsiveness.

(B) Tell the directorate the project cannot comply.

(C) Analyse the requirement and its impact with the product owner and team, then present options (e.g. phase 2 release before the mandate date) through change control.

(D) Escalate to the sponsor to request more budget.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · INTEGRATIVE CHALLENGE: UNIVERSITY LMS MIGRATION

A university is migrating 600 courses to a new learning management system before the next academic year (9 months). Faculty are sceptical, the vendor contract is fixed-price for configuration, and student-data privacy rules apply. The registrar wants AI-generated course summaries for students.

1. For each ZPMP stage, name one decision or artefact you would prioritise.

2. Which two risks would you analyse quantitatively, and why?

3. How would you handle the AI-summary request responsibly?

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Model answer: module answer key.

Your **ZPMP capstone** asks you to apply the whole loop to a real or realistic project of your choice — ideally from your workplace, with confidential details removed. You will produce a compact portfolio of artefacts and a short reflective report.

Capstone deliverables

1. **Frame** One-page business case and charter with measurable success criteria.
2. **Shape** Stakeholder register and prioritised backlog or WBS, with a change-control approach.
3. **Plan** Schedule or roadmap, budget, and a risk-adjusted forecast (range with confidence).
4. **Deliver** Delivery approach (predictive, agile or hybrid) with justification and quality plan.
5. **Steer** Risk register (top 10), dashboard design with RAG criteria, and an EVM or burnup snapshot.
6. **Lead** RACI and communication plan; one conflict or negotiation you would expect and how you would handle it.
7. **Scale** Change impact assessment, AI-use guidelines for the project, and governance/ESG considerations.
8. **Close** Closure checklist, benefits realisation plan and a 600-word reflection on what you learned.

TEMPLATE

Capstone planner

Plan your capstone: one row per Value Loop stage.

STAGE	ARTEFACT(S) I WILL PRODUCE	DATA / INPUT I NEED	WHO CAN REVIEW IT	TARGET DATE
Steer	*Top-10 risk register, one-page dashboard*	*Past schedule data, team risk workshop*	*Colleague in PMO*	*Week 3*

Key terms

ZPMP Value Loop
ZyeemZia's eight-stage project framework — Frame, Shape, Plan, Deliver, Steer, Lead, Scale, Close — applied iteratively from idea to realised value.

Capstone
A culminating project in which learners apply the full ZPMP Value Loop to a real or realistic project and present the resulting artefacts.

KNOWLEDGE CHECK

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18. In the walk-through, a Steer finding about older patients led to an SMS-first design. Which stage did this feed back into first?

    (A) Close
    (B) Scale
    (C) Shape
    (D) Frame

19. Which ZPMP stage focuses on adoption, AI use and governance?

    (A) Plan
    (B) Deliver
    (C) Scale
    (D) Lead

20. A mandatory new requirement arrives mid-project. What is the FIRST step?

    (A) Add it to the plan immediately, since it is mandatory anyway
    (B) Analyse its impact and present options via change control
    (C) Refuse it until the current phase has been completed
    (D) Request extra budget and time from the sponsor to cover it

21. Why does the Close stage feed the next Frame?

    (A) Because projects never truly end in organisations
    (B) Because the PMO requires the stages to be linked
    (C) Lessons and benefits data improve the next business case
    (D) It does not; each project starts from a clean slate

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 8 · CLOSE

Answer key

Knowledge checks

1. B Terminated projects still need orderly closure.

2. A Acceptance is the customer's formal agreement; quality control is the internal check.

3. D Operational risks continue after the project and must be owned by operations.

4. C It reviews the procurement process to improve future procurement.

5. B A lesson is learned when it changes behaviour or practice.

6. B Benefits arrive in operations and need an owner there.

7. D Focused, owned actions are more likely to happen.

8. B Improvement is measured relative to a baseline.

9. C It is what the project made possible; benefits come from the outcomes of using it.

10. B Disclosure and recusal manage the conflict transparently.

11. A Omitting material information misleads — an honesty issue.

12. B Respect requires acting against behaviour that demeans others.

13. D Valuable gifts during procurement create undue influence regardless of custom.

14. B Analyse and document before acting or escalating.

15. A Passive waiting lets problems grow.

16. B Formal change control boards are characteristic of predictive governance.

17. A Specific context, action and measured result.

18. C The finding changed the backlog — a Shape activity.

19. C Scale covers change management, AI-powered PM and governance.

20. B Impact analysis and options precede decisions.

21. C Learning and benefit evidence make future framing more realistic.

Scenarios & case studies

> **Case study · ERP go-live at a garment factory in Ashulia**
> Missed: formal acceptance, a defined support model and operational owner, hypercare with exit criteria, transfer of warranties, knowledge transfer to IT and compliance staff, and contract closure. Now: the sponsor appoints an operational owner; the vendor and factory hold an acceptance review against the original criteria and agree any punch-list; tickets are triaged and assigned; warranty and support renewal decided before April; compliance-report training run quickly. Prevention: include closure tasks in the schedule and budget from day one, make sponsor sign-off on closure a gate, and do not release the PM until the closure checklist is complete.

> **Scenario · The keen sponsor**
> **B** is best. Correct. Keep closure lightweight but formal: acceptance protects the client, the vendor and the team.

> **Scenario · The missing benefit**
> **C** is best. Correct. Benefit realisation is owned by the business, but the former PM can help diagnose — often the cause is adoption or process, not the system.

> **Scenario · The green report**
> **C** is best. Correct. Honest status plus a constructive path forward serves the sponsor and the organisation.

> **Case study · Ethics at a city flyover project**
> Issues: (1) Dinner — potential undue influence and conflict of interest (fairness); (2) Adjusted test results — falsified safety data (honesty, responsibility), with serious public-safety risk. Order: treat safety first — protect the engineer, secure the original records, inform the sponsor and the quality/safety authority as required by contract and law, and stop or restrict work on the affected piers pending independent retesting. Document everything. Political pressure does not change the duty; a delayed opening is far better than a structural failure. Dinner: decline politely, referring to company policy; if the team celebrates, the organisation pays.

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Scenario · Exam-style practice

D is best. Correct. An unapproved scope change has started; pause it and route it through impact analysis and formal approval.

Scenario · A mid-project curveball

C is best. Correct. Understand the mandate, assess impact on scope, schedule, cost and risk, and bring options to the right decision makers.

Case study · Integrative challenge: university LMS migration

Example answer — Frame: business case and charter with adoption-based success measures. Shape: faculty and student stakeholder map; backlog of course waves. Plan: wave plan with P80 date before semester start; contingency for migration rework. Deliver: hybrid — fixed-price vendor milestones plus agile course-redesign waves; quality rubric. Steer: EVM on vendor milestones, burnup on course waves, RAG criteria. Lead: designer motivation (autonomy, purpose), RACI for course sign-off. Scale: ADKAR for faculty; AI policy check; PMO stage gate before semester. Close: acceptance per wave, hypercare in week 1 of term, benefits plan (student satisfaction, support tickets). Quantitative: data-migration rework and faculty availability, since they drive schedule spread. AI summaries: run as a controlled pilot using approved tools, no personal student data in prompts, faculty review before publishing, clear labelling, and feedback collection.

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REFERENCE

Glossary

A

Acceptance criteria Conditions a deliverable or story must satisfy to be accepted, often written as Given/When/Then. (Shape)

ADKAR A model of individual change: Awareness, Desire, Knowledge, Ability, Reinforcement. (Scale)

B

BATNA Best Alternative To a Negotiated Agreement: the course of action a party will take if negotiations fail. (Lead)

Benefit A measurable improvement that results from an outcome and is valued by the organisation or society. (Frame)

Benefits realisation Measuring and managing the outcomes of a project after delivery to ensure intended benefits are achieved. (Close)

Burnup chart A chart showing completed work and total scope over time, making scope change visible. (Steer)

Business case A documented justification for an investment that compares options, benefits, costs and risks and recommends a way forward. (Frame)

C

Capstone A culminating project in which learners apply the full ZPMP Value Loop to a real or realistic project and present the resulting artefacts. (Close)

CAR format Context, Action, Result — a concise structure for describing professional achievements. (Close)

Conflict of interest A situation in which personal interests could improperly influence, or appear to influence, professional judgement. (Close)

Contingency reserve Budget or time within the baseline set aside for identified risks. (Plan)

Contract closure Completing all contract obligations, settling claims and payments and formally ending the contractual relationship. (Close)

Cost baseline The approved, time-phased budget, including contingency but excluding management reserve, used to measure cost performance. (Plan)

Cost of quality The total cost of conformance (prevention and appraisal) and non-conformance (internal and external failure). (Deliver)

Cost performance index (CPI) EV ÷ AC — the cost efficiency of work performed; below 1.0 means over budget. (Steer)

Crashing Shortening the schedule by adding resources to critical activities, usually at extra cost. (Plan)

Critical path The longest sequence of dependent activities through a schedule network; it determines the shortest possible project duration. (Plan)

Cumulative flow diagram A stacked area chart showing the number of items in each workflow stage over time. (Deliver)

Cycle time The elapsed time from when work starts on an item until it is finished. (Deliver)

D

Data minimisation Sharing only the data strictly needed for a task, with personal or confidential details removed or masked. (Scale)

Definition of done A shared quality standard that every increment of work must meet before it is considered complete. (Shape)

Dis-benefit A measurable negative consequence of a change, perceived as adverse by one or more stakeholders. (Close)

E

Earned value (EV) The budgeted value of the work actually completed to date. (Steer)

Elicitation Drawing out stakeholders' real needs through techniques such as interviews, observation, workshops and prototypes. (Shape)

Emotional intelligence The ability to recognise, understand and manage one’s own emotions and to recognise and influence others’ emotions. (Lead)

ESG Environmental, social and governance considerations used to assess the wider impact and responsibility of projects and organisations. (Scale)

Expected monetary value (EMV) Probability multiplied by monetary impact; summed across risks or branches to compare options and size reserves. (Steer)

F

Fast-tracking Shortening the schedule by overlapping activities normally done in sequence, at a higher risk of rework. (Plan)

Firm fixed price contract A contract in which the seller delivers defined scope for an agreed price and carries most of the cost risk. (Deliver)

Fist-of-five A quick group technique in which people show zero to five fingers to signal their level of support for a proposal. (Lead)

Formal acceptance Documented confirmation by an authorised stakeholder that deliverables meet agreed acceptance criteria. (Close)

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Free float The amount of time an activity can be delayed without delaying the early start of any successor. (Plan)

G

Generative AI copilot An AI assistant that generates text, summaries, code or other content from natural-language prompts to support a person's work. (Scale)

GROW model A coaching structure: Goal, Reality, Options, Will (way forward). (Lead)

H

Hallucination Output from a generative model that is fluent and plausible but factually wrong or invented. (Scale)

Honesty (professional) Understanding and communicating the truth accurately, including unwelcome news, and not misleading by omission. (Close)

Hybrid approach A deliberate combination of predictive and adaptive practices within one project. (Frame)

Hypercare A defined period of intensified support immediately after go-live, ending when agreed stability criteria are met. (Close)

I

Increment A usable step toward the product goal that meets the definition of done. (Deliver)

Intrinsic motivation Motivation that comes from the satisfaction of the work itself — interest, growth and meaning. (Lead)

L

Lead time The elapsed time from when an item is requested or committed to until it is delivered to the customer. (Deliver)

Leading indicator A measure that changes before an outcome and helps predict it, allowing early action. (Scale)

Life cycle The series of phases a project passes through from start to completion. (Frame)

M

Management reserve Budget outside the cost baseline for unforeseen work; its use requires management approval. (Plan)

Model drift The decline in an AI model's performance over time as real-world data diverges from its training data. (Scale)

MoSCoW A prioritisation method that sorts requirements into Must have, Should have, Could have and Won't have this time. (Shape)

O

OKR Objectives and Key Results: a qualitative objective paired with a few measurable outcomes that show progress toward it. (Frame)

Operations Ongoing, repetitive work that sustains the organisation's existing products and services. (Frame)

Organisational change management (OCM) Structured methods for helping people move from current to future ways of working so that project outputs are adopted and benefits realised. (Scale)

P

Parametric estimating Estimating by multiplying a quantity by a unit rate derived from historical data, e.g. cost per square metre. (Plan)

PERT estimate A weighted three-point estimate calculated as (Optimistic + 4 × Most likely + Pessimistic) ÷ 6. (Plan)

PMO Project (or portfolio/programme) management office: a function that standardises governance, supports or directs projects and improves delivery capability. (Scale)

Portfolio The full set of projects, programs and operational investments an organisation manages to achieve its strategy, selected and balanced as a whole. (Frame)

Power–interest grid A two-by-two tool that groups stakeholders by their power over and interest in the project to guide engagement strategy. (Shape)

Pre-mortem A workshop in which the team imagines the project has failed and works backwards to identify the reasons, surfacing risks early. (Steer)

Product backlog An ordered, evolving list of everything that might be needed in a product, owned by the product owner. (Shape)

Product Owner The accountable person in Scrum who maximises product value and owns the ordering of the product backlog. (Deliver)

Program A group of related projects and activities managed together so their combined benefits are greater than if they were managed separately. (Frame)

Project A temporary, purposeful effort that creates a unique product, service, capability or change to move an organisation toward a better future state. (Frame)

Project charter A document issued by the sponsor that formally authorises a project and gives the project manager authority to apply resources. (Frame)

Psychological safety A shared belief within a team that it is safe to take interpersonal risks such as speaking up, asking questions or admitting mistakes. (Lead)

R

RACI A responsibility assignment matrix showing who is Responsible, Accountable, Consulted and Informed for each activity or decision. (Lead)

Reference-class forecasting Estimating by comparing a project with the actual outcomes of a group of similar past projects. (Scale)

Residual risk The risk that remains after responses have been implemented. (Steer)

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Retrospective A regular team meeting to inspect how the team worked and agree concrete improvements for the next cycle. (Close)

Risk An uncertain event or condition that, if it occurs, affects at least one project objective — negatively (threat) or positively (opportunity). (Steer)

Risk breakdown structure (RBS) A hierarchical grouping of risk sources used as a prompt to identify risks systematically. (Steer)

S

Salience model A stakeholder classification based on power, legitimacy and urgency; those with all three are 'definitive'. (Shape)

SBI feedback A feedback structure describing the Situation, the observable Behaviour and its Impact. (Lead)

Scope creep Uncontrolled expansion of scope without adjusting time, cost or resources and without proper approval. (Shape)

Scrum Master The accountable person in Scrum for team effectiveness, coaching the team and organisation and removing impediments. (Deliver)

Secondary risk A new risk that arises as a direct result of implementing a risk response. (Steer)

Servant leadership A leadership approach that focuses on serving the team — removing obstacles, growing people and enabling them to deliver value. (Lead)

Situational-judgement question An exam item presenting a realistic scenario and asking for the best action among plausible alternatives. (Close)

Sprint Goal The single objective for a sprint that gives the developers focus and flexibility in what they deliver. (Deliver)

Stage gate A decision point between phases where governance reviews progress and decides whether to continue, change or stop. (Scale)

Stakeholder Any person, group or organisation that can affect, be affected by, or believe they are affected by a project. (Shape)

Story point A unitless, team-relative measure of the size of a backlog item, combining effort, complexity and uncertainty. (Plan)

T

Tailoring Deliberately adapting the approach, processes and documents to fit the context of a project, team and organisation. (Frame)

Thomas–Kilmann modes Five conflict-handling approaches — competing, collaborating, compromising, avoiding and accommodating — mapped on assertiveness and cooperativeness. (Lead)

Throughput The number of work items finished per unit of time. (Deliver)

Time and materials contract A contract that pays agreed rates for time and materials used, with most cost risk on the buyer. (Deliver)

To-complete performance index (TCPI) The cost efficiency that must be achieved on remaining work to meet a target (BAC or EAC). (Steer)

Total float The amount of time an activity can be delayed without delaying the project finish date. (Plan)

Tuckman model A model of team development through forming, storming, norming, performing and adjourning. (Lead)

U

User story A short description of a need from the user's point of view, used as a placeholder for conversation and refined with acceptance criteria. (Shape)

V

Value The worth, importance or usefulness of a result to stakeholders, after considering cost, risk and time. (Frame)

Velocity The amount of work, usually in story points, a team completes to the definition of done in one iteration. (Plan)

W

Watermelon reporting Status that appears green externally while the underlying reality is amber or red. (Steer)

WIP limit A cap on the number of work items allowed in a workflow stage at one time. (Deliver)

Work breakdown structure (WBS) A hierarchical, deliverable-oriented decomposition of the total scope of a project. (Shape)

Working agreement A team-created set of norms describing how members will collaborate, communicate and make decisions. (Lead)

Z

ZPMP Value Loop ZyeemZia's eight-stage project framework — Frame, Shape, Plan, Deliver, Steer, Lead, Scale, Close — applied iteratively from idea to realised value. (Close)

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CAPSTONE

Capstone: Run a project through the ZPMP Value Loop

Pick a real project from your work or community, or a realistic one (for example: launching a mobile banking feature in Dhaka, opening a new clinic, or migrating a university to a new learning platform). Take it through all eight stages of the ZPMP Value Loop and produce a compact project pack that a sponsor could approve and a team could execute.

Deliverables

☐ One-page project charter with business case, objectives (OKRs) and success criteria

☐ Stakeholder register with power–interest map and engagement plan

☐ Scope: product backlog or WBS with acceptance criteria and a definition of done

☐ Estimates and a schedule: critical path or release plan, with assumptions

☐ Budget with contingency and management reserve, plus an earned value baseline

☐ Risk register: top 10 risks and opportunities with owners and responses

☐ Team charter, RACI and communication plan

☐ Change-management plan (ADKAR or Kotter) and an AI-use plan with privacy rules

☐ Simulated status report at the halfway point using EVM and flow metrics

☐ Closing report with benefits realisation plan and lessons learned

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Value & alignment	○ Objectives vague	○ Clear objectives and business case	○ Measurable OKRs tied to benefits and owners
Approach & tailoring	○ Life cycle not justified	○ Life cycle chosen with reasons	○ Hybrid design tailored to uncertainty and constraints
Scope & requirements	○ Incomplete scope	○ Complete backlog/WBS	○ Prioritised, testable, with change control
Planning & forecasting	○ Single-point dates	○ Estimated schedule and budget	○ Ranges, reserves and a forecasting method
Risk & control	○ Few risks listed	○ Register with responses	○ Quantified top risks, triggers and EVM thresholds
People & communication	○ Roles unclear	○ RACI and comms plan	○ Engagement strategy and team health practices
Change, AI & ethics	○ Not addressed	○ Basic plan	○ Adoption plan, responsible AI use and ethical risks covered
Closing & learning	○ No closure plan	○ Closing checklist	○ Benefits tracking and lessons that change practice

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CAPSTONE

Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The online certification exam draws 40 of these questions at random, with a 60-minute limit and a 80% pass mark. Practise here first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Frame — Foundations of Modern Project Management & Value

1. A sponsor asks you to start work immediately on a new customer portal. Funding is verbally agreed, but no document authorises the project or names you as project manager. What should you do FIRST?

   (A) Begin requirements workshops with key users to save time

   (B) Work with the sponsor to create and sign a project charter

   (C) Develop a detailed project management plan for sponsor approval

   (D) Select a delivery vendor so that work can start quickly

2. Midway through a project, market research shows the expected customer demand that justified the investment has fallen sharply. Deliverables are on track. What should the project manager do NEXT?

   (A) Continue as planned, because scope, time and cost are all still on track

   (B) Raise it with the sponsor and review the business case: continue, change or stop

   (C) Accelerate delivery so the product reaches the market before demand falls further

   (D) Reduce the remaining testing effort to lower costs and protect the expected return

3. A company must deliver a new tax-reporting module with requirements fixed by regulation, and a customer mobile app whose features are still being explored. Which approach should the project manager recommend?

   (A) Fully predictive delivery for both components, with one integrated baseline and milestone plan

   (B) Fully agile delivery for both components, using short sprints and no fixed milestones

   (C) Postpone the app until the regulatory module is complete, then reassess the app features

   (D) Hybrid: predictive for the regulatory module, adaptive for the app, integrated milestones

4. A project delivered a new HR system on time and on budget. Six months later, only 20% of managers use it and the expected time savings have not appeared. What does this situation BEST illustrate?

   (A) The project was a complete success because it met its time and cost constraints

   (B) Outputs were delivered, but the intended outcomes and benefits were not realised

   (C) The schedule baseline was poorly estimated, so the system was not ready for users

   (D) The procurement strategy chose the wrong system for the managers' working needs

5. The portfolio board must choose one of two projects. Project X has a higher ROI; Project Y has a lower but positive NPV and directly supports two of this year's strategic key results. As the PM preparing the analysis, what is the BEST recommendation?

   (A) Recommend Project X, because ROI is the most objective and comparable measure of return

   (B) Recommend funding both projects at half budget so that each can make some progress

   (C) Make no recommendation and leave the board to decide without supporting analysis

   (D) Recommend Project Y, showing its strategic fit with its financial measures and risks

6. A newly appointed PM in a large bank is told to use the organisation's full 40-document project methodology for a six-week, low-risk internal dashboard. What should the PM do?

   (A) Propose a tailored, lighter set of practices and agree it with the governance body

   (B) Follow every step and document exactly, as the methodology requires of all projects

   (C) Set the methodology aside and run the work informally, as the risk is low

   (D) Deliver first and ask the team to complete all required documents after go-live

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7. During project initiation, a stakeholder asks what measurable result will show the project succeeded. Where should this be documented FIRST?

   (A) In the lessons-learned register, so future projects can reuse the measure
   (B) In the project charter, as success criteria linked to the business case
   (C) In the team's working agreement, so the team commits to the result
   (D) In the first sprint retrospective notes, once the team has started work

8. An organisation keeps launching projects that compete for the same specialists, and several deliver overlapping features. Which management level is MOST likely to resolve this?

   (A) Portfolio management, which selects and balances investments against strategy
   (B) Operations management, which runs and coordinates day-to-day business processes
   (C) Each project manager, negotiating directly with the other projects' managers
   (D) Quality control, which inspects deliverables against the agreed standards

Shape — Stakeholders, Scope & Requirements

9. Two months into an ERP project, you discover that the internal audit department was never consulted and has compliance requirements for the finance module. What should you do FIRST?

   (A) Submit a change request covering all possible audit requirements for the finance module
   (B) Add internal audit to the stakeholder register and meet them to learn their needs
   (C) Escalate to the sponsor that the project is now at risk of a compliance failure
   (D) Continue as planned and address any audit concerns in a phase after go-live

10. A senior regulator has high power over your project but shows little interest in it. Using the power–interest grid, what engagement approach is BEST?

    (A) Keep them satisfied with concise, relevant updates and consult them when affected
    (B) Manage them closely with weekly detailed meetings and full access to all project data
    (C) Monitor them with minimal effort, as their low interest means low influence
    (D) Keep them informed by adding them to every team email and distribution list

11. A requirements workshop is dominated by two senior managers while the frontline users who will use the system stay silent. What should the facilitator do?

    (A) Let the senior managers decide, since they hold authority and know the strategy
    (B) End the workshop and send everyone a survey to collect their views in private
    (C) Ask the senior managers to leave so the frontline users can speak more freely
    (D) Use silent brainstorming and breakouts, then interview or observe the users

12. The product owner brings a story estimated at three sprints of work: 'As a customer I want the full loan application journey.' What should the team do?

    (A) Split it by technical layer: database, API and user interface, one per sprint
    (B) Accept it as one story and track its progress across the three sprints
    (C) Split it into thin vertical slices that each deliver testable user value
    (D) Ask the product owner for a complete written specification before estimating

13. On a predictive project with an approved scope baseline, a department head asks the team directly to add a new report. The developers estimate two days of work. What should the project manager do?

    (A) Let the team add it, since two days of work is within normal tolerance
    (B) Refuse the request, because the scope baseline has already been approved
    (C) Ask for a change request and assess its impact via integrated change control
    (D) Schedule it into the next phase quietly, without involving the sponsor

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14. A team member has added an attractive analytics screen that no stakeholder requested, delaying a planned feature. What does this BEST describe, and what should the PM do?

(A) Scope creep; raise a change request against the requesting stakeholder

(B) Progressive elaboration; record it as normal refinement of the requirements

(C) Gold plating; agree with the team that work must trace to approved items

(D) Healthy innovation; recognise it publicly to encourage creative initiative

15. Feature P has a cost of delay of 24 and a job size of 12. Feature Q has a cost of delay of 15 and a job size of 5. Using WSJF, what should the product owner do?

(A) Sequence P first, because its cost of delay (24) is higher than Q's (15)

(B) Treat them as equal priority, because their scores differ by too little

(C) Sequence P first, because its larger size makes it the more important job

(D) Sequence Q first, because its WSJF score (3.0) is higher than P's (2.0)

16. Your WBS for a new clinic has level-2 elements named 'Plan', 'Design', 'Build', 'Test' and 'Deploy', with tasks underneath. What is the BEST feedback?

(A) Restructure it around deliverables and move activities into the schedule

(B) Accept it, because a phase-based structure is a valid way to organise a WBS

(C) Add more levels until every element represents less than one hour of work

(D) Replace the WBS with a Gantt chart, which shows the same work over time

Plan — Estimation, Scheduling & Budgeting

17. An activity has an optimistic estimate of 6 days, most likely 9 days and pessimistic 18 days. The sponsor asks for the expected duration. Using PERT, what should you report?

(A) 9 days, as the most likely value is the best single estimate

(B) 11 days, the simple average of the three estimates

(C) 18 days, to protect the sponsor against any overrun

(D) About 10 days, with a standard deviation of about 2 days

18. A non-critical activity has 5 days of total float and 0 days of free float. A supplier says it will be 3 days late. What is the MOST accurate impact?

(A) The project end date will move by 3 days, as will the successor's start

(B) There is no impact on any other activity, because float absorbs the delay

(C) The end date is unaffected, but the successor's early start moves by 3 days

(D) The activity immediately joins the critical path and drives the end date

19. The sponsor wants the project finished two weeks earlier. The critical path includes design and build, which could partly overlap. What should the project manager do FIRST?

(A) Analyse compression options on the critical path and present the trade-offs to the sponsor

(B) Approve overtime for the whole team so that every activity finishes sooner

(C) Remove user acceptance testing from the schedule to recover the two weeks

(D) Tell the sponsor the request is impossible because the schedule baseline is already approved

20. A Scrum team's velocity over the last four sprints was 30, 34, 28 and 32. The remaining release backlog is 186 points. A stakeholder asks when the release will be ready. What is the BEST answer?

(A) Commit to exactly 6 sprints, using the average velocity of the last four sprints

(B) Forecast a range of about 6–7 sprints from observed velocity, updated each sprint

(C) Ask the team to raise velocity to 40 points so the release finishes sooner

(D) Decline to forecast, because agile teams do not commit to delivery dates

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21. A stakeholder suggests funding a new feature from the project's contingency reserve because 'the risks it covered have not happened yet'. What should the project manager do?

(A) Approve the use, since the money is unused and the feature adds value
(B) Move the money into management reserve until a decision is needed
(C) Explain that contingency covers identified risks; process it as a change request
(D) Fund the feature from management reserve instead, without informing the sponsor first

22. A Monte Carlo schedule simulation shows a 25% probability of meeting the deterministic finish date and P80 three months later. The sponsor must announce a public launch date. What should the project manager recommend?

(A) Announce the deterministic date, because it is what the approved plan shows
(B) Announce the earliest date seen in any simulation run to show ambition
(C) Avoid announcing any public date until the remaining schedule uncertainty is removed
(D) Announce a date near P80 and manage the deterministic date as an internal target

23. Two activities scheduled in parallel both need the only certified electrician, causing an overload. The activities are on the critical path. Which technique applies, and what is a likely consequence?

(A) Resource smoothing, which never changes the project end date
(B) Fast-tracking, which reduces demand on the shared resource
(C) Crashing, which removes the dependency on the electrician
(D) Resource levelling, which may extend the project end date

Deliver — Agile, Hybrid & Quality Execution

24. During a sprint, a senior manager asks a developer to start an urgent new feature. What should the developer do?

(A) Start the feature immediately, because the request comes from a senior manager
(B) Refer it to the product owner, who decides priority while the sprint goal is protected
(C) Ask the team to cancel the sprint and replan it so the urgent feature can be included
(D) Add it to the sprint backlog without telling anyone and absorb it with extra effort

25. A Kanban team's Test column is at its WIP limit and developers say they are idle because they cannot pull new work. What should the team do?

(A) Raise the WIP limit in Test so that more work can flow through
(B) Start new development work and park it until Test has capacity
(C) Help finish items in Test by swarming on the bottleneck
(D) Remove WIP limits entirely, since they are causing idle time

26. A team's average WIP is 30 items and average throughput is 10 items per week. Management wants average cycle time to fall to about 1.5 weeks without adding people. What is the BEST approach?

(A) Ask the team to work overtime until cycle time falls
(B) Reduce WIP to about 15 items by limiting new starts
(C) Add more columns to the board to make work visible
(D) Measure lead time instead, as it better reflects delays

27. Two days before the sprint ends, developers propose skipping regression tests required by the definition of done so they can complete every planned item. What should the Scrum Master encourage?

(A) Keep the definition of done; return unfinished items to the product backlog
(B) Skip the tests this once and repay the resulting technical debt next sprint
(C) Mark the items as done, with a clear note in the backlog that tests are still pending
(D) Extend the sprint by two days so the tests and planned items both finish

28. A buyer needs a supplier for a research-like data platform whose scope is highly uncertain, but wants the supplier motivated to control costs. Which contract type is MOST suitable?

(A) Cost plus incentive fee
(B) Firm fixed price
(C) Fixed price with economic price adjustment
(D) Purchase order for a fixed quantity

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29. A garment factory's defect data shows that three causes account for most rejected pieces. Which tool should the quality lead use to communicate where to focus improvement efforts?

(A) A Pareto chart
(B) A scatter diagram
(C) A flowchart
(D) A control chart

30. Your organisation's PMO requires a six-month task-level Gantt chart from an agile team building a customer app alongside a predictive hardware rollout. What should the project manager do?

(A) Have the team produce the detailed six-month Gantt chart in the format the PMO requires
(B) Refuse the request and explain that agile teams do not plan work beyond the next sprint
(C) Provide an integrated milestone roadmap with release forecasts, agreed with the PMO
(D) Produce the task-level Gantt chart yourself so the team is not distracted from delivery

Steer — Steer — Risk, Performance & Control

31. During a status meeting, a developer mentions that the third-party payment API the team depends on will be retired by its provider ‘sometime next year’. Your release is planned for nine months from now. What should the project manager do FIRST?

(A) Record it as a risk, analyse its probability and impact, and assign an owner
(B) Log it as an issue and escalate it to the sponsor for an immediate decision
(C) Start replacing the API now so that the release date is fully protected
(D) Note it but take no action, since the retirement date is not yet confirmed

32. At month 5 of a project, PV = 200,000, EV = 180,000 and AC = 225,000. Which statement BEST describes the project’s status?

(A) Behind schedule and over budget (SPI 0.90, CPI 0.80)
(B) Ahead of schedule and over budget (SPI 1.11, CPI 0.80)
(C) Behind schedule and under budget (SPI 0.90, CPI 1.25)
(D) On schedule and over budget (SPI 1.00, CPI 0.80)

33. A project has BAC = 800,000, EV = 300,000 and AC = 400,000. The sponsor asks for a forecast assuming current cost performance continues. What is the estimate at completion?

(A) About 900,000
(B) About 1,000,000
(C) About 1,066,667
(D) About 600,000

34. A project has BAC = 500,000, EV = 200,000 and AC = 250,000. The sponsor insists on finishing within the original budget. What cost performance must the team achieve on the remaining work?

(A) A TCPI of 0.83
(B) A CPI of at least 0.80
(C) A TCPI of 1.20
(D) A TCPI of 1.00

35. A risk owner reports that the warning sign defined for a high-priority risk has just appeared, and an approved contingency plan exists within the PM’s authority. What should the project manager do NEXT?

(A) Schedule a risk workshop to reassess all risks before taking action
(B) Ask the sponsor for permission before executing the contingency plan
(C) Implement the contingency plan and update the risk register and issue log
(D) Wait to see whether the risk actually materialises before using reserves

36. A team’s burndown chart has been almost flat for three sprints, but the team reports finishing many stories every sprint. What is the MOST likely explanation, and the best next step?

(A) The team is inflating its estimates; ask it to re-estimate stories with fewer points
(B) Scope grows as fast as work is done; use a burnup and review it with the product owner
(C) The team is not working hard enough; increase daily monitoring of individual output
(D) The charting software is miscalculating progress; stop using the charts until it is fixed

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37. A programme dashboard has shown every project green for six months. Last week two projects turned red overnight, each reporting a slip of more than two months. What should the PMO do to prevent this in future?

(A) Require PMs to justify any red status in writing to the board before reporting

(B) Introduce objective, forecast-based RAG criteria and encourage early bad news

(C) Remove the red category so projects escalate through amber before failing

(D) Report status quarterly rather than monthly to reduce reporting pressure

38. Two vendors are being considered. Vendor A costs 50,000 more but has a 10% chance of a 200,000 delay penalty. Vendor B has a 40% chance of the same penalty. Based on expected monetary value, which is the better choice?

(A) Vendor B, because it is 50,000 cheaper upfront

(B) Vendor A, because it has the lowest probability of delay regardless of cost

(C) They are equal once the penalty risk is taken into account

(D) Vendor A, because its expected cost is 70,000 versus 80,000 for Vendor B

Lead — Lead — People, Teams & Communication

39. Two senior engineers have argued for a week about which database to use, and the team is now split into camps. The decision is important and the deadline is three weeks away. What should the project manager do FIRST?

(A) Meet both to understand their criteria, then facilitate an evidence-based joint decision

(B) Choose the database yourself, based on your own analysis, to end the argument

(C) Ask them to compromise by using both databases, one for each part of the system

(D) Escalate the disagreement to the sponsor and ask for a binding decision by the end of this week

40. A new team member is very enthusiastic but has never used the team's testing framework. According to situational leadership, which style should the PM use for this task?

(A) Directing — clear instructions and close follow-up

(B) Delegating — let them work it out independently

(C) Supporting — focus on encouragement with little instruction

(D) Avoiding — step back until they ask for help

41. A project team grows from 8 to 12 members. By how many communication channels does the team's communication complexity increase?

(A) 4

(B) 28

(C) 66

(D) 38

42. In a retrospective, team members say they hesitate to raise problems because a previous manager publicly criticised people for mistakes. What is the BEST action for the project manager?

(A) Reassure the team that the previous manager has left, so the problem is now solved

(B) Share your own mistakes, thank people for raising problems and use blameless reviews

(C) Make raising problems mandatory and track who contributes in each retrospective

(D) Arrange a team-building outing so that people get to know each other better

43. A functional manager refuses to release a database specialist your project needs next month, saying 'my team is fully booked'. What should the project manager do FIRST?

(A) Escalate to the sponsor and ask them to instruct the manager to release the specialist now

(B) Explore the manager's constraints and options such as partial allocation or other timing

(C) Hire an external database contractor immediately so that the schedule is not delayed

(D) Remove the database work from scope and report the change to the steering committee

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44. The project manager is negotiating a hardware contract. The supplier insists on a 15% price increase. The PM has a confirmed offer from another qualified supplier at a 5% increase. How should this alternative be used?

(A) As the PM's BATNA, setting a walk-away point and strengthening the negotiation
(B) Keep it hidden, because revealing alternatives weakens a negotiating position
(C) As a reason to end the negotiation now and sign with the other supplier
(D) As leverage to threaten reporting the supplier to the industry regulator

45. A steering committee member says your 8-page weekly status report is 'too long to read', and decisions are being delayed. What should you do?

(A) Redesign it as a one-page, decision-first summary with links, agreed with key readers
(B) Keep the report as it is, since all of the information it contains is needed for governance
(C) Stop sending written reports and rely on discussion at steering committee meetings
(D) Send the report daily so the information is fresher and easier to act on

Scale — Scale — AI-Powered Project Management, Change & Governance

46. A project manager uses an AI assistant to draft a status summary. The draft states that the project is '12% under budget', but the PM is not sure where that figure came from. What should the PM do BEFORE sending it?

(A) Send it as drafted, since AI tools are usually accurate with simple numbers
(B) Remove all the numbers from the report to avoid any risk of inaccuracy
(C) Verify the figure against the cost reports and correct the draft if needed
(D) Ask the AI assistant to confirm the figure and cite where it came from

47. A team member wants to upload a spreadsheet of customer names, phone numbers and complaints to a free online AI tool to get a summary of complaint themes. The organisation has an approved enterprise AI assistant. What should the project manager advise?

(A) Use the free tool, because generating summaries does not store any data
(B) Avoid AI altogether and have the team read every single complaint manually
(C) Use the free tool, but upload only half of the data to limit exposure
(D) Use the approved enterprise tool and mask personal data before analysis

48. Six weeks after a new procurement system went live, usage is low. Surveys show users understand why the change matters and want it, but feel unsure how to complete key tasks. Which ADKAR element should the next intervention target?

(A) Awareness — more communication about why the change is needed
(B) Knowledge and Ability — hands-on training and floor support
(C) Desire — incentives to use the new system
(D) Reinforcement — recognition for heavy users

49. An organisation wants a PMO that sets mandatory reporting standards and reviews project compliance, but leaves day-to-day management with project managers in business units. Which PMO type fits?

(A) Supportive
(B) Directive
(C) Controlling
(D) Portfolio-only

50. A project's forecast shows it will finish 2% over budget. The governance board has set a cost tolerance of ±5%. What should the project manager do?

(A) Request an urgent exception meeting with the board to approve the overrun
(B) Manage the variance within the project and report it in the normal cycle
(C) Stop non-essential work until the board formally approves the overrun
(D) Re-baseline the budget immediately to reflect the new cost forecast

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51. A senior supervisor publicly criticises a new digital process, and adoption on her team is the lowest in the organisation. What should the project manager do FIRST?

(A) Report her behaviour to HR as a breach of the change programme's standards

(B) Meet her privately to understand her concerns and involve her in solving them

(C) Exclude her team from the rollout until adoption elsewhere is fully complete

(D) Double the training sessions for her team to build skills and confidence

52. An AI model that prioritises maintenance work orders performed well at launch, but after a year planners report that its recommendations are increasingly wrong. What is the MOST likely cause and best response?

(A) Scope creep; freeze the requirements and re-baseline the model's features

(B) A hallucination problem; rewrite the prompts and add output guardrails

(C) Model drift; monitor performance on current data and plan retraining

(D) Team resistance; run change management workshops with the planners

Close — Close — Closing, Benefits, Ethics & Certification Readiness

53. A software system has been deployed and users are working with it. The vendor is asking for final payment, but no formal acceptance has been recorded. What should the project manager do FIRST?

(A) Approve final payment, since users are already working with the system

(B) Withhold payment until every outstanding enhancement request is completed

(C) Archive the project documents and pass the vendor's invoice to finance for payment

(D) Hold a formal acceptance review against the agreed criteria and obtain sign-off

54. Halfway through a project, the organisation's strategy changes and the sponsor cancels the project. What should the project manager do?

(A) Stop all activity immediately and reassign the team, without further documentation

(B) Continue the work until the original end date to protect the investment already made

(C) Ask the team to keep working informally in case the sponsor later reverses the decision

(D) Carry out closure: document what was done, close contracts, capture lessons, archive

55. During closure, the team identifies that late involvement of the security team caused a three-week delay. What is the BEST way to make sure this lesson is applied?

(A) Record it in the final project report with a clear description of the impact and cause

(B) Update templates or gate criteria to require early security review, with an owner

(C) Email the lesson to all project managers and ask them to apply it on new projects

(D) Discuss it openly at the closing celebration so the whole team remembers it

56. Six months after closure, the business owner reports that a planned 30% reduction in processing time has not been achieved. Who is primarily responsible for managing the realisation of this benefit?

(A) The business owner of the benefit, with support from others as needed

(B) The project manager of the closed project, who planned the benefit

(C) The vendor that built the system, under its warranty obligations

(D) The PMO, which owns and tracks all benefits across the organisation

57. A vendor bidding on your project offers you free tickets to an international cricket match while the evaluation is under way. What should you do?

(A) Accept, as long as you continue to score the vendor fairly and objectively

(B) Accept and share the tickets with your team so that no one person benefits

(C) Accept, and disclose the gift after the contract is awarded

(D) Decline the offer, follow the gift policy and report it if required

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58. Before a board meeting, the sponsor asks the PM to report the project as amber instead of red, promising that the problems will be fixed next month. What should the PM do?

(A) Report the true status and offer a recovery plan with the decisions needed

(B) Report amber as asked, since the sponsor is accountable for the project

(C) Report green to avoid alarming the board before the fixes are in place

(D) Avoid attending the board meeting and let the sponsor present the status

59. On an agile project, a stakeholder at the sprint review asks for a new report that is not in the backlog. What should the project manager do FIRST?

(A) Add the report to the current sprint immediately to satisfy the stakeholder

(B) Raise a formal change request and submit it to the change control board

(C) Work with the product owner to add it to the product backlog for prioritisation

(D) Tell the stakeholder that the release scope is fixed and cannot be changed now

60. A project manager is preparing for a job interview. Which portfolio entry is MOST likely to demonstrate competence?

(A) ‘Responsible for project documentation, meetings, risk logs and stakeholder reporting’

(B) ‘Excellent communicator and team player with strong leadership skills’

(C) ‘Familiar with many project management tools, methods and frameworks’

(D) ‘Led a 10-person team to migrate 600 courses on time; support tickets fell 35%’

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EXAM PREPARATION

Practice exam answers & explanations

1. B The charter formally authorises the project and the PM's use of resources. Workshops, detailed planning and vendor selection all assume authority and budget that do not yet exist.

2. B The business case is the yardstick for continued investment; when its assumptions change, decision-makers must review it. Delivering on plan does not create value if the justification has gone. Speeding up or cutting testing ignores the real issue.

3. D Each component needs the approach that matches its uncertainty, while integrated milestones manage dependencies. Fully predictive freezes uncertain features; fully agile ignores the fixed regulatory deadline; postponing delays value unnecessarily.

4. B Success in modern project management is judged by outcomes, benefits and value. Meeting time and cost constraints is necessary but not sufficient. Nothing suggests estimating or procurement problems.

5. D Investment decisions should weigh strategic fit, financial value and risk together. ROI alone ignores strategy and the timing of money; halving budgets usually means neither project delivers; withholding analysis fails the decision-makers.

6. A Tailoring fits the approach to the context while keeping governance informed and in agreement. Applying everything wastes effort; ignoring governance creates compliance risk; writing documents after the fact adds no value.

7. B Success criteria belong in the charter at initiation so everyone agrees on what success means from the start. The other documents serve different purposes and come later.

8. A Choosing and balancing which projects to fund — and stopping overlap — is the role of portfolio management. Individual PMs and operations lack the cross-project view; quality control inspects products, not investment choices.

9. B First understand the stakeholder and their actual requirements; only then can impact be assessed and a change request raised if needed. Escalating before understanding is premature, and ignoring compliance creates serious risk.

10. A High power and low interest calls for keeping the stakeholder satisfied without overloading them. Weekly detailed meetings or mass emails overload; minimal monitoring underestimates their power.

11. D Structured facilitation balances participation, and follow-up elicitation captures users' real needs. Deferring to rank misses user needs; abandoning the session wastes it; excluding managers damages relationships and loses expertise.

12. C Vertical slices keep stories small, valuable and testable within one sprint. Layer splits deliver no user value on their own; multi-sprint stories hide progress; heavy specification delays learning.

13. C Changes to a baselined scope go through change control, however small, so impacts are understood and an authorised decision is made. Quietly adding work is scope creep; refusing outright ignores possible value; hiding it breaks governance.

14. C Adding unrequested extras is gold plating, which adds cost and risk. It is not stakeholder-driven scope creep, nor normal elaboration; rewarding it would encourage more unapproved work.

15. D WSJF = cost of delay ÷ job size: P = 24 ÷ 12 = 2.0, Q = 15 ÷ 5 = 3.0. The higher score goes first. Looking at cost of delay or size alone ignores the ratio WSJF relies on.

16. A A deliverable-oriented WBS supports the 100% rule, ownership and acceptance; activities belong in the schedule. Over-decomposition adds overhead, and a Gantt chart shows timing rather than scope decomposition.

17. D PERT = (6 + 4×9 + 18) ÷ 6 = 60 ÷ 6 = 10 days; σ = (18 − 6) ÷ 6 = 2 days. The most likely value ignores the risk of overrun, 11 days is the simple triangular mean, and reporting the pessimistic value overstates the estimate.

18. C A 3-day delay within 5 days of total float does not delay the project, but zero free float means its successor starts later, consuming shared float. The activity becomes critical only if the delay uses all its float.

19. A Compression decisions should be based on the critical path and made with visible trade-offs. Blanket overtime may target non-critical work, cutting testing shifts risk to users, and declaring it impossible without analysis is premature.

20. B 186 ÷ 34 ≈ 5.5 and 186 ÷ 28 ≈ 6.6, so about 6–7 sprints. A range reflects uncertainty and is refined as data arrives. A single number hides uncertainty; velocity targets inflate estimates; agile teams do forecast.

21. C Contingency protects against identified risks; new scope requires change control and an authorised funding decision. Spending contingency leaves risks unfunded, and using management reserve requires management approval.

22. D A public commitment should carry high confidence; P80 means about 80% of simulated outcomes finished by then. The deterministic date has only a 25% chance; the earliest run is very unlikely; refusing a date does not meet the sponsor's need.

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23. D When a resource limit is hard and the activities are critical, levelling delays one of them and can push out the end date. Smoothing only moves work within float, fast-tracking increases overlap, and crashing adds resources rather than resolving a single-specialist limit.

24. B The product owner owns the order of work, and the developers manage the sprint plan to protect the sprint goal. Seniority does not bypass this; only the product owner can cancel a sprint; silent additions destroy transparency.

25. C Swarming on the bottleneck improves flow — 'stop starting, start finishing'. Raising or removing limits hides the constraint, and starting new work increases WIP and lengthens cycle time.

26. B By Little's Law, cycle time = WIP ÷ throughput: 30 ÷ 10 = 3 weeks; at the same throughput, 15 ÷ 10 = 1.5 weeks. Overtime is unsustainable, extra columns do not change flow, and changing the metric does not change performance.

27. A The definition of done is the quality commitment for the increment; scope can flex but quality cannot. Skipping tests or marking incomplete work as done misrepresents progress, and sprint length should stay fixed.

28. A Cost-reimbursable contracts suit high uncertainty, and an incentive fee shares overruns and savings, motivating cost control. Fixed-price types need clear scope; a simple purchase order fits standard goods.

29. A A Pareto chart ranks causes by frequency so the team can focus on the vital few. A scatter diagram tests relationships between variables, a flowchart shows process steps, and a control chart shows process stability over time.

30. C The PMO's real need is visibility and integration; a milestone roadmap with range-based release forecasts meets it using artefacts suited to adaptive work. A task-level six-month plan would be fiction, refusing is untrue, and a plan made without the team will not be owned or followed.

31. A The event is uncertain and in the future, so it is a risk to be recorded, analysed and owned. Logging it as an issue misclassifies it; replacing the API immediately skips analysis; ignoring it leaves a potentially serious threat unmanaged.

32. A SPI = EV ÷ PV = 180,000 ÷ 200,000 = 0.90 (behind). CPI = EV ÷ AC = 180,000 ÷ 225,000 = 0.80 (over budget). The other options invert one or both ratios.

33. C CPI = 300,000 ÷ 400,000 = 0.75, so EAC = BAC ÷ CPI = 800,000 ÷ 0.75 ≈ 1,066,667. 900,000 is the atypical formula AC + (BAC − EV); the other values do not follow from any standard formula here.

34. C TCPI = (BAC − EV) ÷ (BAC − AC) = 300,000 ÷ 250,000 = 1.20 — every unit of money must produce 20% more work than planned. 0.80 is the current CPI, and 0.83 and 1.00 do not reflect the remaining work and funds.

35. C Triggers exist so that pre-approved responses can be executed promptly. A workshop or seeking fresh permission wastes the planning already done, and waiting defeats the purpose of having a trigger.

36. B A flat burndown with steady delivery usually means scope is growing. A burnup shows completed work and total scope separately, enabling a scope conversation with the product owner. The other options assume bad faith or tool failure without evidence.

37. B This is watermelon reporting. Objective thresholds tied to forecasts, plus a culture that welcomes bad news early, keep status honest. Adding justification burdens to red, removing red or reporting less often all make hidden problems more likely.

38. D A: 50,000 + (0.10 × 200,000) = 70,000. B: 0 + (0.40 × 200,000) = 80,000. A is better by 10,000 in expected cost. Upfront price or probability alone ignores the full picture, and the values are not equal.

39. A The issue is important and there is time, so collaborating on interests and objective criteria gives the best, most committed outcome. Forcing a choice loses buy-in, a forced compromise may produce a poor technical result, and escalation is premature.

40. A High enthusiasm with low competence calls for directing. Delegating and supporting assume skills they do not yet have, and waiting leaves them to struggle.

41. D 8 people: 8 × 7 ÷ 2 = 28 channels. 12 people: 12 × 11 ÷ 2 = 66 channels. The increase is 66 − 28 = 38. The other options are the head-count change or one of the totals.

42. B Psychological safety is rebuilt through consistent leader behaviour and blameless practices. Simply stating that things have changed, tracking compliance or a one-off social event does not create lasting safety.

43. B Negotiate on interests: understanding why the manager says no often reveals workable options. Escalation, hiring or cutting scope may be needed later but skip the negotiation the PM can do first.

44. A A strong BATNA sets the walk-away point and strengthens the negotiating position. Hiding it gives no benefit, ending talks immediately may forgo a better deal, and threatening regulators is inappropriate.

45. A Tailoring communication to audience needs is the sender's responsibility. Keeping the format ignores the feedback, dropping written reports removes a record, and increasing frequency worsens the overload.

46. C AI-generated figures must be checked against authoritative data; the PM remains accountable. Assuming accuracy, removing all numbers or asking the same tool to confirm itself does not verify anything.

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47. D Approved tools plus data minimisation protect customers and the organisation. Free consumer tools may retain data, avoiding AI altogether wastes a legitimate capability, and uploading half the data still exposes personal information.

48. B Awareness and desire are already present; the first barrier is knowing how and being able to perform. More why-messaging, incentives or recognition will not fix a skills gap.

49. C A controlling PMO requires compliance and assures quality with moderate control. A supportive PMO only advises; a directive PMO manages projects directly; 'portfolio-only' is not one of the standard types.

50. B Within tolerance, the PM manages by exception and reports normally. An urgent board meeting, stopping work or re-baselining are reactions reserved for forecasts outside tolerance or approved changes.

51. B Resistance often contains useful information, and influential sceptics can become strong champions. Disciplinary action, exclusion or more training skip understanding the root cause.

52. C Performance decline as real-world data changes is model drift, handled through monitoring and retraining. Scope creep, prompt rewriting and change workshops do not address degraded model accuracy.

53. D Formal acceptance against agreed criteria should precede contract closure and payment. Paying without acceptance invites disputes, demanding new enhancements is outside the contract, and archiving comes after acceptance.

54. D Early termination still requires orderly closure. Walking away loses knowledge and creates contractual risk; continuing or working informally ignores a legitimate governance decision.

55. B A lesson is learned only when practice changes; embedding it in templates or gate criteria with an owner makes it stick. A report entry, an email or a conversation alone is easily forgotten.

56. A Benefits arise in operations and are owned by a named business owner, who can call on the former PM or others for help. The PM's accountability ended with the project; vendors and PMOs support but do not own the benefit.

57. D Accepting a valuable gift during an evaluation creates undue influence or its appearance. Fair scoring, sharing with others or later disclosure do not remove the conflict.

58. A Honesty requires accurate reporting; adding a recovery plan keeps the conversation constructive. Changing the status at anyone's request misleads the board, and avoidance leaves the board uninformed.

59. C In agile delivery, new requests go to the product owner, who prioritises the backlog. Changing the current sprint disrupts the team's commitment, a change control board is a predictive mechanism, and declaring scope fixed contradicts adaptive practice.

60. D Context, action and a measured result provide evidence of capability. Task lists, generic adjectives and tool familiarity do not show outcomes.

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NOTES

My notes

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